

AP EAMCET-2019-Engineering-21 Apr-Shift-2
Mathematics

1. Let A and B be finite sets and P_A and P_B respectively denotes their power sets. If P_B has 112 elements more than those in P_A , then the number of functions from A to B which are injective is

- (1) 224
- (2) 56
- (3) 120
- (4) 840

2. Let $D = \left\{ x \in \mathbb{R} : f(x) = \sqrt{\frac{x - |x|}{x - [x]}} \text{ is defined} \right\}$ and C be the range

of the real function $g(x) = \frac{2x}{4 + x^2}$. Then $D \cap C =$

- (1) $\left[-\frac{1}{2}, \frac{1}{2} \right]$
- (2) $\left(0, \frac{1}{2} \right]$
- (3) $\mathbb{R}^+$
- (4) $\mathbb{R}^+ - \mathbb{Z}^+$

3. Which of the following is divisible by $x^2 - y^2$, $\forall x \neq y$?

(1) $x^n - y^n, \forall n \in \mathbb{N}$

(2) $x^n + y^n, \forall n \in \mathbb{N}$

(3) $(x^n - y^n)(x^{2n+1} + y^{2n+1}), \forall n \in \mathbb{N}$

(4) $(x^n - y^n)(x^m + y^m), \forall m, n \in \mathbb{N}$

4. If $A = \begin{bmatrix} p & q & r \\ r & p & q \\ q & r & p \end{bmatrix}$ and $A A^T = I$ then, $p^3 + q^3 + r^3 =$

(1) ± 1

(2) pqr

(3) $3pqr$

(4) $3pqr \pm 1$

5. Match the items of List-I with the items of List-II and choose the correct option.

List-I

List-II

A. If A is non singular matrix of order I

Null matrix

3 and $|A| = a$, then $\left| \left(\text{adj} A^{-1} \right)^{-1} \right| =$

B. A is non singular matrix of order 3 II a^2
and B is any matrix of order 3 such
that $AB = 0$, then B is

C. $\begin{vmatrix} 1 & x & x^2 \\ \cos(a-b)y & \cos ay & \cos(a+b)y \\ \sin(a-b)y & \sin ay & \sin(a+b)y \end{vmatrix}$ III b

does not depend on

D. A is a square matrix of order 3 and IV a
 $B = A - A^T$, then $|B|$ is

V 0

The correct answer is

	A	B	C	D
(1)	II	IV	III	I
	A	B	C	D
(2)	III	I	IV	V
	A	B	C	D
(3)	II	V	III	I
	A	B	C	D
(4)	II	I	IV	V

6. The solution of the linear system of equations

$$\begin{bmatrix} 2 & 2 & 3 \\ 7 & 1 & 1 \\ 0 & 6 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3y+11 \\ 6z-1 \\ 5y+11 \end{bmatrix} + \begin{bmatrix} x \\ x \\ 4z \end{bmatrix} + \begin{bmatrix} z \\ 3x \\ 4y \end{bmatrix} \text{ is}$$

(1) $x = 4, y = -3, z = 2$

(2) $x = 2, y = 1, z = 1$

(3) $x = 1, y = -1, z = 2$

(4) $x = 2, y = -4, z = 3$

7. If a, b are the least and the greatest values respectively of $|z_1 + z_2|$, where $z_1 = 12 + 5i$ and $|z_2| = 9$, then $a^2 + b^2 =$

(1) 468

(2) 500

(3) 250

(4) 450

8. If a complex number z is such that

$$(7+i)(z+\bar{z}) - (4+i)(z-\bar{z}) + 116i = 0, \text{ then } z \cdot \bar{z} =$$

(1) 400

(2) 300

(3) 200

(4) 100

9. Let the point P represents $z = x + iy, x, y \in \mathbb{R}$ in the Argand plane. Let the curves C_1 and C_2 be the loci of P satisfying the conditions

(i) $\frac{2z+i}{z-2}$ is purely imaginary and

(ii) $\text{Arg}\left(\frac{z+i}{z+1}\right) = \frac{\pi}{2}$ respectively. Then the point of intersection of the curves C_1 and C_2 , other than the origin, is

(1) (1, 2)

(2) $\left(\frac{2}{7}, \frac{-5}{7}\right)$

(3) (-3, 4)

(4) $\left(\frac{5}{37}, -\frac{30}{37}\right)$

10. If $z = \cos 6^\circ + i \sin 6^\circ$, $\sum_{n=1}^{20} \text{Im}(z^{2n-1}) =$

(1) 0

(2) -1

(3) $\frac{-3}{4\sin 6^\circ}$

(4) $\frac{3}{4\sin 6^\circ}$

11. If α, β are the real roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$, then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always

(1) two positive roots

(2) two negative roots

(3) one positive and one negative roots

(4) two real roots

12. If $\frac{x-P}{x^2-3x+2}$ takes all real values for $x \in \mathbb{R}$, then the range of P is

(1) $1 \leq P \leq 2$

(2) $1 < P < 2$

(3) $P < 1$ or $P > 2$

(4) $P \geq 2$ or $P \leq 1$

$$13. \left\{ x \in \mathbb{R} : \frac{\sqrt{6+x-x^2}}{2x+5} \geq \frac{\sqrt{6+x-x^2}}{x+4} \right\} =$$

(1) $[-2, 3]$

(2) $(-\infty, -4] \cup \left[\frac{-5}{2}, -1 \right]$

(3) $[-2, -1] \cup \{3\}$

(4) $(-\infty, -4] \cup [-2, -1]$

14. Let θ be an acute angle such that the equation $x^3 + 4x^2 \cos \theta + x \cot \theta = 0$ has multiple roots. Then the value of θ (in radians) is

(1) $\frac{\pi}{3}$

(2) $\frac{\pi}{8}$

(3) $\frac{\pi}{12}$ or $\frac{5\pi}{12}$

(4) $\frac{\pi}{6}$ or $\frac{5\pi}{12}$

15. Six persons A, B, C, D, E and F are to be seated at a circular table facing towards the centre. Then the number of ways that can be done if A must have either E and F on his immediate right and E must have either F and D on his immediate right, is

- (1) 18
- (2) 30
- (3) 12
- (4) 24

16. Number of ways of forming a committee of 6 members out of 5 Indians, 5 Americans and 5 Australians such that there will be at least one member from each country in the committee is

- (1) 3375
- (2) 4375
- (3) 3875
- (4) 4250

17. If 'a' is the middle term in the expansion of $(2x - 3y)^8$ and b, c are the middle terms in the expansion of $(3x + 4y)^7$, then the

value of $\frac{b+c}{a}$, when $x = 2$ and $y = 3$, is

(1) $\frac{1}{2}$

(2) $\frac{2}{3}$

(3) 1

(4) 2

18. The first negative coefficient in the terms occurring in the expansion of $(1+x)^{\frac{21}{5}}$ is

(1) $\frac{-6160}{15625}$

(2) $\frac{-416}{3125}$

(3) $\frac{-616}{5^7}$

(4) $\frac{-616}{5^6}$

19. When $|x| < \frac{1}{2}$, the coefficient of x^4 in the expansion of

$\frac{3x^2 - 5x + 3}{(x-1)(2x+1)(x+3)}$ is

(1) $\frac{722}{27}$

(2) $\frac{724}{27}$

(3) $\frac{-722}{27}$

(4) $\frac{-724}{27}$

20. Let $x = a \sin^{\alpha} \theta \cos^{\alpha+1} \theta$, $y = a \sin^{\alpha+1} \theta \cos^{\alpha} \theta$, $\left(\theta \neq \frac{n\pi}{2} \right)$. If

$\frac{(x^2 + y^2)^m}{(xy)^n}$ is independent of θ , then the relation between α, m

and n is

(1) $2m\alpha = n(2\alpha + 1)$

(2) $m + n = \alpha$

(3) $3 = 2m\alpha = 2n\alpha + m$

(4) $2m = (2n + 1)\alpha$

21. Assertion (A): If $\sqrt{4\sin^4 \theta + \sin^2 2\theta} + 4\cos^2 \left(\frac{\pi}{4} - \frac{\theta}{2} \right) = 2$, then θ

lies in 3rd quadrant of 4th quadrant.

Reason (R): $\sqrt{\sin^2 \theta} = \sin \theta$

(1) Both (A) and (R) are correct (R) is a correct explanation of (A)

(2) Both (A) and (R) are correct (R) is not correct explanation of (A)

(3) (A) is correct, but (R) is false

(4) (A) is incorrect, but (R) is false

22. If $x = \frac{\sin^3 \theta}{\cos^2 \theta}$ and $y = \frac{\cos^3 \theta}{\sin^2 \theta}$, where $\sin \theta + \cos \theta = \frac{1}{2}$, then

$x + y =$

(1) $\frac{48}{9}$

(2) $\frac{34}{9}$

(3) $\frac{65}{18}$

(4) $\frac{79}{18}$

23. If $4(\sin 2x \sin 4x + \sin^2 x) = 3$, then $x =$

(1) $\frac{2n\pi}{3} \pm \frac{\pi}{9}, n \in \mathbb{Z}$

(2) $\frac{n\pi}{3} \pm \frac{\pi}{9}, n \in \mathbb{Z}$

(3) $\frac{n\pi}{3} + (-1)^n \frac{\pi}{9}, n \in \mathbb{Z}$

(4) $\frac{n\pi}{3} + (-1)^n \frac{2\pi}{9}, n \in \mathbb{Z}$

24. If $\sum_{k=1}^n \tan^{-1}\left(\frac{1}{k^2 + k + 1}\right) = \tan^{-1}(\theta)$, then $\theta =$

(1) $\frac{n}{n+2}$

(2) $\frac{n}{n+1}$

(3) 1

(4) $\frac{n}{n-1}$

25. $e^{\left(\operatorname{sech}^{-1} \frac{1}{2} + \tanh^{-1} + \sinh^{-1} \frac{1}{2}\right)} =$

$$(1) \frac{2+3\sqrt{3}+2\sqrt{5}+3\sqrt{15}}{2}$$

$$(2) \frac{3+2\sqrt{3}+3\sqrt{5}+2\sqrt{15}}{2}$$

$$(3) \frac{2+3\sqrt{3}+4\sqrt{5}+5\sqrt{15}}{2}$$

$$(4) \frac{2+3\sqrt{3}-4\sqrt{5}+5\sqrt{15}}{2}$$

26. In $\triangle ABC$, if $a : b : c = 3 : 5 : 7$, then $\cos A + \cos B =$

$$(1) \frac{13}{7}$$

$$(2) \frac{11}{7}$$

$$(3) \frac{12}{7}$$

$$(4) \frac{10}{7}$$

27. If ABCD is a cyclic quadrilateral with $AB = 6$, $BC = 4$, $CD = 5$, $DA = 3$ and $\angle ABC = \theta$, then $\cos \theta =$

$$(1) \frac{3}{13}$$

(2) $\frac{18}{76}$

(3) $\frac{16}{76}$

(4) $\frac{78}{86}$

28. Let a triangle ABC be inscribed in a circle of radius 2 units. If the 3 bisectors of the angles A, B and C are extended to cut the circle at A_1 , B_1 and C_1 respectively, then the value of

$$\left[\frac{AA_1 \cos \frac{A}{2} + BB_1 \cos \frac{B}{2} + CC_1 \cos \frac{C}{2}}{\sin A + \sin B + \sin C} \right]^2 =$$

(1) 4

(2) 16

(3) 25

(4) 1

29. Let D and E be the midpoints of the sides AC and BC of a triangle ABC respectively. If O is an interior point of the

triangle ABC such that $\overrightarrow{OA} + 2\overrightarrow{OB} + 3\overrightarrow{OC} = \overrightarrow{0}$, then the area (in sq units) of the triangle *ODE* is

- (1) 6
- (2) 5
- (3) 3/4
- (4) 0

30. The vector equation of the plane passing through the points $(1, -2, 5)$, $(0, -5, -1)$ and $(-3, 5, 0)$ is

- (1) $\vec{r} = (1 - \lambda - 4\mu)\vec{i} - (2 + 3\lambda - 7\mu)\vec{j} + (5 - 6\lambda - 5\mu)\vec{k}$
- (2) $\vec{r} = (1 + \lambda + 4\mu)\vec{i} - (2 - 3\lambda + 7\mu)\vec{j} + (5 - 6\lambda - 5\mu)\vec{k}$
- (3) $\vec{r} = (1 - \lambda + 4\mu)\vec{i} - (2 + 3\lambda + 7\mu)\vec{j} + (5 - 6\lambda + 5\mu)\vec{k}$
- (4) $\vec{r} = (1 - \lambda - 4\mu)\vec{i} - (2 + 3\lambda - 7\mu)\vec{j} + (5 - 6\lambda - 5\mu)\vec{k}$

31. The angle made by the vector $2\vec{i} - \vec{j} + \vec{k}$ with the plane represented by $\vec{r} \cdot (\vec{i} + \vec{j} + 2\vec{k}) = 7$ is

- (1) 30°
- (2) 60°
- (3) 45°
- (4) 75°

32. If $\vec{a}, \vec{b}, \vec{c}$ are non-zero, non collinear vectors and

$$\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}, \text{ then } \vec{a} + \vec{b} + \vec{c} =$$

- (1) $3\vec{a}$
- (2) 0
- (3) $3(\vec{a} + \vec{b})$
- (4) $3(\vec{b} \times \vec{c})$

33. If $\vec{V} = 2\vec{i} + \vec{j} - \vec{k}$, $\vec{W} = \vec{i} + 3\vec{k}$ and $\vec{U}$ is a unit vector, then the maximum value of $[\vec{U}\vec{V}\vec{W}]$ is

- (1) $\sqrt{57}$
- (2) $\sqrt{59}$
- (3) $\sqrt{60}$
- (4) $\sqrt{10} + \sqrt{6}$

34. Assertion (A): If $\vec{a}, \vec{b}$ are two non collinear vectors, then the vector component of $\vec{b}$ along the line perpendicular to $\vec{a}$ is
$$\frac{\vec{a} \times (\vec{b} \times \vec{a})}{|\vec{a}|^2}$$

Reason (R): $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ and vector component of $\vec{b}$ and $\vec{c}$ is $\left(\vec{b} \cdot \frac{\vec{c}}{|\vec{c}|} \right) \frac{\vec{c}}{|\vec{c}|}$

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct (R) is not correct explanation of (A)
- (3) (A) is correct, but (R) is false
- (4) (A) is incorrect, but (R) is true

35. The standard deviations of $x_i (i = 1, 2, \dots, 10)$ and $y_i (i = 1, \dots, 10)$ are respectively ' a ' and ' b '. $\bar{x}, \bar{y}$ are the means of these two sets of observations respectively. If $z_i = (x_i - \bar{x})(y_i - \bar{y})$ and

$\sum_{i=1}^{10} z_i = c$, then the standard deviation of the observations $(x_i - y_i)$, $(i = 1, 2, \dots, 10)$ is

(1) $\sqrt{a^2 + b^2 + \frac{c}{5}}$

(2) $\sqrt{a^2 + b^2 - \frac{c}{5}}$

(3) $\sqrt{a^2 + b^2 - \frac{c^2}{5}}$

(4) $\sqrt{a^2 + b^2 + \frac{c^2}{5}}$

36. For a group of 100 students, the mean $\bar{x}_1$ and the standard deviation σ_1 of their marks were found to be 40 and 15 respectively. Later it was observed that the scores 40 and 50 were misread as 30 and 60 respectively. If the mean and the standard deviation with the corrected observation of the scores, are $\bar{x}_2$ and σ_2 respectively, then

(1) $\bar{x}_1 = \bar{x}_2; \sigma_1 = \sigma_2$

(2) $\bar{x}_1 = \bar{x}_2; \sigma_1 < \sigma_2$

(3) $\bar{x}_1 = \bar{x}_2; \sigma_1 > \sigma_2$

(4) $\bar{x}_1 > \bar{x}_2; \sigma_1 = \sigma_2$

37. If two unbiased six-faced dice are thrown simultaneously until a sum of either 7 or 11 occurs, then the probability that 7 comes before 11 is

(1) $\frac{1}{4}$

(2) $\frac{3}{4}$

(3) $\frac{5}{9}$

(4) $\frac{5}{18}$

38. If A and B throw two dices 100 times each simultaneously, then the probability that both of them will get even number as the total at the same time in all the throws is

(1) $\left(\frac{1}{6}\right)^{100}$

(2) $\left(\frac{1}{4}\right)^{100}$

$$(3) \left(\frac{1}{2}\right)^{100}$$

$$(4) \left(\frac{3}{4}\right)^{100}$$

39. The probabilities of having a defective toy in three cartons A, B, and C are $\frac{1}{3}, \frac{1}{4}, \frac{2}{5}$ respectively. If a carton is selected at random and a toy drawn randomly from it is found to be defective, then the probability that it is found to be defective, then the probability that it is drawn from the carton B is

$$(1) \frac{15}{47}$$

$$(2) \frac{20}{47}$$

$$(3) \frac{20}{59}$$

$$(4) \frac{15}{59}$$

40. A die is thrown twice. If getting a number greater than four on the die is considered a success, then the variance of the probability distribution of the number of successes is

(1) $\frac{2}{3}$

(2) $\frac{1}{3}$

(3) $\frac{4}{9}$

(4) $\frac{8}{9}$

41. If X is a poisson variate that $2P(X=1) = 5P(X=5) + 2P(X=3)$, then the standard deviation of X is

(1) 4

(2) 2

(3) $1/2$

(4) $\sqrt{2}$

42. If the sum of the distances from a variable point P to the given points $A(1,0)$ and $B(0,1)$ is 2, then the locus of P is

(1) $3x^2 + 3y^2 - 4x - 4y = 0$

$$(2) 16x^2 + 7y^2 - 64x - 48y = 0$$

$$(3) 3x^2 + 2xy + 3y^2 - 4x - 4y = 0$$

$$(4) 16x^2 + 38xy + 7y^2 - 64x - 48y = 0$$

43. If the equation of the curve C is transformed to $9x^2 + 25y^2 = 225$ by the rotation of the coordinate axes about the origin through an $\frac{\pi}{4}$ in the positive direction then the equation of the curve C , before the transformation is

$$(1) 17x^2 + 16xy + 17y^2 = 225$$

$$(2) 17x^2 + 23y^2 = 391$$

$$(3) 17x^2 - 16xy + 17y^2 = 225$$

$$(4) 23x^2 + 17y^2 = 391$$

44. A straight line $4x + y - 1 = 0$ through the point $A(2, -7)$ meets the line BC whose equation is $3x - 4y + 1 = 0$ at the point B . Then the equation of the line AC such that $AB = AC$, is

$$(1) 89x - 52y - 162 = 0$$

$$(2) 52x + 89y + 519 = 0$$

$$(3) 4x - y - 15 = 0$$

(4) $4x + 3y + 13 = 0$

45. In a ΔABC , $2x + 3y + 1 = 0$, $x + 2y - 2 = 0$ are the perpendicular bisectors of its sides AB and AC respectively and if $A = (3, 2)$, then the equation of the side BC is

(1) $x + y - 3 = 0$

(2) $x - y - 3 = 0$

(3) $2x - y - 2 = 0$

(4) $2x + y - 2 = 0$

46. If the perpendicular bisector of the line segment joining $A(\alpha, 3)$ and $B(2, -1)$ has y-intercept 1, then $\alpha =$

(1) 0

(2) ± 1

(3) ± 2

(4) ± 3

47. The number of values of a for which the pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are at right angles to each other, is

- (1) 2
- (2) 1
- (3) Infinitely many
- (4) 0

48. If the pair of lines joining the origin and the points of intersection of the line $ax + by = 1$ and the curve $x^2 + y^2 - x - y - 1 = 0$ are at right angles, then the locus of the point (a, b) is a circle of radius

- (1) 2
- (2) $\sqrt{\frac{3}{2}}$
- (3) $\sqrt{\frac{5}{2}}$
- (4) $\frac{\sqrt{5}}{2}$

49. If the lines $x + 2y - 5 = 0$ and $2x - 3y + 4 = 0$ lie along diameter of a circle of area is 9π , then the equation of the circle is

- (1) $x^2 + y^2 - 2x - 4y - 4 = 0$
- (2) $x^2 + y^2 + 2x - 4y - 4 = 0$

$$(3) x^2 + y^2 + 2x + 4y - 4 = 0$$

$$(4) x^2 + y^2 - 2x + 4y - 4 = 0$$

50. Given that $a > 2b > 0$ and that the line $y = mx - b\sqrt{1+m^2}$ is a common tangent to the circles $x^2 + y^2 = b^2$ and $(x-a)^2 + y^2 = b^2$. Then the positive value of m is

$$(1) \frac{2b}{a-2b}$$

$$(2) \frac{b}{a-2b}$$

$$(3) \frac{\sqrt{a^2 - 4b^2}}{2b}$$

$$(4) \frac{2b}{\sqrt{a^2 - 4b^2}}$$

51. Two circles each of radius 5 units touch each other at $(1, 2)$ and $4x + 3y = 10$ is their common tangent. The equation of that circle among the two given circles, such that some portion of it lies in every quadrant is

$$(1) x^2 + y^2 + 6x + 2y + 15 = 0$$

(2) $x^2 + y^2 + 2x + 6y - 15 = 0$

(3) $x^2 + y^2 + 6x + 2y - 15 = 0$

(4) $x^2 + y^2 - 6x + 2y - 15 = 0$

52. If the angle between the circles $x^2 + y^2 + 4x - 5 = 0$ and

$x^2 + y^2 + 2\lambda y - 4 = 0$ is $\frac{\pi}{3}$, then $\lambda =$

(1) $\pm\sqrt{5}$

(2) ± 2

(3) $\pm\sqrt{3}$

(4) $\pm\sqrt{6}$

53. The equation of a circle passing through the points of intersection of the circles $x^2 + y^2 - 4x - 6y - 12 = 0$,

$x^2 + y^2 + 6x + 4y - 12 = 0$ and having radius $\sqrt{13}$ is

(1) $x^2 + y^2 - 2x - 12 = 0$

(2) $x^2 + y^2 + 2y - 12 = 0$

(3) $x^2 + y^2 - 2y - 13 = 0$

(4) $x^2 + y^2 + 2x - 13 = 0$

54. The normal at a point on the parabola $y^2 = 4x$ passes through $(5,0)$. If two more normals to this parabola also pass through $(5,0)$, then the centroid of the triangle formed by the feet of these three normals is

(1) $\left(\frac{1}{2}, \frac{1}{2}\right)$

(2) $(2,0)$

(3) $(5,0)$

(4) $(0,2)$

55. The equation of the normal to the parabola $y^2 = 4x$ which is perpendicular to

(1) $3x - y = 33$

(2) $3x - y + 33 = 0$

(3) $3x + y = 33$

(4) $3x + y + 33 = 0$

56. Let P be any point on the ellipse $7x^2 + 16y^2 = 112$, S be a focus, L be the corresponding directrix and PM be the perpendicular

distance from P to directrix L. Then $\frac{SP}{PM} =$

- (1) $\frac{1}{4}$
- (2) $\frac{1}{2}$
- (3) $\frac{3}{4}$
- (4) $\frac{1}{\sqrt{2}}$

57. If tangent are drawn to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ at the ends of latus recta, then the area of the quadrilateral thus formed is

- (1) 27
- (2) $\frac{15}{4}$
- (3) $\frac{13}{2}$
- (4) 45

58. A hyperbola with centre at $(0,0)$ has its transverse axis along X-axis whose length is 12. If $(8,2)$ is a point on the hyperbola, then its eccentricity is

- (1) $\frac{8}{7}$
- (2) $\frac{2\sqrt{2}}{7}$
- (3) $\frac{3}{\sqrt{7}}$
- (4) $\frac{9}{7}$

59. In a triangle ABC, if the mid points of sides AB, BC, CA are $(3,0,0), (0,4,0), (0,0,5)$ respectively, then $AB^2 + BC^2 + CA^2 =$

- (1) 50
- (2) 200
- (3) 300
- (4) 400

60. The angle between a line with direction ratios 2, 2, 1 and the line joining the points $(3, 1, 4)$ and $(7, 2, 12)$ is

$$(1) \cos^{-1}\left(\frac{2}{3}\right)$$

$$(2) \cos^{-1}\left(\frac{3}{4}\right)$$

$$(3) \tan^{-1}\left(\frac{-2}{3}\right)$$

$$(4) \cos^{-1}\left(\frac{1}{3}\right)$$

61. The equation of the plane in normal form which passes through the points $(-2, 1, 3)$, $(1, 1, 1)$ and $(2, 3, 4)$

$$(1) \left(\frac{2}{3}\right)x + \left(-\frac{2}{3}\right)y + \left(\frac{1}{3}\right)z = \frac{1}{3}$$

$$(2) \left(-\frac{2}{3}\right)x + \left(\frac{2}{3}\right)y + \left(-\frac{1}{3}\right)z = \frac{1}{3}$$

$$(3) \left(\frac{-4}{\sqrt{173}}\right)x + \left(\frac{11}{\sqrt{173}}\right)y + \left(\frac{-6}{\sqrt{173}}\right)z = \frac{1}{\sqrt{173}}$$

$$(4) \left(\frac{4}{\sqrt{173}}\right)x + \left(\frac{-11}{\sqrt{173}}\right)y + \left(\frac{6}{\sqrt{173}}\right)z = \frac{1}{\sqrt{173}}$$

62. If $\alpha = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x}$ and $\beta = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}}$, then

(1) $\alpha = 5\beta$

(2) $\alpha = 2\beta$

(3) $\beta = 2\alpha^2$

(4) $\beta = \frac{1}{6}\alpha$

63. $\lim_{n \rightarrow \infty} \left(\frac{1}{3 \cdot 7} + \frac{1}{7 \cdot 11} + \frac{1}{11 \cdot 15} + \dots + (n \text{ terms}) \right) =$

(1) $\frac{1}{12}$

(2) $\frac{1}{4}$

(3) $\frac{1}{3}$

(4) 0

64. $\lim_{x \rightarrow \infty} \left[\sqrt{x^2 + ax + b} - x \right] (a < 0 < b)$

(1) depends on both a and b

(2) depends only on b

(3) depends only on a

(4) does not depend on a and b

65. If α and β are such that the function $f(x)$ defined by

$$f(x) = \begin{cases} \alpha x^2 - \beta, & \text{for } |x| < 1 \\ \frac{-1}{|x|}, & \text{for } |x| \geq 1 \end{cases} \quad \text{is differentiable everywhere, then}$$

the ordered pair $(\alpha, \beta) =$

(1) $\left(\frac{-1}{2}, \frac{3}{2}\right)$

(2) $\left(\frac{1}{2}, -\frac{3}{2}\right)$

(3) $\left(\frac{1}{2}, \frac{3}{2}\right)$

(4) $\left(-\frac{1}{2}, \frac{3}{2}\right)$

66. If $y = \sin^2\left(\cot^{-1}\sqrt{\frac{1+x}{1-x}}\right)$, then $\frac{dy}{dx} =$

(1) $\frac{-1}{2}$

(2) $\frac{1}{1+x}$

(3) $\frac{1}{1-x}$

(4) 1

67. If $a \neq b, x \neq n\pi, n \in \mathbb{Z}$ and $y^2 = a^2 \cos^2 x + b^2 \sin^2 x$, then

$$\frac{d^2 y}{dx^2} + y =$$

(1) $\left(\frac{ab}{y}\right)^2$

(2) $\frac{1}{y} \left(\frac{ab}{y}\right)^2$

(3) $\frac{(ab)^2}{y}$

(4) $\frac{ab}{y^3}$

68. If $2y = 3x - 1$ is a tangent drawn to the curve $y^2 = ax^3 + b$ at

$(1, 1)$ where a, b are constants, then $(a, b) =$

(1) $(1, 0)$

(2) $(0, 1)$

(3) $(1, -1)$

(4) $(-1, 1)$

69. A ladder of 5 meters long rests against a vertical wall with the lower end on the horizontal ground. The lower end of the ladder is pulled along the ground away from the wall at the rate 3 m/sec. The height of the upper end (in meters) while it is descending at the rate of 4 m/sec, is

- (1) 1
- (2) 2
- (3) 3
- (4) 4

70. Suppose $f''(x)$ exists for all real x . If $f(2)=2$, $f(3)=5$ and $f(4)=10$, then which one among the following statements is definitely true?

- (1) $f''(x) < 1$ for some $x \in (2, 4)$
- (2) $f''(x) > 1$ for some $x \in (2, 4)$
- (3) $f''(x) = 1$ for some $x \in (2, 4)$
- (4) $f''(x) = 0$ for some $x \in (2, 4)$

71. If p and q are respectively the global maximum and global minimum of the function $f(x) = x^2 e^{2x}$ on the interval $[-2, 2]$,

then $pe^{-4} + qe^4 =$

- (1) 0
- (2) $4e^8$
- (3) 4
- (4) $4e^8 + 1$

72. $\int \frac{x + \sin x}{1 + \cos x} dx =$

- (1) $\log_e(1 + \cos x) + c$
- (2) $x \sin^2 \frac{x}{2} + c$
- (3) $\tan \frac{x}{2} + c$
- (4) $x \tan \frac{x}{2} + c$

73. $I = \int x^2 \left[\sqrt{2} \left(\sin \frac{\pi}{4} + x \right) + e^x \right] dx =$

- (1) $(x^2 + 2x - 2) \sin x + (-x^2 + 2x + 2) \cos x + (x^2 - 2x + 2) e^x + c$

$$(2) (-x^2 + 2x - 2)\sin x + (x^2 + 2x - 2)\cos x + (x^2 - 2x + 2)e^x + c$$

$$(3) (x^2 + 2x + 2)\sin x + (-x^2 - 2x - 2)\cos x + (x^2 - 2x + 2)e^x + c$$

$$(4) (x^2 - 2x - 2)\sin x + (-x^2 + 2x - 2)\cos x + (x^2 - 2x + 2)e^x + c$$

$$74. \int \frac{dx}{(x+1)^2(x^2+1)}$$

$$(1) \log_e \sqrt{x+1} + \frac{1}{2} \log_e \sqrt{x^2+1} - \frac{1}{x+1} + c$$

$$(2) \log_e \sqrt{x+1} - \frac{1}{2} \log_e \sqrt{x^2+1} - \frac{1}{2(x+1)} + c$$

$$(3) \frac{1}{2} \log_e \sqrt{x+1} - \frac{1}{4} \log_e \sqrt{x^2+1} + \frac{1}{2(x+1)} + c$$

$$(4) \frac{1}{4} \log_e \sqrt{x+1} + \frac{1}{2} \log_e \sqrt{x^2+1} + \frac{1}{2(x+1)} + c$$

$$75. \text{ For } n \geq 2, \text{ if } I_n = \int (\sin x + \cos x)^n dx \text{ then } nI_n - 2(n-1)I_{n-2} =$$

$$(1) (\sin x + \cos x)^{n+1} (\sin x - \cos x) + c$$

$$(2) (\sin x + \cos x)^n (\sin x - \cos x) + c$$

$$(3) (\sin x + \cos x)^{n-1} (\sin x - \cos x) + c$$

$$(4) (\sin x - \cos x)^{n-1} (\sin x + \cos x) + c$$

$$76. \lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \dots + \sqrt{n}}{n^{3/2}} =$$

(1) 0

(2) $2/3$

(3) 1

(4) $3/2$

$$77. \int_0^{\infty} e^{-x} \sin^m x dx =$$

(1) $\frac{24}{85}$

(2) $\frac{124}{285}$

(3) $\frac{136}{529}$

(4) $\frac{144}{629}$

78. The area (in sq units) bounded by the curve $y = x^2 + 2x + 1$ and the tangent to it at $(1, 4)$ and the Y-axis is

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) 1

(4) $\frac{2}{3}$

79. The differential equation formed by eliminating a and b from the equation $y = e^x (a \cos x + b \sin x)$ is

(1) $2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} - 2y = 0$

(2) $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 2y = 0$

(3) $2 \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 2y = 0$

(4) $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$

80. If $y = A(x)e^{-\int P dx}$ is a solution of $\frac{dy}{dx} + P(x)y = Q(x)$, then

$A'(x) =$

$$(1) e^{\int P dx}$$

$$(2) Q(x)e^{-\int P dx}$$

$$(3) \int Q(x)e^{\int P dx} dx$$

$$(4) Q(x)e^{\int P dx}$$

AP EAMCET-2019-Engineering-21 Apr-Shift-2 Physics

81. N divisions on the main scale of a vernier callipers coincide with $(N+1)$ division of the vernier scale. If each division of main scale is ' a ' units, then the least count of the callipers is,

$$(1) \frac{a}{N}$$

$$(2) a$$

$$(3) \frac{a}{N+1}$$

$$(4) \frac{Na}{N+1}$$

82. When a body is in SHM, then match the following.

List-I

List-II

(A) Velocity is maximum

(I) At extreme position

(B) KE is $\left(\frac{3}{4}\right)^{th}$ of the total energy

(II) At mean position

(C) PE is $\left(\frac{3}{4}\right)^{th}$ of the total energy

(III) At half of the amplitude

(D) Acceleration is maximum

(IV) At $\frac{\sqrt{3}}{2}$ times of the amplitude

The correct answer is

	A	B	C	D
(1)	III	I	IV	II

	A	B	C	D
(2)	I	III	IV	II

	A	B	C	D
(3)	II	III	IV	I

	A	B	C	D
(4)	II	I	IV	III

83. Ship 'A' is moving Westwards with a speed of 20 kmh^{-1} and another ship 'B' which is at 200 km South of "A" is moving Northwards with a speed of 10 kmh^{-1} . The time after which the distance between them is shortest and the shortest distance between them

- (1) 4 h, $80\sqrt{5} \text{ km}$
- (2) $50\sqrt{2} \text{ h}$, $\sqrt{10} \text{ km}$
- (3) $100\sqrt{2} \text{ h}$, $2\sqrt{10} \text{ km}$
- (4) $80\sqrt{5} \text{ h}$, 4 km

84. A body is projected at an angle of 60° with the horizontal such that the vertical component of its initial velocity is 40 ms^{-1} . The magnitude of velocity of the projectile at one quarter of its time of flight is nearly

(Acceleration due to gravity = 10 ms^{-2})

- (1) 3.54 ms^{-1}
- (2) 35.40 ms^{-1}

(3) 30.54 ms^{-1}

(4) 34.5 ms^{-1}

85. A block of mass 48 kg kept on a smooth horizontal surface is pulled by a rope of length 4 m by a horizontal force of 25 N applied to the other end. If the linear density of the rope is 0.5 kg m^{-1} , the force acting on the block is

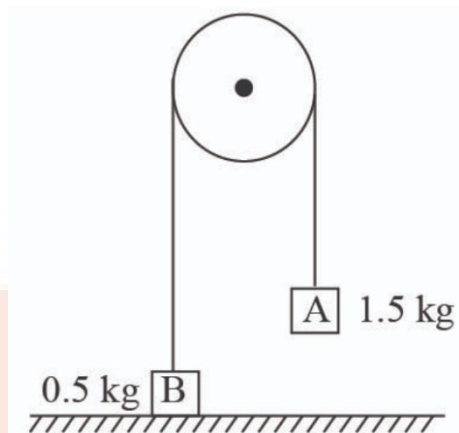
(1) 24 N

(2) 25 N

(3) 12 N

(4) 13 N

86. Two blocks A and B of masses 1.5 kg and 0.5 kg respectively are connected by a massless inextensible string spring over a frictionless pulley as shown in the figure. Block A is lifted block B touches the ground and then block A is released. The initial height of block A is 80 cm when block B just touches the ground. The maximum height reached by block B from the ground after the block A falls on the ground is



- (1) 80 cm
- (2) 120 cm
- (3) 140 cm
- (4) 160 cm

87. A body is released from a height of 30 m vertically downwards. The speed of the body at which potential energy is twice that of kinetic energy is

(Acceleration due to gravity = 10 ms^{-2})

- (1) $20\sqrt{2} \text{ ms}^{-1}$
- (2) $10\sqrt{2} \text{ ms}^{-1}$
- (3) 10 ms^{-1}
- (4) 20 ms^{-1}

88. The work done by a force $\vec{F} = -5x^4\hat{i}$ N in displacing a body from $x = 2$ m to $x = -2$ m is

- (1) 6 J
- (2) 8 J
- (3) 64 J
- (4) 0 J

89. A wheel having moment of inertia 2 kg m^2 about the same axis. The torque required to stop the wheel in one minute is

- (1) $\frac{\pi}{10} \text{ Nm}$
- (2) $\frac{\pi}{18} \text{ Nm}$
- (3) $-\frac{\pi}{12} \text{ Nm}$
- (4) $\frac{3\pi}{8} \text{ Nm}$

90. The angular retardation of a rotating flywheel is proportional to the angle through which it rotates. If its kinetic energy gets reduced by ΔE while it rotates through an angle ' θ ' then,

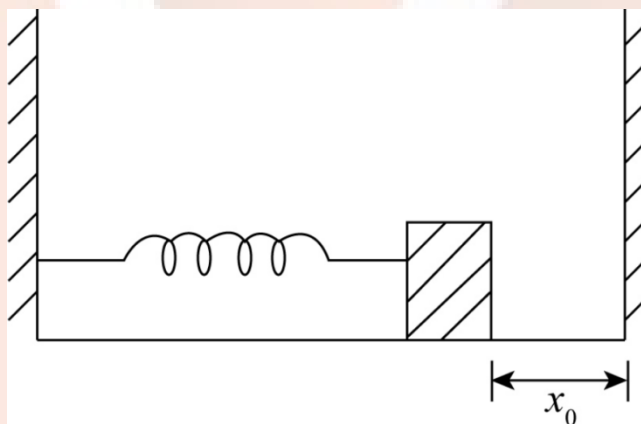
- (1) $\Delta E \propto \theta^2$

$$(2) \Delta E \propto \sqrt{\theta}$$

$$(3) \Delta E \propto \theta$$

$$(4) \Delta E \propto \theta^{\frac{3}{2}}$$

91. One end of a spring force constant ' k ' is fixed to a vertical wall and the other to a block of mass ' m ' resting on a smooth horizontal surface. There is another wall at a distance ' x_0 ' from the block. The spring is then compressed by $2x_0$ and released. The time taken by the block to strike the other wall is



$$(1) \frac{1}{6}\pi\sqrt{\frac{m}{k}}$$

$$(2) \sqrt{\frac{m}{k}}$$

$$(3) \frac{2\pi}{3}\sqrt{\frac{m}{k}}$$

$$(4) \frac{\pi}{4} \sqrt{\frac{m}{k}}$$

92. The amplitude of a damped oscillator becomes half in one minute. The amplitude after 3 minutes will be $\frac{1}{x}$ times the original. Then x is

- (1) 4
- (2) 8
- (3) 6
- (4) 12

93. A uniform rod of length ' l ' and density ' ρ ' is revolving about a vertical axis passing through its one end. If ' ω ' is the angular velocity of the rod then the centrifugal force per unit area of the rod is

- (1) $\frac{\rho\omega^2 l^2}{4}$
- (2) $\frac{\rho\omega^2 l^2}{12}$
- (3) $\frac{\rho\omega^2 l^2}{2}$

$$(4) \frac{\rho \omega^2 l^2}{8}$$

94. Two capillary tubes of same length each of 50 cm but of different radii 4 mm and 2 mm are connected in series. When water flows, the pressure difference between the ends of the arrangement is P . Then the pressure difference between the ends of the first tube is

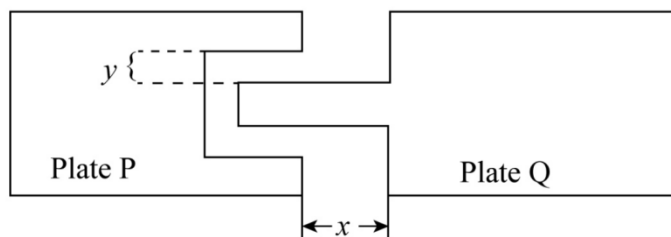
$$(1) \frac{P}{2}$$

$$(2) \frac{P}{17}$$

$$(3) \frac{P}{4}$$

$$(4) \frac{P}{8}$$

95. Two metal plates P and Q of same material are arranged as shown in the figure. If both the plates are uniformly heated through same range of temperature, then



- (1) Both x and y increase
- (2) Both x and y decrease
- (3) x decrease and y increase
- (4) x increase and y decrease

96. A liquid of mass 250 g is kept warm in vessel using an electric heater. The liquid is maintained at 57°C when the power supplied by the heater is 30 W and surrounding temperature is 27°C . As the heater is switched off, it took 10 s time for the temperature of the liquid to fall from 47°C to 46.9°C . The specific heat capacity of the liquid is

- (1) $8000 \text{ Jkg}^{-1}\text{K}^{-1}$
- (2) $9000 \text{ Jkg}^{-1}\text{K}^{-1}$
- (3) $6000 \text{ Jkg}^{-1}\text{K}^{-1}$
- (4) $12000 \text{ Jkg}^{-1}\text{K}^{-1}$

97. The amount of heat that must be supplied to 35 g of oxygen at room temperature to raise its temperature by 80°C at constant volume is

(molecular mass of oxygen is 32 and $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)

(1) 2.84 kJ

(2) 1.68 kJ

(3) 1.81 kJ

(4) 2.88 kJ

98. A Carnot engine of efficiency 40% takes heat from a source maintained at a temperature of 500 K. It is desired to have an engine of efficiency 60%. Then, the source temperature for the same sink temperature must be

(1) 650 K

(2) 750 K

(3) 550 K

(4) 850 K

99. For a gas the value of $\frac{R}{C_v} = 0.4$, so the gas is,

(R - Universal gas constant)

- (1) monoatomic
- (2) diatomic
- (3) triatomic
- (4) polyatomic

100. A cylindrical tube open at both ends has a frequency f in air. The tube is dipped vertically in water so that 60% of the tube is in water. Then the fundamental frequency of air column is

- (1) $\frac{f}{2}$
- (2) $\frac{5f}{4}$
- (3) $\frac{3f}{4}$
- (4) $2f$

101. A train approaching a railway crossing at a speed of 120 kmph sounds a whistle of frequency 567 Hz, when it is 288 m away from the crossing. The frequency heard by the observer standing on the road perpendicular to the track from the crossing at a distance of 384 m is

(Speed of sound in air = 340 ms^{-1})

- (1) 632 Hz
- (2) 612 Hz
- (3) 512 Hz
- (4) 472 Hz

102. A small object is enclosed in a transparent solid sphere of radius 8 cm. The object is situated at 2 cm from the centre of the sphere. If its image appears to be at 3.2 cm from the nearest side then the refractive index of the material of the sphere is

- (1) 1.62
- (2) 1.45
- (3) 1.55
- (4) 1.50

103. A Young double slit experimental setup is immersed in water of refractive index 1.33. It has slit separation 1 mm and the distance between slits and screen is 1.33 m. If the wavelength of incident light on slits is $6300 \text{ \AA}$ then the fringe width on screen is

- (1) 6.3 mm
- (2) 0.63 mm

(3) 0.63 m

(4) 6.3 m

104. In a hydrogen atom, an electron of mass 9.1×10^{-31} kg revolves about a proton in circular orbit of radius 0.53 Å. The radial acceleration and angular velocity of electron are respectively

(1) $9 \times 10^{22} \text{ ms}^{-2}$, $4.1 \times 10^{16} \text{ s}^{-1}$

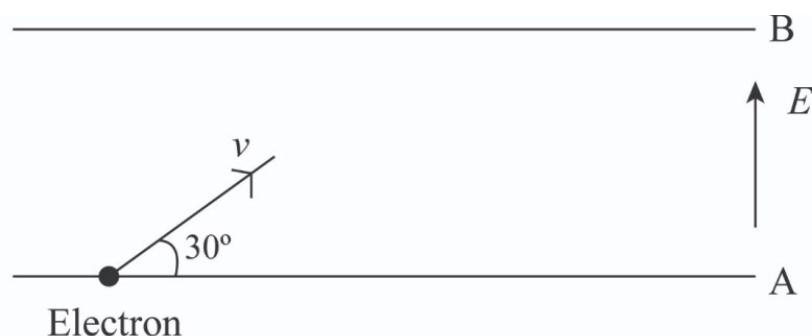
(2) $4.1 \times 10^{16} \text{ ms}^{-2}$, $9 \times 10^{22} \text{ s}^{-1}$

(3) $9 \times 10^{16} \text{ ms}^{-2}$, $4.1 \times 10^{22} \text{ s}^{-1}$

(4) $4.1 \times 10^{22} \text{ ms}^{-2}$, $9 \times 10^{16} \text{ s}^{-1}$

105. Two long parallel plates A and B are separated by a distance of 4 cm with an electric field of 45.5 Vm^{-1} between the plates A to plate B, as shown in the figure. An electron is projected from plate A with velocity v at angle of 30° with the surface of plate A. The maximum value of v so that the electron does not hit plate B is

(Assume gravity free space, charge of electron = $1.6 \times 10^{-19} \text{ C}$ and mass of electron = $9.1 \times 10^{-31} \text{ kg}$)



- (1) 400 kms^{-1}
- (2) 3200 kms^{-1}
- (3) 800 kms^{-1}
- (4) 1600 kms^{-1}

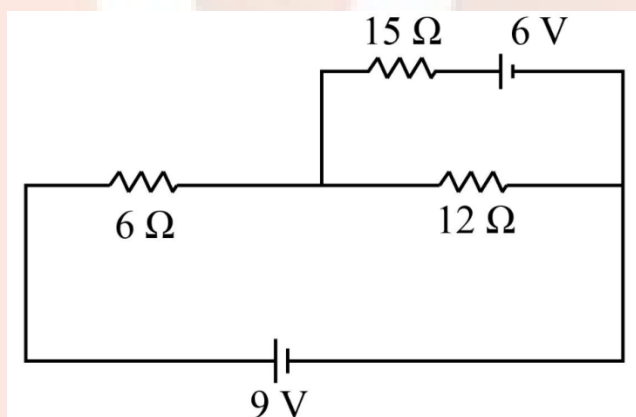
106. Three point charges of 2 mC each are kept at the vertices of an equilateral triangle of side 50 cm . If the system is supplied energy at the rate of 2 kW , the time taken to move one of the charges to the midpoint of the line joining the other two charges is

- (1) 18 s
- (2) 36 s
- (3) 72 s
- (4) 144 s

107. Two identical charged sphere separated by a distance repel each other with a force F . If 10% of electrons are transferred from one sphere to the other then the force between them become

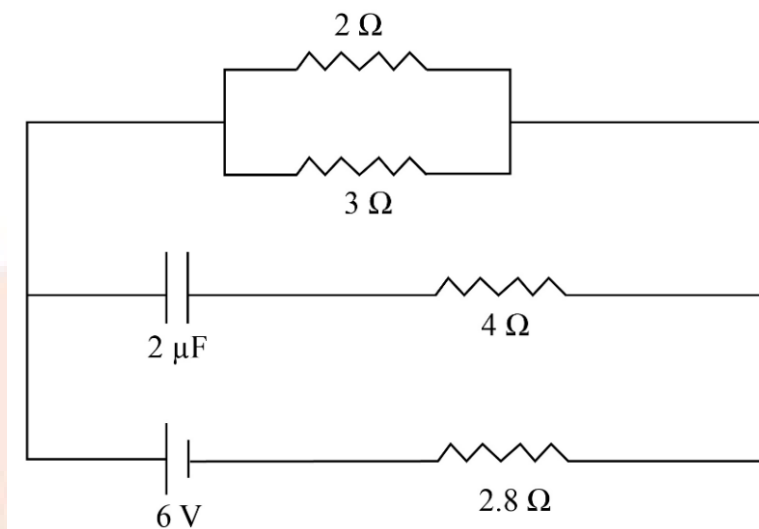
- (1) F
- (2) $1.21F$
- (3) $0.99F$
- (4) $0.81F$

108. In the given circuit, the electric currents through $15\ \Omega$ and $6\ \Omega$ respectively are



- (1) 0 A, 0.5 A
- (2) 0 A, 1 A
- (3) 0.5 A, 1 A
- (4) 1 A, 0 A

109. In the given circuit, the current through $2\ \Omega$ resistor is



- (1) $9\ \text{A}$
- (2) $0.9\ \text{A}$
- (3) $\frac{1}{9}\ \text{A}$
- (4) $\frac{1}{0.9}\ \text{A}$

110. A circular coil connected to a battery of emf E produced a certain magnetic induction field at its centre. The coil is unwound, stretched to double its length rewound into a coil of $\frac{1}{3}^{\text{rd}}$ of the original radius and connected to a battery of emf E' to produce same field at the centre. Then E' is

(1) $\frac{2E}{9}$

(2) $\frac{3E}{7}$

(3) $\frac{9E}{4}$

(4) $\frac{E}{6}$

111. Assertion (A) : The work done by the electrostatic force is zero when a point charges moves in a circular path around another charge.

Reason (R) : The dot product of force and displacement vectors gives work done.

- (1) Both (A) and (R) are correct and (R) is a correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not correct explanation of (A)
- (3) (A) is correct but (R) is incorrect
- (4) (A) is incorrect but (R) is correct

112. The magnetic susceptibility of a paramagnetic substance at -173°C is 1.5×10^{-2} . To have the susceptibility 0.5×10^{-2} , the change in temperature in $^{\circ}\text{C}$ is

- (1) 100
- (2) 180
- (3) 200
- (4) 220

113. A coil of mean area 500 cm^2 and having 1000 turns is held with its plane perpendicular to a uniform field of 0.4 G. If the coil is turned through 180° in $\frac{1}{10}$ second, then the average induced emf is,

$(1 \text{ G} = 10^{-4} \text{ T})$

- (1) 0.04 V
- (2) 0.4 V
- (3) 4 V
- (4) 40 V

114. An emf of 15 V is applied to a circuit containing 5 H inductance and $10\ \Omega$ resistance. The ratio of currents at time $t = \infty$ and $t = 1\text{ s}$ is

(1) $\frac{e}{e^2 - 1}$

(2) $\frac{e^2}{e - 1}$

(3) $\frac{e}{1 - e^2}$

(4) $\frac{e^2}{e^2 - 1}$

115. The amplitude of the electric field in parallel beam of plane electromagnetic waves of intensity 53.1 W/m^2 is

(Permittivity of free space = $8.85 \times 10^{-12}\text{ C}^2\text{N}^{-1}\text{m}^{-2}$)

(1) 400 NC^{-1}

(2) 50 NC^{-1}

(3) 100 NC^{-1}

(4) 200 NC^{-1}

116. The α -particle moves in a circular path of radius 1 cm in a uniform magnetic field of 0.125 T. The de Broglie wavelength with the α particle is

- (1) 1.65×10^{-12} m
- (2) 3.3×10^{-12} m
- (3) 4.95×10^{-12} m
- (4) 6.6×10^{-12} m

117. When a hydrogen atom emits a photon during a transition from $n = 4$ to $n = 2$, its recoil speed is about

- (1) 4.28 ms^{-1}
- (2) 0.814 ms^{-1}
- (3) 2.07 ms^{-1}
- (4) 0.407 ms^{-1}

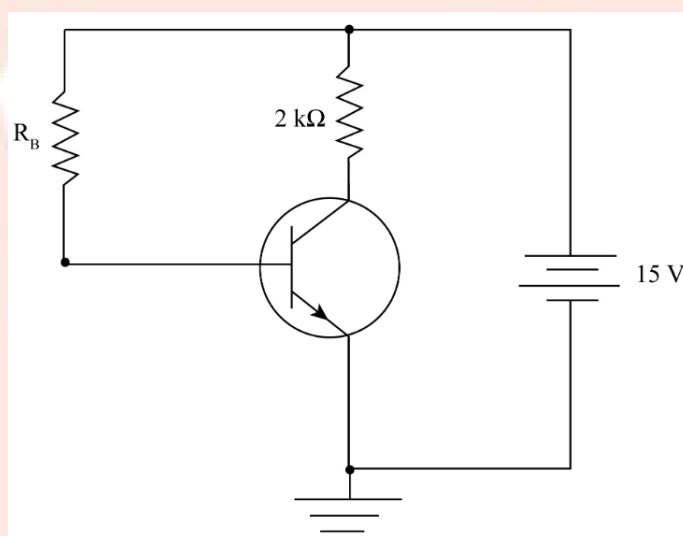
118. If the binding energy of N^{14} is 7.5 MeV per nucleon and that of N^{15} is 7.7 MeV per nucleon, then the energy required to remove a neutron from N^{15} is

- (1) 5.25 MeV
- (2) 0.2 MeV

(3) 10.5 MeV

(4) 0.4 MeV

119. In the following common emitter circuit, $\beta = 100$ and $V_{CE} = 7 \text{ V}$. If V_{BE} is negligible, then the base circuit is



(1) 0.01 mA

(2) 0.04 mA

(3) 0.02 mA

(4) 0.03 mA

120. The frequency suitable for beyond the horizon communication using sky waves is

(1) 10^{12} Hz

(2) 10^9 Hz

(3) 10^7 Hz

(4) 10^4 Hz



AP EAMCET-2019-Engineering-21 Apr-Shift-2 Chemistry

121. Which of the following equations does represent the velocity (v) of the ejected electrons when a metal is made to strike with light of frequency ν and threshold frequency of the metal is ν_0 ? (m_e = mass of electron and h is Planck's constant)

(1) $v = \sqrt{\frac{h(\nu - \nu_0)}{m_e}}$

(2) $v = \sqrt{\frac{2h(\nu - \nu_0)}{m_e}}$

(3) $v = \sqrt{\frac{h(\nu - \nu_0)}{2m_e}}$

(4) $v = \sqrt{h(\nu - \nu_0)m_e}$

122. An element with mass number 181 contains 32% more neutrons as compared to protons. What is the symbol of that element?

(1) Pt

(2) Pd

(3) Au

(4) Hg

123. The electron gain enthalpy $\Delta_{eg}H$ of Cl (g) is -349 kJ mol^{-1} . If the ground state energy of Cl (g) is $x \text{ kJ mol}^{-1}$, the ground state energy (in kJ mol^{-1}) of Cl^- (g) is

(1) $x + 349$

(2) x

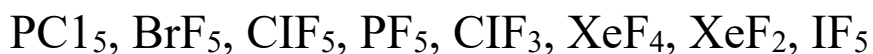
(3) $x - 349$

(4) $\frac{x - 349}{17}$

124. Identify the correct set of molecules with different geometries and central atoms with different hybridisations.



125. Observe the following molecules.



The number of molecules having square pyramidal geometry from the above is

- (1) 4
- (2) 5
- (3) 3
- (4) 6

126. If the kinetic energy and RMS speed of a gas at a certain temperature are 4.0 kJ mol^{-1} and $5.0 \times 10^4 \text{ cm} \cdot \text{s}^{-1}$ respectively.

The molecular weight of the gas is

- (1) 16
- (2) 32
- (3) 64
- (4) 44

127. In how many of the following compounds of sulphur, the oxidation state of sulphur atom is +6?

$\text{H}_2\text{S}_2\text{O}_8$, H_2SO_5 , H_2SO_3 , H_2SO_4 , $\text{H}_2\text{S}_2\text{O}_7$, SO_2Cl_2 , SOCl_2 .

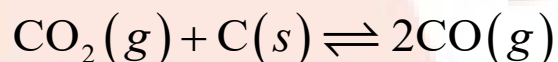
- (1) 3
- (2) 5
- (3) 4

(4) 6

128. What is the nature of reaction at 298 K, if the entropy change and enthalpy change for a chemical reaction are 7.4 cal K^{-1} and $-2.5 \times 10^3 \text{ cal}$ respectively.

- (1) Reversible
- (2) Spontaneous
- (3) Non-Spontaneous
- (4) Irreversible

129. The value of K_c for the equilibrium, reaction



at T(K) is 0.036. If the equilibrium concentration of $\text{CO}_2(g)$ is 0.004 M, the concentration of $\text{CO}(g)$ in mol L^{-1} is,

- (1) 3.6×10^{-2}
- (2) 2.0×10^{-2}
- (3) 1.2×10^{-2}
- (4) 1.2×10^{-3}

130. 50 mL of 0.02 M NaOH solution is mixed with 50 mL of 0.06 M acetic acid solution, the pH of resulting solution is (pKa of acetic acid is 4.76. $\log 5 = 0.70$)

- (1) 5.06
- (2) 4.06
- (3) 5.46
- (4) 4.46

131. Which of the following is water-gas shift reaction ?

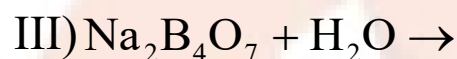
- (1) $\text{C}(s) + \text{H}_2\text{O} \xrightarrow{1270 \text{ K}} \text{CO}(g) + \text{H}_2(g)$
- (2) $\text{CH}_4(g) + \text{H}_2\text{O}(g) \xrightarrow{1270 \text{ K}} \text{CO}(g) + 3\text{H}_2(g)$
- (3) $\text{CO}(g) + \text{H}_2\text{O}(g) \xrightarrow[from \text{ chromate}]{673 \text{ K}} \text{CO}_2(g) + \text{H}_2(g)$
- (4) $2\text{H}_2\text{O}(l) \xrightarrow[traces \text{ of } acid/base]{electrolysis} 2\text{H}_2(g) + \text{O}_2(g)$

132. Magnesium is burnt in air to form A and B. When B is hydrolyzed, C and D are formed. D is the reactant in the manufacture of nitric acid by Ostwald's process. What is C?

- (1) NH_3
- (2) $\text{Mg}(\text{OH})_2$
- (3) MgO

(4) NO

133. Which of the following reaction can be used to prepare diborane ?



(1) I, II, III

(2) II, III only

(3) III, IV only

(4) I, II, IV

134. Identify the correct statements.

I) Germanium exists only in traces.

II) The order of electronegativity of Si, Ge, Sn is $\text{Sn} > \text{Ge} > \text{Si}$.

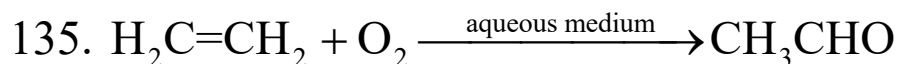
III) All group 14 elements are available in solid state.

(1) I, II only

(2) II, III only

(3) I, III only

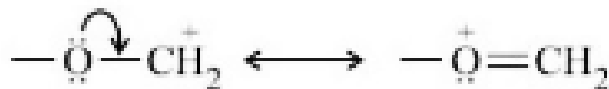
(4) I, II, III



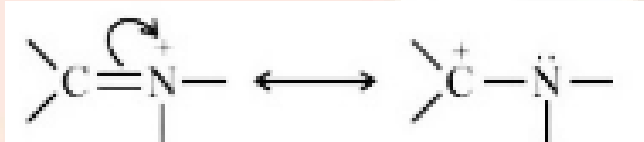
Where is the catalyst used in the above green reaction ?

- (1) Pd(II)
- (2) Pt
- (3) ZnO
- (4) Rh

136. In the following resonance structures the curved arrow indicates that electrons are shifted from



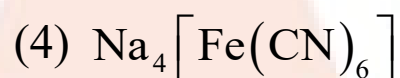
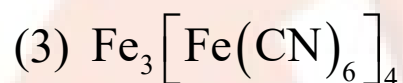
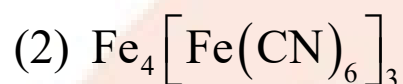
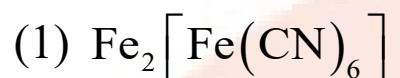
A)



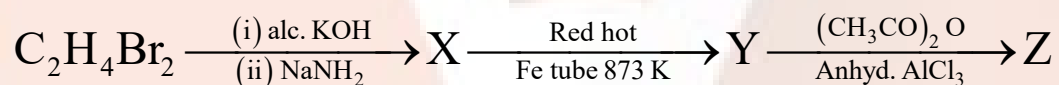
B)

- (1) atom to adjacent bond in both (A) and (B)
- (2) π bond to adjacent atom in both (A) and (B)
- (3) π bond to adjacent atom in (A) and atom to adjacent bond in (B)
- (4) atom to adjacent bond in (A) and it bond to adjacent atom in (B)

137. In the detection of nitrogen of an organic compound by Lassaigne's test, prussian blue colour is obtained. This is due to the formation of which of the following complexes?



138. Identify Z in the following sequence of reactions.



(1) Acetophenone

(2) Anisole

(3) Toluene

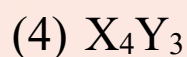
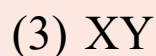
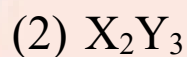
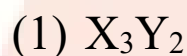
(4) Chlorobenzene

139. Which of the following is not present in the nitration mixture ?

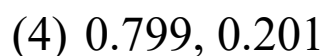
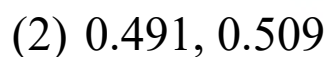
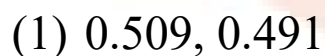




140. A compound is formed from elements X and Y. The atoms of Y (anions) form ccp lattice. The atoms of X (cations) occupy half of the octahedral voids and half of tetrahedral voids. What is the formula of the compound ?



141. The vapour pressures of chloroform (CHCl_3), dichloromethane (CH_2Cl_2) at 298 K are 200 mm Hg and 415 mm Hg respectively. An ideal solution is prepared by mixing 59.75 g of CHCl_3 and 21.25 g of CH_2Cl_2 , the mole fractions of chloroform and dichloromethane in vapour phase respectively are



142. The elevation in boiling point of an aqueous solution of NaCl is 0.01°C . If its van't Hoff factor is 1.92, the molality of NaCl solution is

(K_b for water = $0.52\text{ K kg mol}^{-1}$)

- (1) 0.01 m
- (2) 0.001 m
- (3) 0.005 m
- (4) 0.02 m

143. CuSO_4 solution is electrolysed for 15 minutes to deposit 0.4725 g of copper at the cathode.

The current in amperes required is

(Faraday = $96,500\text{ C mol}^{-1}$, atomic weight of copper = 63)

- (1) 0.804
- (2) 1.608
- (3) 1.206
- (4) 0.402

144. The rate constants for a reaction at 400 K and 500 K are $2.60 \times 10^{-5} \text{ s}^{-1}$ and $2.60 \times 10^{-3} \text{ s}^{-1}$ respectively. The activation energy of the reaction in kJ mol^{-1} is

- (1) 38.3
- (2) 57.4
- (3) 114.9
- (4) 76.6

145. Match the following.

List-I	List-II
Reaction	Catalyst
A) $\text{N}_2(g) + 3\text{H}_2(g) \rightarrow 2\text{NH}_3(g)$	I) Ni
B) $2\text{H}_2(g) + \text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(l)$	II) Pt
C) $\text{CO}(g) + 3\text{H}_2(g) \rightarrow \text{CH}_4(g) + \text{H}_2\text{O}(g)$	III) $\text{ZnO-Cr}_2\text{O}_3$
D) $\text{CO}(g) + 2\text{H}_2(g) \rightarrow \text{CH}_3\text{OH}(g)$	IV) Fe

- | | | | | |
|-----|-----|----|---|----|
| | A | B | C | D |
| (1) | III | II | I | IV |

- | | | | | |
|-----|----|-----|----|---|
| | A | B | C | D |
| (2) | IV | III | II | I |

- | | | | | |
|-----|----|----|---|-----|
| | A | B | C | D |
| (3) | IV | II | I | III |

A	B	C	D
(4) IV	I	III	II

146. The pair of metals refined by “vapour phase refining” is

- (1) Ni, Cu
- (2) Sn, Ni
- (3) Zr, Ni
- (4) Cu, Zr

147. White phosphorus, when heated with conc NaOH solution in an inert atmosphere of CO_2 forms phosphine and a sodium salt of oxoacid of phosphorus, 'X'. The oxidation state of phosphorus in 'X' is

- (1) +3
- (2) +4
- (3) +1
- (4) +5

148. The number of P-OH bonds present in pyrophosphoric acid and hypophosphoric acid is respectively

- (1) 4, 3

(2) 2, 4

(3) 3, 4

(4) 4, 4

149. Sodium nitrite is reacted with H_2SO_4 to form NaHSO_4 , HNO_3 , water and X. Gold is dissolved in aqua regia to form water, AuCl_4^- and Y. X and Y are respectively.

(1) NO , NO_2

(2) NO_2 , NO

(3) NO , NO

(4) N_2O , NO

150. Which of the following complex ions is most stable?

(1) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$

(2) $[\text{Co}(\text{CN})_6]^{3-}$

(3) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

(4) $[\text{CoF}_6]^{3-}$

151. The copper (II) halide which does not exist is

- (1) CuF_2
- (2) CuBr_2
- (3) CuI_2
- (4) CuCl_2

152. Match the following.

List-I

- A) Addition polymer
- B) Condensation polymer
- C) Acrilan
- D) Rubber

List-II

- I) Bakelite
- II) 2-methyl-1, 3-butadiene
- III) 2,3-dimethyl-1, 3-butadiene
- IV) Vinyl cyanide
- V) Polythene

- | | A | B | C | D |
|----|---|---|----|-----|
| 1. | V | I | IV | II |
| | A | B | C | D |
| 2. | V | I | II | III |
| | A | B | C | D |
| 3. | I | V | IV | II |
| | A | B | C | D |
| 4. | I | V | II | III |

153. Identify the correct statements from the following.

I) When DNA is hydrolysed adenine and thymine are obtained in equal quantities.

II) When RNA is hydrolysed adenine and uracil are obtained in equal quantities.

III) Amylose is branched polymer with $\alpha 1 \rightarrow 4$ and $\alpha 1 \rightarrow 6$ glycosidic linkages.

IV) Addison disease is due to the abnormal functioning of adrenal cortex.

(1) I, II, III only

(2) I, II, III, IV

(3) I, II, IV only

(4) I, IV only

154. Identify the correct pair from the following.

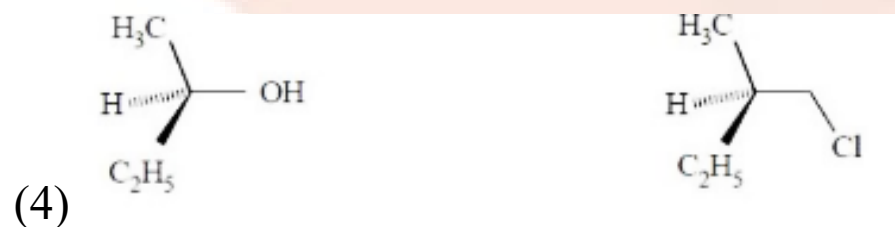
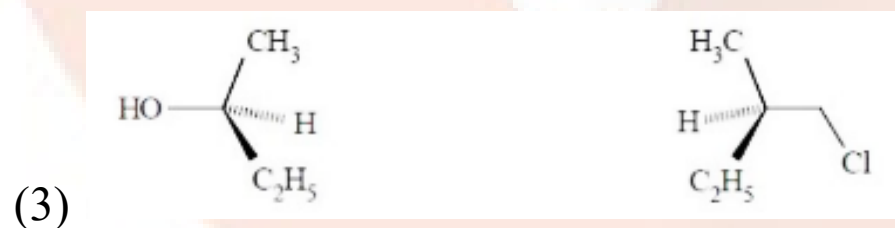
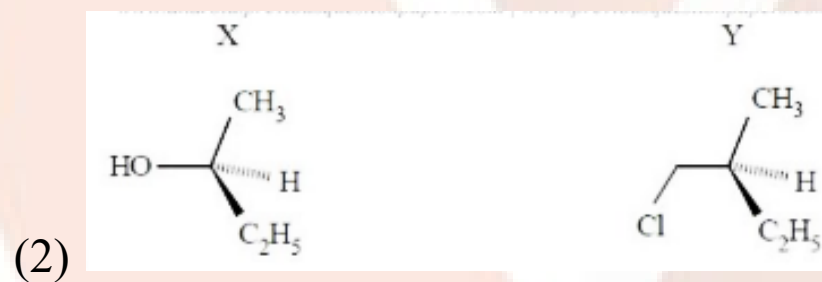
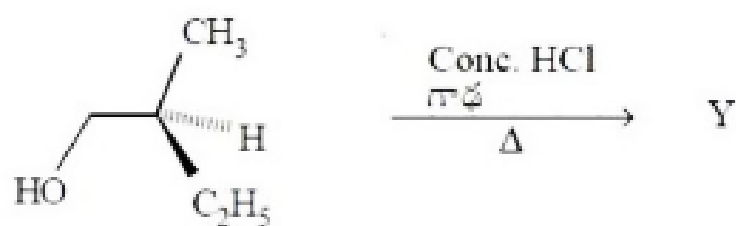
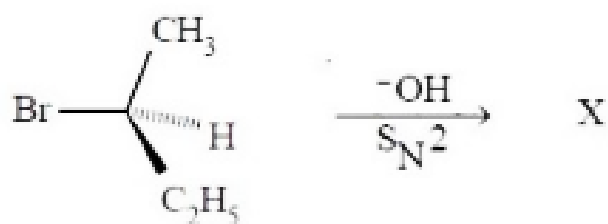
(1) Codeine - analgesic ; Equanil - tranquilizer

(2) Chloramphenicol - analgesic ; Nardil - antibiotic

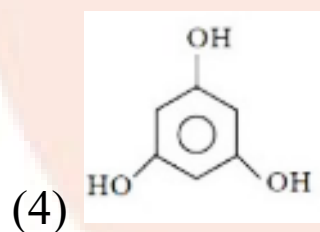
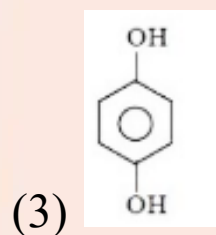
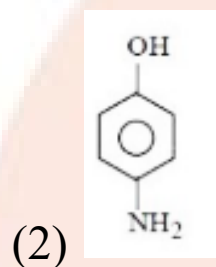
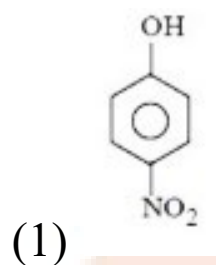
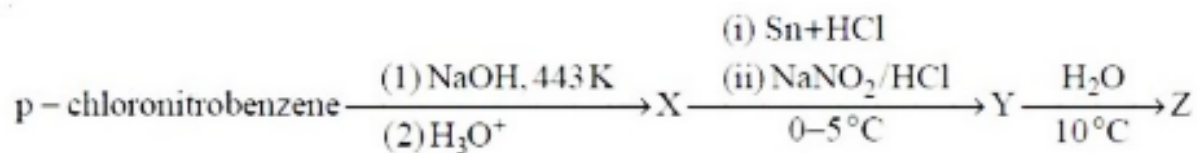
(3) Histamine – tranquilizer ; Salvarsan - antibiotic

(4) Norethindrone - antacid ; alitame - artificial sweetening agent

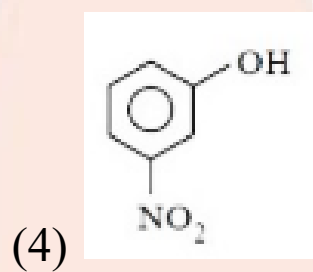
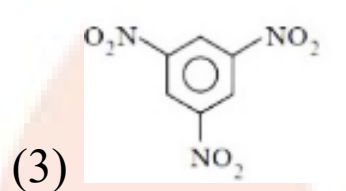
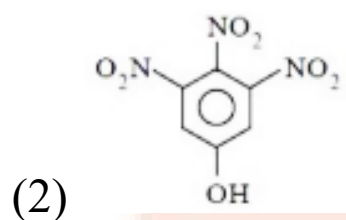
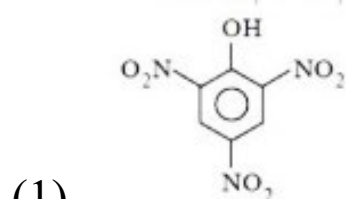
155. What are X and Y in the following reactions?



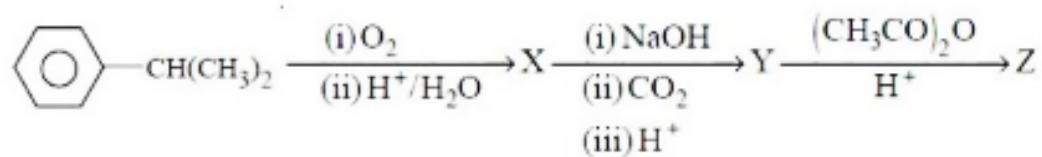
156. What is Z in the following sequence of reactions?



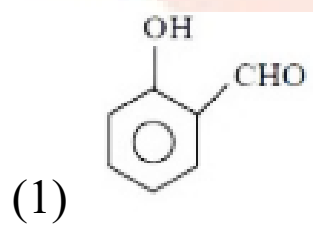
157. All organic compounds A ($\text{C}_6\text{H}_7\text{N}$) on reaction with NaNO_2/HCl at 273-278 K followed by warming with water gave B. B reacts with conc. HNO_3 to give C. What is C?

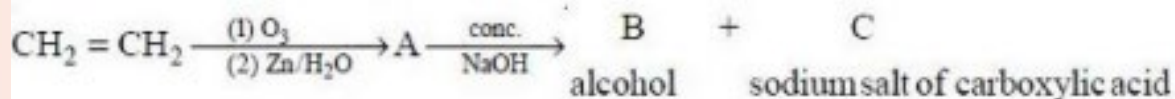
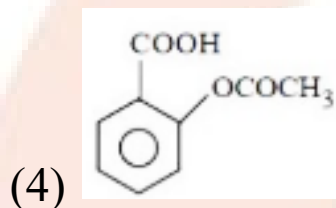
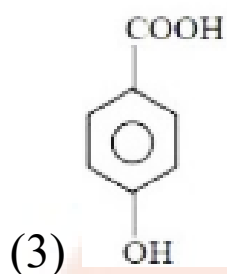


158.



What is Z in the above sequence of reactions ?

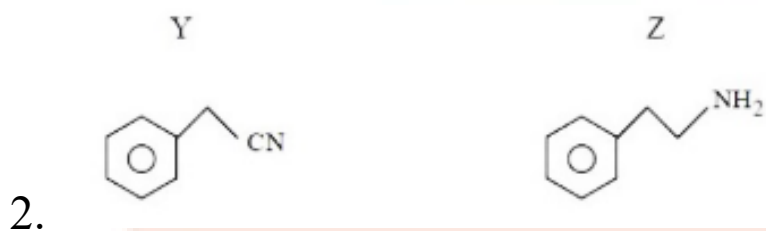




The reaction of A to give B and C is an example of

- (1) HVZ reaction
- (2) Stephen reaction
- (3) Etard reaction
- (4) Cannizzaro reaction

160. An organic compound X (C_7H_7Cl) when reacted with KCN / C_2H_5OH gave major product Y. Z is formed when Y is reduced with $LiAlH_4$. What are Y and Z?



AP EAMCET-2019-Engineering-21 Apr-Shift-2
Answer Key

1	4	33	2	65	3	97	3	129	3
2	2	34	1	66	1	98	2	130	4
3	3	35	2	67	2	99	2	131	3
4	4	36	3	68	1	100	2	132	2
5	4	37	2	69	3	101	2	133	4
6	1	38	2	70	2	102	4	134	3
7	2	39	4	71	3	103	2	135	1
8	3	40	3	72	4	104	1	136	4
9	4	41	4	73	1	105	4	137	2
10	4	42	3	74	2	106	3	138	1
11	4	43	3	75	3	107	3	139	3
12	1	44	2	76	2	108	1	140	1
13	3	45	2	77	4	109	2	141	2
14	3	46	3	78	1	110	1	142	1
15	1	47	1	79	4	111	1	143	2
16	2	48	3	80	4	112	3	144	4
17	4	49	1	81	3	113	1	145	3
18	3	50	4	82	3	114	4	146	3

19	3	51	3	83	1	115	4	147	3
20	1	52	1	84	3	116	1	148	4
21	3	53	4	85	1	117	2	149	3
22	4	54	2	86	2	118	3	150	3
23	2	55	1	87	2	119	2	151	3
24	1	56	3	88	3	120	3	152	1
25	2	57	1	89	2	121	2	153	4
26	3	58	2	90	1	122	1	154	1
27	1	59	4	91	3	123	3	155	1
28	2	60	1	92	2	124	4	156	3
29	4	61	3	93	3	125	3	157	1
30	1	62	2	94	2	126	2	158	4
31	4	63	1	95	2	127	3	159	4
32	2	64	3	96	4	128	2	160	2

AP EAMCET-2019-Engineering-21 Apr-Shift-2

Solutions – Mathematics

1. Consider $N(A) = m$ and $N(B) = n$,

P_B has 112 elements more than P_A ,

$$N(P_B) = 112 + N(P_A)$$

$$N(P_B) - N(P_A) = 112$$

$$2^n - 2^m = 16 \times 7$$

Rewrite the above equation.

$$2^m (2^{n-m} - 1) = 16 \times 7$$

$$2^m (2^{n-m} - 1) = 2^4 (2^3 - 1)$$

From the above equation,

$$m = 4$$

$$n - m = 3$$

$$n - 4 = 3$$

$$n = 7$$

The number of injective function from A to B,

$$= {}^{N(B)}P_{N(A)}$$

$$= {}^7P_4$$

$$= \frac{7!}{3!}$$

$$= 840$$

2. The given function is,

$$f(x) = \sqrt{\frac{x - |x|}{x - [x]}}$$

Now,

$$x - |x| \geq 0 \text{ and } x - [x] > 0$$

$$x > |x| \text{ and } x > [x]$$

Therefore, $x \in \mathbb{R}^+ - (\text{all integers})$

Again,

$$g(x) = \frac{2x}{4 + x^2}$$

$$y = \frac{2x}{4 + x^2}$$

Rewrite the above equation.

$$yx^2 - 2x + 4y = 0$$

$$x = \frac{2 \pm \sqrt{4 - 16y^2}}{2y}$$

From the above equation,

$$4 - 16y^2 \geq 0 \text{ and } y \neq 0.$$

$$1 - 4y^2 \geq 0 \text{ and } y \neq 0.$$

$$\text{Therefore, } y \in \left[-\frac{1}{2}, \frac{1}{2}\right] - \{0\}.$$

$$\text{Range of } g(x) \text{ is } \left[-\frac{1}{2}, \frac{1}{2}\right].$$

$$\text{The } D = \mathbb{R}^+ - (\text{all integers}) \text{ and } C = \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$\text{Thus, } D \cap C = \left(0, \frac{1}{2}\right].$$

3. The expression $x^n - y^n$ is divisible by $(x + y)(x - y)$, if $n =$ even.

Similarly, $x^n - y^n$ is divisible by $(x - y)$, if $n =$ odd and $(x^{2n+1} + y^{2n+1})$ is divisible by $(x + y)$.

4. It is given that,

$$AA^T = I$$

It is an orthogonal matrix and the determinant of orthogonal matrix is ± 1 .

$$\begin{vmatrix} p & q & r \\ r & p & q \\ q & r & p \end{vmatrix} = \pm 1$$

Solve the above determinant,

$$p^3 - pqr - pqr + q^3 + r^3 = \pm 1$$

$$p^3 + q^3 + r^3 - 3pqr = \pm 1$$

$$p^3 + q^3 + r^3 = 3pqr \pm 1$$

5. (A)

The non-singular matrix of order 3, $|A| = a$,

$$\begin{aligned} \left| (adj A^{-1})^{-1} \right| &= \frac{1}{|adj A^{-1}|} \\ &= \frac{1}{|A^{-1}|^2} \quad \{ |adj A| = |A|^2, \text{ if order of } A \text{ is } 3 \} \\ &= \frac{1}{\left(\frac{1}{|A|} \right)^2} \\ &= |A|^2 \\ &= a^2 \end{aligned}$$

(B)

It is given that for non-singular matrix A of order B, $AB = 0$

$$|AB| = 0$$

$$|A||B| = 0$$

$$|B| = 0 \quad (\because |A| \neq 0)$$

And matrix B should be null matrix.

(C)

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ \cos(a-b)y & \cos ay & \cos(a+b)y \\ \sin(a-b)y & \sin ay & \sin(a+b)y \end{vmatrix}$$

By simplify the above equation, we get

$$\sin(by) - x \sin(2by) + x^2 \sin(by)$$

Thus, it doesn't depend on a .

(D)

It is given that,

$$B = A - A^T$$

$$B^T = (A - A^T)^T$$

$$B^T = A^T - A$$

$$B^T = -(A - A^T)$$

$$B^T = -B$$

B is a skew symmetric matrix.

$$\text{Thus, } |B| = 0$$

6. It is given that,

$$\begin{bmatrix} 2 & 2 & 3 \\ 7 & 1 & 1 \\ 0 & 6 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3y+11 \\ 6z-1 \\ 5y+11 \end{bmatrix} + \begin{bmatrix} x \\ x \\ 4z \end{bmatrix} + \begin{bmatrix} z \\ 3x \\ 4y \end{bmatrix}$$

Solve the above matrix,

$$\begin{bmatrix} 2x+2y+3z \\ 7x+y+z \\ 6y+5z \end{bmatrix} = \begin{bmatrix} 3y+x+z+11 \\ 3x+x+6z-1 \\ 5y+4y+4z+11 \end{bmatrix}$$

$$\begin{bmatrix} 2x+2y+3z \\ 7x+y+z \\ 6y+5z \end{bmatrix} = \begin{bmatrix} x+3y+z+11 \\ 4x+6z-1 \\ 9y+4z+11 \end{bmatrix}$$

On comparing,

$$x - y + 2z = 11$$

$$3x + y - 5z = -1$$

$$-3y + z = 11$$

From above three equations,

$$x = 4, y = -3, z = 2$$

7. It is given that,

$$z_1 = 12 + 5i$$

The modulus is,

$$|z_1| = \sqrt{12^2 + 5^2}$$

$$|z_1| = \sqrt{144 + 25}$$

$$|z_1| = 13$$

The greatest value of $|z_1 + z_2|$,

$$a = 13 + 9$$

$$a = 22$$

The lowest value of $|z_1 + z_2|$

$$b = 13 - 9$$

$$b = 4$$

8. Consider a complex equation, $z = x + iy$

Its conjugate is $\bar{z} = x - iy$.

Now,

$$(7 + i)(z + \bar{z}) - (4 + i)(z - \bar{z}) + 116i = 0$$

$$(7 + i)(2x) - (8iy - 2y) + 116i = 0$$

$$(14x + 2y) + i(2x - 8y + 116) = 0$$

The equations obtained are

$$2x - 8y + 116 = 0$$

$$14x + 2y = 0$$

From above two equations,

$$x = -2, y = 14$$

The modulus is,

$$\begin{aligned}|z| &= \sqrt{x^2 + y^2} \\ &= \sqrt{(-2)^2 + 14^2} \\ &= \sqrt{200}\end{aligned}$$

$$\text{Thus, } z\bar{z} = |z|^2 = (\sqrt{200})^2 = 200$$

9. It is given that,

$$z = x + iy, x, y \in \mathbb{R}$$

And

$$\frac{2z + i}{z - 2} = \frac{2(x + iy) + i}{(x - 2) + iy}$$

$$\operatorname{Re}\left(\frac{2z + i}{z - 2}\right) = \frac{2x + i(2y + 1)}{(x - 2) + iy} \times \frac{(x - 2) - iy}{(x - 2) - iy}$$

$$0 = 2x(x - 2) + y(2y + 1) = 0$$

$$0 = x^2 + y^2 - 2x + \frac{y}{2} \quad \dots\dots (I)$$

And

$$\frac{z + i}{z - i} = \frac{x + i(y + 1)}{(x + 1) + iy} \times \frac{(x + 1) - iy}{(x + 1) - iy}$$

Now,

$$\arg\left(\frac{z+i}{z-i}\right) = \tan^{-1}\left[\frac{(x+1)(y+1)-xy}{x(x+1)+y(y+1)}\right] = \frac{\pi}{2}$$

$$x^2 + y^2x + x + y = 0 \quad \dots\dots (II)$$

Equation of common chord of circles (i) and (ii) is

$$S_1 - S_2 = 0$$

$$-3x - \frac{y}{2} = 0$$

$$y + 6x = 0 \quad \dots\dots (III)$$

From equation (II) and (III), the intersection of the circles is,

$$x^2 + 36x^2 + x - 6x = 0$$

$$x = 0, \frac{5}{37}$$

$$\text{So the values of } y \text{ are } 0, -\frac{30}{37}$$

This implies the point of intersection of the curves C_1 and C_2

other than the origin is $\left(\frac{5}{37}, -\frac{30}{37}\right)$.

$$10. \sum_{n=1}^{20} z^{2n-1} = z + z^3 + z^5 + \dots + z^{39}$$

$$\text{Now, } z = \cos 6^\circ + i \sin 6^\circ = e^{i6^\circ},$$

Substitute the value in the series,

$$\begin{aligned}
 \sum_{n=1}^{20} z^{2n-1} &= e^{i6^\circ} + e^{i18^\circ} + e^{i30^\circ} + \dots + e^{i(39 \times 6^\circ)} \\
 &= e^{i6^\circ} (1 + e^{i12^\circ} + e^{i24^\circ} + \dots + e^{i228^\circ}) \\
 &= e^{i6^\circ} \left[\frac{e^{i240} - 1}{e^{i12^\circ} - 1} \right]
 \end{aligned}$$

Simplify the above equation,

$$\begin{aligned}
 \sum_{n=1}^{20} \operatorname{Im}(z^{2n-1}) &= e^{i6^\circ} \left[\frac{e^{i240} - 1}{e^{i12^\circ} - 1} \right] \\
 &= \frac{e^{i6^\circ} \left[\frac{-1}{2} - \frac{\sqrt{3}}{2}i - 1 \right]}{e^{i6^\circ} (2i \sin 6^\circ)} \\
 &= \frac{\left[-\frac{3}{2} - \frac{\sqrt{3}}{2}i \right]}{2i \sin 6^\circ} \\
 &= -\frac{\sqrt{3}}{4 \sin 6^\circ} + \frac{\sqrt{3}}{4 \sin 6^\circ}i
 \end{aligned}$$

$$\text{Thus, } \sum_{n=1}^{20} \operatorname{Im}(z^{2n-1}) = \frac{3}{4 \sin 6^\circ}$$

11. It is given that, α, β are the roots of $x^2 + px + q = 0$

$$\alpha + \beta = -p \text{ and } \alpha\beta = q.$$

Since, α^4, β^4 are roots of $x^2 - rx + s = 0$

Therefore, $\alpha^4 + \beta^4 = r$ and $\alpha^4 \beta^4 = s$

Now, $x^2 - 4qx + 2q^2 - r = 0$

$$\begin{aligned} D &= (4q)^2 - 4(2q^2 - r) \\ &= 16q^2 - 8q^2 + 4r \\ &= 8q^2 + 4r \end{aligned}$$

Here,

$$r = \alpha^4 + \beta^4 \geq 0$$

This implies $8q^2 \geq 0$.

Thus, $D \geq 0$.

Therefore, the equation has two real roots.

12. Consider,

$$y = \frac{x - P}{x^2 - 3x + 2} \text{ and it is provided that}$$

$$x^2 - 3x + 2 \neq 0$$

$$x \neq 1, 2$$

Now solve the above equation.

$$yx^2 - (3y+1)y + (2y+P) = 0 \quad \because x \in \mathbb{R}, \text{ so } D \geq 0$$

$$(3y+1)^2 - 4y(2y+P) \geq 0$$

$$y^2 + (6-4P)y + 1 \geq 0 \quad \because y \in \mathbb{R}, \text{ so } D \leq 0$$

$$(6-4P)^2 - 4 \leq 0$$

$$(4P-6+2)(4P-6-2) \leq 0$$

$$(P-1)(P-2) \leq 0$$

$$P \in [1, 2]$$

13. It is given that,

$$\frac{\sqrt{6+x-x^2}}{2x+5} \geq \frac{\sqrt{6+x-x^2}}{x+4}$$

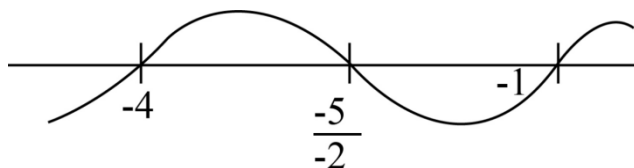
Rewrite the above equation.

$$\frac{\sqrt{6+x-x^2}}{2x+5} \left(\frac{1}{2x+5} - \frac{1}{x+4} \right) \geq 0$$

$$\sqrt{6+x-x^2} \left(\frac{x+4-2x-5}{(2x+5)(x+4)} \right) \geq 0$$

$$\sqrt{6+x-x^2} \left(\frac{-(x+1)}{(2x+5)(x+4)} \right) \geq 0$$

$$(x+1)(2x+5)(x+4) \leq 0$$



Now,

$$x \in (-\infty, -4) \cup \left[-\frac{5}{2}, -1\right] \quad \text{..... (I)}$$

For expression to be exist,

$$6 + x - x^2 \geq 0$$

$$x^2 - x - 6 \leq 0$$

$$(x - 3)(x + 2) \leq 0$$

$$x \in [-2, 3] \quad \text{..... (II)}$$

From equation (I) and (II),

$$x \in [-2, -1] \cup \{3\}$$

14. The given equation is,

$$x^3 + 4x^2 \cos \theta + x \cot \theta = 0$$

$$x(x^2 + 4x \cos \theta + \cot \theta) = 0$$

The root of the equation is $x = 0$

$x^2 + 4x \cos \theta + \cot \theta$ has multiple solutions.

Rewrite the above equation.

$$(4\cos\theta)^2 - 4(1)\cot\theta = 0$$

$$16\cos^2\theta = 4\cot\theta$$

$$4\cos^2\theta = \frac{\cos\theta}{\sin\theta}$$

$$4\cos\theta = \frac{1}{\sin\theta}$$

$$2\sin\theta\cos\theta = \frac{1}{2}$$

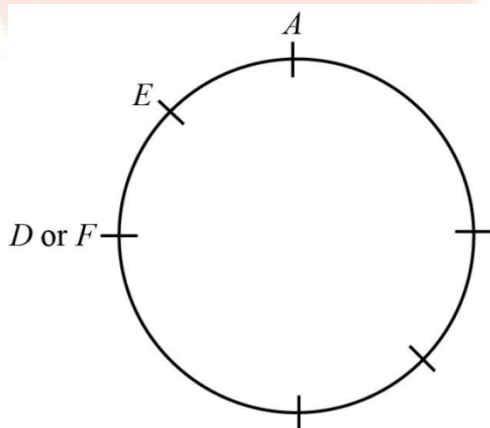
$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}$$

15. There are two cases,

(i) E is to the immediate right of A.

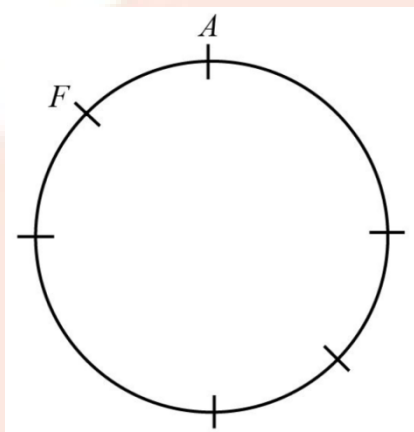


The seat on immediate right side of E can be filled in 2 ways that is either D or F.

Remaining 3 seats can be filled in $3!$ ways

Total number of ways $= 2 \times 3! = 12$

(ii) F is at the immediate right to A.



Number of ways E can fill $= 3$ ways

E cannot take sit on the left of A as there will be no space.

Number of ways D can fill $= 1$ ways

The remaining seats can be filled in 2 ways.

The total numbers of ways are $3 \times 1 \times 2 = 6$

The required number of ways are $12 + 6 = 18$

16. The number of ways a could form in which at least one member from each country in the committee,

Indians	Americans	Australians
1	1	4
1	2	3
1	3	2
1	4	1
2	1	3
2	2	2
2	3	1
3	1	2
3	2	1
4	1	1

$$N = \left({}^5C_1 \times {}^5C_1 \times {}^5C_4 + {}^5C_1 \times {}^5C_2 \times {}^5C_3 + {}^5C_3 \times {}^5C_1 \times {}^5C_2 + \right. \\ \left. {}^5C_1 \times {}^5C_4 \times {}^5C_1 + {}^5C_2 \times {}^5C_1 \times {}^5C_3 + {}^5C_2 \times {}^5C_2 \times {}^5C_2 + \right. \\ \left. {}^5C_2 \times {}^5C_3 \times {}^5C_1 + {}^5C_3 \times {}^5C_1 \times {}^5C_2 + {}^5C_3 \times {}^5C_2 \times {}^5C_1 + \right. \\ \left. {}^5C_4 \times {}^5C_1 \times {}^5C_1 \right)$$

Simplify,

$$N = 4375$$

17. It is given that $x = 2, y = 3$

By the expansion of $(2x - 3y)^8$ middle term is 5th term.

$$a = {}^8C_4 (2x)^4 (3y)^4 (-1)^4$$

$$a = \frac{8!}{4!4!} \times 2^4 \times 3^4 \times x^2 y^2$$

$$= 70 \times 2^4 \times 3^4 \times 2^4 \times 3^4$$

Similarly, in the expansion of $(3x + 4y)^7$, 4th and 5th terms are middle terms.

$$b = {}^7C_3 (3x)^4 (4y)^3$$

$$= 35 \times 3^7 \times 2^{10}$$

$$c = {}^7C_4 (3x)^3 (4y)^4$$

$$= 35 \times 3^7 \times 2^{11}$$

Now

$$\frac{b+c}{2} = \frac{35 \times 3^7 \times 2^{10} + 35 \times 3^7 \times 2^{11}}{70 \times 2^4 \times 3^4 \times 2^4 \times 3^4}$$

$$\frac{b+c}{2} = \frac{2^{10}}{2^9} = 2$$

18. The expansion of

$$(1+x)^{\frac{21}{5}} = 1 + \frac{21}{5}x + \frac{\frac{21}{5}\left(\frac{21}{5}-1\right)}{2!} + \frac{\frac{21}{5}\left(\frac{21}{5}-1\right)\left(\frac{21}{5}-2\right)}{3!}x^3 + \dots$$

Simplify the above expansion.

$$(1+x)^{21/5} = 1 + \frac{21}{5}x + \frac{21 \times 16}{5^2 \times 2^2}x^2 + \frac{21 \times 16 \times 11}{5^3 \times 3!}x^3 + \frac{21 \times 16 \times 11 \times 6}{5^4 \times 4!} + \frac{21 \times 16}{5^5 \times 5!}x^5 + \dots$$

The first negative coefficient in the terms occurring in the expansion of $(1+x)^{21/5}$ is

$$\begin{aligned} &= -\frac{21 \times 16 \times 11 \times 6 \times 1 \times 4}{5^6 \times 6!} \\ &= \frac{-88704}{5^7 \times 144} \\ &= \frac{-616}{5^7} \end{aligned}$$

19. Rewrite the given expression,

$$\begin{aligned} F &= -\left(x^2 - \frac{5}{3}x + 1\right)(1-x)^{-1}(1+2x)^{-1}\left(1+\frac{x}{3}\right)^{-1} \\ &= \left\{ -\left(x^2 - \frac{5}{3}x + 1\right)(1+x+x^2+x^3+x^4) \right. \\ &\quad \left. \left(1-2x+4x^2-8x^3+16x^4\right)\left(1-\frac{x}{3}+\frac{x^2}{9}-\frac{x^3}{27}+\frac{x^4}{81}\right) \right\} \end{aligned}$$

On neglecting higher degree terms,

$$= -\left(1+x+x^2+x^3+x^4 - \left(\frac{5}{3}x + \frac{5}{3}x^2 + \frac{5}{3}x^3 + \frac{5}{4}x^4\right) + x^2 + x^3 + x^4\right)$$

Simplify the above expression.

$$= - \left(1 - \frac{2}{3}x + \frac{x^2}{3} + \frac{x^3}{3} + \frac{x^4}{3} \right) \left(1 - \frac{7x}{3} + \frac{43}{9}x^2 - \frac{259}{27}x^3 + \frac{1555}{81}x^4 \right)$$

The coefficient of x^4 is,

$$\begin{aligned} &= \left(-\frac{1555}{81} + \frac{518}{81} + \frac{43}{27} - \frac{7}{9} + \frac{1}{3} \right) \\ &= -\frac{2166}{81} \\ &= -\frac{722}{27} \end{aligned}$$

20. It is given that $x = a \sin^\alpha \theta \cos^{\alpha+1} \theta$ and $y = a \sin^{\alpha+1} \theta \cos^\alpha \theta$

Now, $\frac{(x^2 + y^2)^m}{(xy)^2}$

Substitute the values in the above expression.

$$x = a \sin^\alpha \theta \cos^{\alpha+1} \theta \text{ and } y = a \sin^{\alpha+1} \theta \cos^\alpha \theta$$

$$\begin{aligned} &= \frac{\left\{ \left(a \sin^\alpha \theta \cos^{\alpha+1} \theta \right)^2 + \left(a \sin^{\alpha+1} \theta \cos^\alpha \theta \right)^2 \right\}^m}{\left(a \sin^\alpha \theta \cos^{\alpha+1} \theta \cdot a \sin^{\alpha+1} \theta \cos^\alpha \theta \right)^n} \end{aligned}$$

Simplify the above equation.

$$\begin{aligned}
 x &= \frac{a^{2m} (\sin^{2\alpha} \theta \cos^{2\alpha} \theta)^m (\sin^2 \theta + \cos^2 \theta)^n}{a^{2n} (\sin^{2\alpha+1} \theta \cos^{2\alpha+1} \theta)} \\
 &= \frac{a^{2m-2n} (\sin^{2\alpha} \theta \cos^{2\alpha} \theta)^{m-n}}{(\sin \theta \cos \theta)^n} \\
 &= a^{2m-2n} (\sin \theta \cos \theta)^{2\alpha(m-n)-n}
 \end{aligned}$$

For independent value of θ

$$2\alpha(m-n) - n = 0$$

$$2\alpha m - 2\alpha n = n$$

$$2\alpha m = n(2\alpha + 1)$$

21. It is given that,

$$\sqrt{4\sin^4 \theta + \sin^2 2\theta} + 4\cos^2 \left(\frac{\pi}{2} - \frac{\theta}{2} \right) = 2$$

Simplify the above expression.

$$\sqrt{4\sin^4 \theta + 4\sin^2 \theta \cos^2 \theta} + 2 \left\{ 1 + \cos 2 \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \right\} = 2$$

$$\sqrt{4\sin^2 \theta (\sin^2 \theta + \cos^2 \theta)} + 2 \{1 + \sin \theta\} = 2$$

$$\sqrt{4\sin^2 \theta} + 2 + 2\sin \theta = 2$$

$$|2\sin \theta| + 2\sin \theta = 0$$

$$\sin \theta < 0$$

Therefore, θ will lie in third or fourth quadrant.

22. It is given that,

$$x = \frac{\sin^3 \theta}{\cos^2 \theta}$$

$$y = \frac{\cos^3 \theta}{\sin^2 \theta}$$

The summation of x and y is,

$$\begin{aligned} x + y &= \frac{\sin^3 \theta}{\cos^2 \theta} + \frac{\cos^3 \theta}{\sin^2 \theta} \\ &= \frac{\sin \theta (1 - \cos^2 \theta)}{\cos^2 \theta} + \frac{\cos \theta (1 - \sin^2 \theta)}{\sin^2 \theta} \\ &= \sin \theta \left(\frac{1}{\cos^2 \theta} - 1 \right) + \cos \theta \left(\frac{1}{\sin^2 \theta} - 1 \right) \\ &= \frac{\sin \theta}{\cos^2 \theta} + \frac{\cos \theta}{\sin^2 \theta} - (\sin \theta + \cos \theta) \\ &= \frac{\sin \theta}{\cos^2 \theta} + \frac{\cos \theta}{\sin^2 \theta} - \frac{1}{2} \\ &= \frac{\sin^3 \theta + \cos^3 \theta}{\cos^2 \theta \sin^2 \theta} - \frac{1}{2} \\ &= \frac{(\sin \theta + \cos \theta)(\sin^2 \theta + \cos^2 \theta - \sin \theta \cos \theta)}{\sin^2 \theta \cos^2 \theta} - \frac{1}{2} \\ &= \frac{\frac{1}{2}(1 - \sin \theta \cos \theta)}{(\sin \theta \cos \theta)^2} - \frac{1}{2} \quad \dots (I) \end{aligned}$$

It is given that

$$\sin \theta + \cos \theta = \frac{1}{2}$$

Squaring both sides,

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{1}{4}$$

$$1 + 2 \sin \theta \cos \theta = \frac{1}{4}$$

$$\sin \theta \cos \theta = \frac{-3}{8}$$

Substitute the values in equation (I).

$$x + y = \frac{\frac{1}{2} \left(1 - \left(\frac{-3}{8} \right) \right)}{\left(\frac{-3}{8} \right)^2} - \frac{1}{2}$$

$$x + y = \frac{79}{18}$$

23. Simplify the given expression,

$$4(\sin 2x \sin 4x + \sin^2 x) = 3$$

$$4(\cos 2x - \cos 6x) + 2(1 - \cos 2x) = 3$$

$$2 \cos 2x - 2 \cos 6x + 2 - 2 \cos 2x = 3$$

$$-2 \cos 6x + 2 = 3$$

$$\cos 6x = -\frac{1}{2}$$

Therefore,

$$6x = 2\pi n \pm \frac{2\pi}{3}$$

$$x = \frac{2n\pi}{6} \pm \frac{2}{6} \times \frac{\pi}{3}$$

$$x = \frac{n\pi}{3} \pm \frac{\pi}{9}$$

24. It is given that,

$$\sum_{k=1}^n \tan^{-1} \left(\frac{1}{k^2 + k + 1} \right) = \tan^{-1} \theta$$

Simplifying the above expression, we get.

$$\sum_{k=1}^n \tan^{-1} \left(\frac{(k+1) - k}{1 + k(k+1)} \right) = \tan^{-1} \theta$$

$$\sum_{k=1}^n \{ \tan^{-1}(k+1) - \tan^{-1} k \} = \tan^{-1} \theta$$

$$\left\{ \begin{array}{l} \tan^{-1} 2 - \tan^{-1} 1 + \tan^{-1} 3 - \tan^{-1} 2 + \tan^{-1} 4 - \\ \tan^{-1} 3 + \dots + \tan^{-1}(n+1) - \tan^{-1} n \end{array} \right\} = \tan^{-1} \theta$$

$$\tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1} \theta$$

$$\tan^{-1} \left(\frac{n+1-1}{1+n+1} \right) = \tan^{-1} \theta$$

$$\frac{n}{2+n} = \theta$$

25. We know that,

$$\operatorname{sech}^{-1} x = \log \left(\frac{1 + \sqrt{1 - x^2}}{x} \right)$$

$$\tanh^{-1} x = \frac{1}{2} \log \left(\frac{1 + x}{1 - x} \right)$$

$$\sinh^{-1} x = \log \left(x + \sqrt{x^2 + 1} \right)$$

Substitute the values in the given equation and simplify.

$$\begin{aligned} e^{\log \left(\frac{1 + \sqrt{1 - 1/4}}{1/2} \right) + \frac{1}{2} \log \left(\frac{1 + 1/2}{1 - 1/2} \right) + \log \left(\frac{1}{2} + \sqrt{\frac{1}{4} + 1} \right)} &= (2 + \sqrt{3})(\sqrt{3}) \left(\frac{\sqrt{5} + 1}{2} \right) \\ &= \frac{(2\sqrt{3} + 3)(\sqrt{5} + 1)}{2} \\ &= \frac{2\sqrt{15} + 2\sqrt{3} + 3\sqrt{5} + 3}{2} \\ &= \frac{3 + 3\sqrt{5} + 2\sqrt{3} + 2\sqrt{15}}{2} \end{aligned}$$

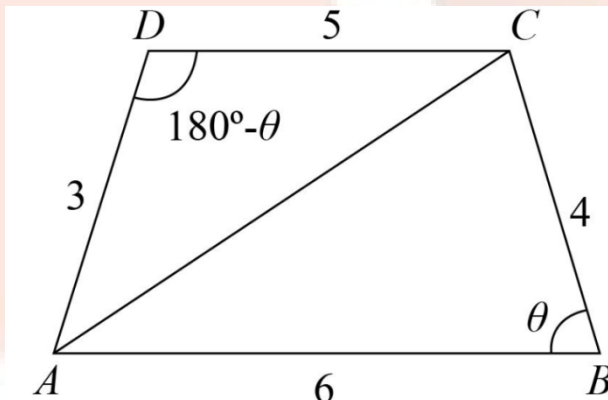
26. The ratio is given as, $a : b : c = 3 : 5 : 7$

Consider $a = 3x$, $b = 5x$, $c = 7x$

Now,

$$\begin{aligned}
 \cos A + \cos B &= \frac{b^2 + c^2 - a^2}{2bc} + \frac{a^2 + c^2 - b^2}{2ac} \\
 &= \frac{(5x)^2 + (7x)^2 - (3x)^2}{2(5x)(7x)} + \frac{(3x)^2 + (7x)^2 - (5x)^2}{2(3x)(7x)} \\
 &= \frac{65x^2}{70x^2} + \frac{33x^2}{42x^2} \\
 &= \frac{65}{70} + \frac{33}{42} \\
 &= \frac{12}{7}
 \end{aligned}$$

27. The figure below represents the quadrilateral.



In triangle ABC,

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos \theta$$

$$AC^2 = 6^2 + 4^2 - 2 \times 6 \times 4 \times \cos \theta$$

$$AC^2 = 52 - 48\cos \theta \quad \text{..... (I)}$$

Similarly in triangle ADC,

$$AC^2 = AD^2 + DC^2 - 2 \times AD \times DC \cos(180^\circ - \theta)$$

$$AC^2 = 3^2 + 5^2 + 2 \times 3 \times 5 \times \cos \theta$$

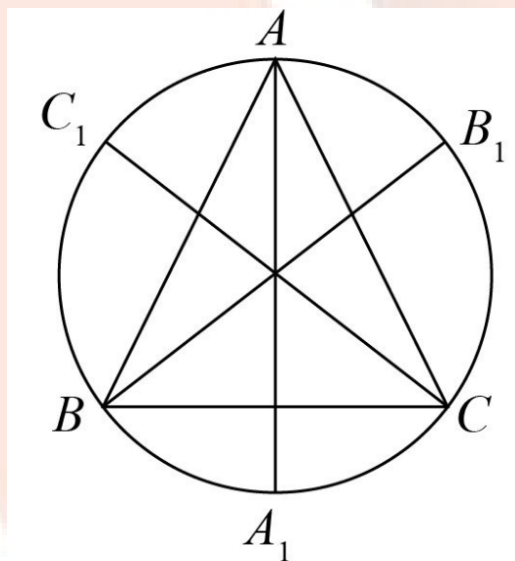
$$AC^2 = 34 + 30 \cos \theta \quad \text{..... (II)}$$

From equation (I) and (II),

$$52 - 48 \cos \theta = 34 + 30 \cos \theta$$

$$\cos \theta = \frac{3}{13}$$

28. Consider the figure below.



The triangle ABC is equilateral.

Therefore, $A = B = C = 60^\circ$

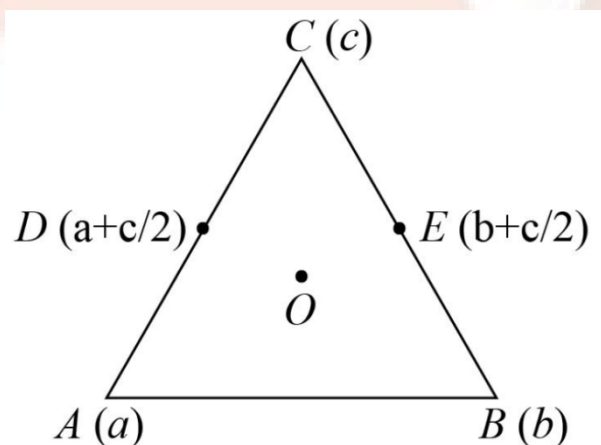
Substitute the value in the given expression.

$$\begin{aligned}
 & \left[\frac{AA_1 \cos \frac{A}{2} + BB_1 \cos \frac{B}{2} + CC_1 \cos \frac{C}{2}}{\sin A + \sin B + \sin C} \right]^2 \\
 &= \left\{ \frac{4 \cos \frac{60^\circ}{2} + 4 \cos \frac{60^\circ}{2} + 4 \cos \frac{60^\circ}{2}}{\sin 60^\circ + \sin 60^\circ + \sin 60^\circ} \right\}^2 \\
 &= \left[\frac{4 \times 3 \frac{\sqrt{3}}{2}}{3 \frac{\sqrt{3}}{2}} \right]^2 \\
 &= 16
 \end{aligned}$$

29. It is given that,

$$\vec{OA} + 2\vec{OB} + 3\vec{OC} = 0$$

The figure below represents the triangle ABC and midpoints.



The area of the triangle ODE,

$$\begin{aligned}
 A_{ODE} &= \frac{1}{2} |OD \times OE| \\
 &= \frac{1}{2} \left| \left(\frac{a+c}{2} \right) \times \left(\frac{b+c}{2} \right) \right| \\
 &= \frac{1}{8} (a \times b + a \times c + c \times b) \quad \dots (I)
 \end{aligned}$$

The given equation can be written as,

$$a + 2b + 3c = 0 \quad \dots (II)$$

Multiply b in equation (II)

$$\begin{aligned}
 a \times b + 2(b \times b) + 3(c \times b) &= 0 \\
 a \times b + 3(c \times b) &= 0 \quad \dots (III)
 \end{aligned}$$

Multiply c in equation (II).

$$\begin{aligned}
 a \times c + 2(b \times c) + 3(c \times c) &= 0 \\
 a \times c + 2(b \times c) &= 0 \quad \dots (IV)
 \end{aligned}$$

Add equation (III) and (IV)

$$\begin{aligned}
 a \times b + 3(c \times b) + a \times c + 2(b \times c) &= 0 \\
 a \times b + a \times c + c \times b &= 0
 \end{aligned}$$

Substitute the above value in equation (I).

$$A_{ODE} = 0$$

30. The vectors of the points are,

$$\vec{a} = \hat{i} - 2\hat{j} + 5\hat{k}$$

$$\vec{b} = -5\hat{j} - \hat{k}$$

$$\vec{c} = -3\hat{j} + 5\hat{k}$$

The vector equation of the plane,

$$r = a + \lambda(b - a) + \mu(c - a)$$

$$r = (\hat{i} - 2\hat{j} + 5\hat{k}) + \lambda[(-5\hat{j} - \hat{k}) - (\hat{i} - 2\hat{j} + 5\hat{k})] + \mu[(-3\hat{j} + 5\hat{k}) - (\hat{i} - 2\hat{j} + 5\hat{k})]$$

$$r = \hat{i} - 2\hat{j} + 5\hat{k} + \lambda(-\hat{i} - \hat{j} - \hat{k}) + \mu(-4\hat{i} + 7\hat{j} - 5\hat{k})$$

$$r = (1 - \lambda - 4\mu)\hat{i} - (2 + 3\lambda - 7\mu)\hat{j} + (5 - 6\lambda - 5\mu)\hat{k}$$

31. The vector is given as, $\vec{n} = \hat{i} + \hat{j} + 2\hat{k}$.

$$\text{The equation of plane is } r \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 7$$

The angle made by the vector $(2\hat{i} - \hat{j} + \hat{k})$ with $\hat{i} + \hat{j} + 2\hat{k}$ is,

$$\cos \theta = \frac{2 \times 1 + (-1) \times 1 + 1 \times 2}{\sqrt{2^2 + 1^2 + 1^2} \sqrt{1^2 + 1^2 + 2^2}}$$

$$\cos \theta = \frac{3}{\sqrt{6}\sqrt{6}} = \frac{1}{2}$$

$$\theta = 60^\circ$$

The angle made by the plane with the vector is

$$90^\circ - 60^\circ = 30^\circ$$

32. Consider $\vec{a} + \vec{b} + \vec{c} = \vec{k}$

Multiply the above equation with $\vec{a}$.

$$\vec{a} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{k} \times \vec{a}$$

$$\vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = \vec{k} \times \vec{a}$$

$$\vec{a} \times \vec{b} - \vec{c} \times \vec{a} = \vec{k} \times \vec{a}$$

$$\vec{a} \times \vec{b} - \vec{c} \times \vec{a} = \vec{k} \times \vec{a}$$

$$\vec{a} \times \vec{b} - \vec{a} \times \vec{b} = \vec{k} \times \vec{a}$$

$$0 = \vec{k} \times \vec{a}$$

$$0 = \vec{k}$$

Therefore, $\vec{a} + \vec{b} + \vec{c} = 0$

33. It is given that,

$$\vec{V} = 2\hat{i} + \hat{j} - \hat{k}$$

$$\vec{W} = \vec{i} + 3\hat{k}$$

The cross product is,

$$\vec{V} \times \vec{W} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix}$$

$$= \hat{i}(3 - 0) - \hat{j}(6 + 1) + \hat{k}(0 - 1)$$

$$= 3\hat{i} - 7\hat{j} - \hat{k}$$

Therefore,

$$\begin{aligned}
 [\vec{U}\vec{V}\vec{W}] &= \vec{U} \cdot (\vec{V} \times \vec{W}) \\
 &= |\vec{U}| |\vec{V} \times \vec{W}| \cos \theta \\
 &= (1) (\sqrt{9 + 49 + 1}) \cos \theta \\
 &= 1 (\sqrt{59}) \cos \theta
 \end{aligned}$$

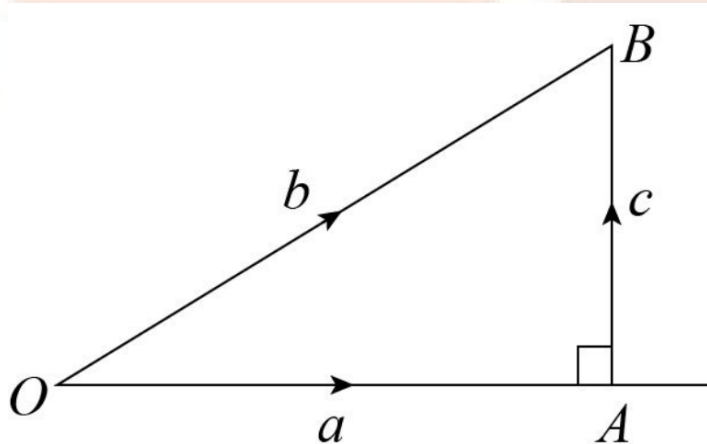
The maximum value of $[\vec{U}\vec{V}\vec{W}]$ is,

$$\begin{aligned}
 [\vec{U}\vec{V}\vec{W}] &= (\sqrt{59}) \cos 0^\circ \\
 &= \sqrt{59}
 \end{aligned}$$

34. The vector component of $\vec{b}$ along the line parallel to $\vec{a}$ is

$$\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \right) \frac{\vec{a}}{|\vec{a}|}$$

The figure below represents the vectors.



$$\overrightarrow{OA} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \right) \vec{a}$$

Now,

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \vec{b} - \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \right) \vec{a} \\ &= \frac{(\vec{a} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}}{|\vec{a}|} \\ &= \frac{a \times (b \times a)}{|\vec{a}|} \end{aligned}$$

Thus, (A) and (R) are true and (R) is the correct explanation.

35. It is given that, $z_i = (x_i - \bar{x})(y_i - \bar{y})$

And $\sum_{i=1}^{10} z_i = c$

Rewrite the above equation.

$$\sum_{i=1}^{10} \{(x_i - \bar{x})(y_i - \bar{y})\} = c$$

$$\sum_{i=1}^{10} \{x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \cdot \bar{y}\} = c$$

$$\sum_{i=1}^{10} x_i y_i - \sum_{i=1}^{10} x_i \bar{y} - \sum_{i=1}^{10} \bar{x} y_i + \sum_{i=1}^{10} \bar{x} \cdot \bar{y} = c$$

$$\sum_{i=1}^{10} x_i y_i - 10\bar{x} \cdot \bar{y} - 10\bar{x} \cdot \bar{y} + 10\bar{x} \cdot \bar{y} = c$$

$$\sum_{i=1}^{10} x_i y_i - 10\bar{x} \cdot \bar{y} = c$$

Now,

$$\begin{aligned} \sigma_{(x_i - y_i)} &= \sqrt{\frac{1}{10} \sum (x_i - y_i)^2 - (\bar{x} - \bar{y})^2} \\ &= \sqrt{\left(\frac{1}{10} \sum x_i^2 + \frac{1}{10} \sum y_i^2 - \frac{1}{5} \sum x_i y_i - \bar{x}^2 - \bar{y}^2 + 2\bar{x}\bar{y} \right)} \\ &= \sqrt{\left(\frac{1}{10} \sum x_i^2 - \bar{x}^2 \right) + \left(\frac{1}{10} \sum y_i^2 - \bar{y}^2 \right) - \frac{1}{5} (\sum x_i y_i - 10\bar{x}\bar{y})} \\ &= \sqrt{a^2 + b^2 - \frac{c}{5}} \end{aligned}$$

36. It is given that,

$$\bar{x}_1 = 40$$

$$\sigma = 15$$

Now, the mean is,

$$\bar{x}_1 = \frac{\sum x_i}{n}$$

$$40 = \frac{\sum x_i}{100}$$

$$\sum x_i = 4000$$

Corrected summation,

$$\sum x'_2 = \sum x_i - 30 - 60 + 40 + 50 = 4000$$

The corrected mean,

$$\bar{x}_2 = \frac{\sum x'_2}{100}$$

$$\bar{x}_2 = 4000 / 100 = 400$$

The standard deviation is,

$$\sigma_1^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$225 = \frac{\sum x_i^2}{100} - \left(\frac{4000}{100} \right)^2$$

$$\sum x_i^2 = 182500$$

Corrected summation,

$$\sum x_2^2 = 182500 - 30^2 - 60^2 + 40^2 + 50^2$$

$$\sum x_2^2 = 182100$$

The corrected standard deviation,

$$\sigma_2^2 = \frac{\sum x_2^2}{n} - \left(\frac{\sum x_2}{n} \right)^2$$

$$\sigma_2^2 = \frac{182500}{100} - \left(\frac{4000}{100} \right)^2$$

$$\sigma_2^2 = 1821 - 1600$$

$$\sigma_2 = 14.86$$

Thus, $\bar{x}_1 = \bar{x}_2$ and $\sigma_1 > \sigma_2$.

37. Consider A and B be the events when the sum is obtained as 7 and 11 respectively.

Number of times sum 7 and 11 occurs,

$$n(A) = \{(2,5), (5,2), (3,4), (4,3), (1,1), (6,1)\} = 6$$

$$n(B) = \{(5,6), (6,5)\} = 2$$

Now, the probability is,

$$P(A) = \frac{6}{36} = \frac{1}{6}$$

$$P(B) = \frac{2}{36} = \frac{1}{18}$$

The probability when neither the sum of 7 or 11 occurs,

$$P(C) = \frac{36 - (6 + 2)}{36} = \frac{7}{9}$$

The probability that 7 comes before 11,

$$\begin{aligned}
 P &= \frac{1}{6} + \frac{1}{6} \times \frac{7}{9} + \left(\frac{7}{9}\right)^2 \left(\frac{1}{6}\right) + \left(\frac{7}{9}\right)^3 \left(\frac{1}{6}\right) \dots \\
 &= \frac{1}{6} \left(\frac{1}{1 - 7/9} \right) \\
 &= \frac{3}{4}
 \end{aligned}$$

38. Consider a be the number of die A and b be the number of die B .

It is given that $a + b$ is even.

Therefore, a and b are both even or both odd.

Thus, the probability of both even or both odd is,

$$P = \frac{1}{2}$$

Thus, for 100th times, the required probability is $\left(\frac{1}{2}\right)^{100}$.

39. Consider E_1, E_2 and E_3 be the events of selecting defective toys in cartons A, B and C .

From Bays rule, the probability that a selected carton is drawn from carton B,

$$\begin{aligned}
 P &= \frac{P(E_2) \cdot P\left(\frac{A}{E_2}\right)}{P(E_1) \cdot P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right)} \\
 &= \frac{\frac{1}{3} \times \frac{1}{4}}{\frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{4} + \frac{1}{3} \times \frac{2}{5}} \\
 &= \frac{\frac{1}{12}}{\frac{1}{9} + \frac{1}{12} + \frac{2}{15}} = \frac{15}{59}
 \end{aligned}$$

40. The numbers that are greater than 4 on the die is 5,6.

The probability of successes is,

$$p = \frac{2}{6} = \frac{1}{3}$$

Now, the variance of probability distribution,

$$\begin{aligned}
 \sigma^2 &= npq \\
 &= 2 \times \frac{1}{3} \times \frac{2}{3} \\
 &= \frac{4}{9}
 \end{aligned}$$

41. It is given that,

$$2P(X=1) = 5P(X=5) + 2P(X=3),$$

The formula for Poisson $P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$

$$\frac{2e^{-\lambda} \lambda}{1!} = \frac{5e^{-\lambda} \lambda^5}{5!} + \frac{2e^{-\lambda} \lambda^3}{3!}$$

$$2 = \frac{\lambda^4}{24} + \frac{\lambda^2}{3}$$

$$2 = \frac{3\lambda^4 + 24\lambda^2}{72}$$

$$144 = 3\lambda^4 + 24\lambda^2$$

$$\lambda^4 + 8\lambda^2 - 48 = 0$$

Consider $\lambda^2 = x$

$$x^2 + 8x - 48 = 0$$

$$(x+12)(x-4) = 0$$

$$x = 4 \text{ and } x \neq -12$$

The standard deviation is,

$$x = \lambda^2 = 4 \text{ and}$$

$$\sqrt{\lambda} = \sqrt{2}$$

42. It is given that, $AP + BP = 2$

Consider the point P be (x, y)

$$\sqrt{(x-1)^2 + y^2} + \sqrt{x^2 + (y-1)^2} = 2$$

$$\sqrt{(x-1)^2 + y^2} = 2 - \sqrt{x^2 + (y-1)^2}$$

Squaring on both sides,

$$(x-1)^2 + y^2 = 4 + x^2 + (y-1)^2 - 4\sqrt{x^2 + (y-1)^2}$$

$$x^2 + y^2 + 1 - 2x = 4 + x^2 + y^2 + 1 - 2y - 4\sqrt{x^2 + (y-1)^2}$$

$$2y - 2x = 4 - 4\sqrt{x^2 + (y-1)^2}$$

$$y - x - 2 = -2\sqrt{x^2 + (y-1)^2}$$

Squaring both the sides,

$$x^2 + y^2 + 4 - 2xy + 4x - 4y = 4x^2 + 4y^2 + 4 - 8y$$

$$3x^2 + 3y^2 - 4x - 4y + 2xy = 0$$

43. The original and new coordinates are,

$$x = x \cos \frac{\pi}{4} - y \sin \frac{\pi}{4} = \frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}}$$

$$y = x \sin \frac{\pi}{4} + y \cos \frac{\pi}{4} = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

The equation of the curve is,

$$9x^2 + 25y^2 = 225$$

$$9\left(\frac{x-y}{\sqrt{2}}\right)^2 + 25\left(\frac{x+y}{\sqrt{2}}\right)^2 = 225$$

After simplification,

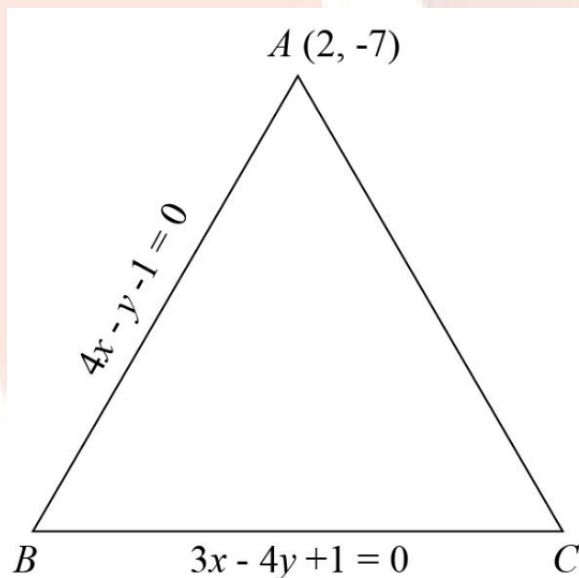
$$9x^2 + 9y^2 + 18xy + 25x^2 + 25y^2 - 50xy = 450$$

$$34x^2 + 34y^2 - 32xy = 450$$

$$17x^2 + 17y^2 - 16xy = 225$$

44. The equation of line given as, $4x + y - 1 = 0$, $3x - 4y + 1 = 0$

The slopes of line are, $m_1 = -4$ and $m_2 = 3/4$



Triangle ABC is an isosceles; $AB = AC$

AB and AC both pass through points $(2, -7)$.

The slope is,

$$m_3 = \left| \frac{-4 - \frac{3}{4}}{1 + (-4)\left(\frac{3}{4}\right)} \right| = \frac{19}{8}$$

The required equation of line AC is,

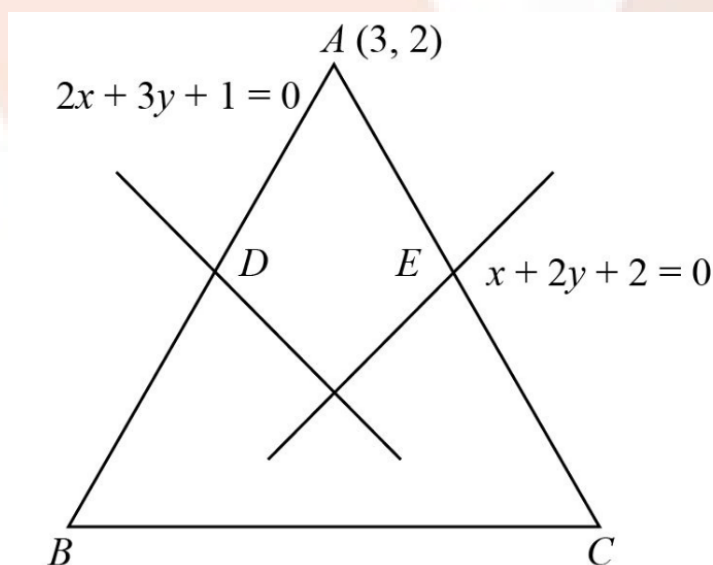
$$\frac{y+7}{x-2} = \frac{\frac{3}{4} - \frac{19}{8}}{1 + \frac{3}{4} \times \frac{19}{8}}$$

After simplification,

$$89y + 623 = -52x + 104$$

$$52x + 89y + 519 = 0$$

45. The figure below represents the triangle and lines bisectors of its sides.



The equation of AB is $3x - 2y + c = 0$

As it passes through $A(3,2)$, the value of $c = -5$

Thus, $3x - 2y - 5 = 0$

The coordinates of D is $(1, -1)$ as D is the mid-point of AB .

Consider the coordinates of B are (α, β) .

$$\frac{3 + \alpha}{2} = 1, \frac{2 + \beta}{2} = -1$$

$$\alpha = -1, \beta = -4$$

$$B(-1, -4)$$

Similarly, the equation of AC is $2x - y + c = 0$

As it passes through $(3,2)$, the value of $c = -4$

Thus, $2x - y - 4 = 0$.

E is a point of intersection of $x + 2y - 2 = 0$ and $2x - y - 4 = 0$

$E = (2, 0)$ and E is the midpoint of AC .

The coordinates of C are,

$$\frac{3 + \alpha}{2} = 2, \frac{2 + \beta}{2} = 0$$

$$\alpha = 1, \beta = -2$$

$$C(1, -2)$$

The equation of BC is,

$$\frac{y + 2}{x - 1} = \frac{-2 + 4}{1 + 1}$$

Further simplification,

$$x - y - 3 = 0$$

46. Consider the equation of perpendicular bisector is $y = mx + c$

The intercept is 1, thus the equation becomes $y = mx + 1$

The mid-point of the points $A(\alpha, 3)$ and $B(2, -1)$ is $\left(\frac{\alpha + 2}{2}, 1\right)$

Since the equation passes through $\left(\frac{\alpha + 2}{2}, 1\right)$.

Thus,

$$1 = m\left(\frac{\alpha + 2}{2}\right) + 1$$

$$2 = m(\alpha + 2) + 2$$

$$m(\alpha + 2) = 0$$

$$m\alpha + 2m = 0 \quad \dots\dots (I)$$

Here, m is the gradient of the perpendicular line.

$$m = -\left(\frac{\alpha - 2}{4}\right)$$

Substitute the value in equation (I).

$$\alpha\left(\frac{2 - \alpha}{4}\right) + 2\left(\frac{2 - \alpha}{4}\right) = 0$$

$$\alpha = \pm 2$$

47. The pair of the lines represented by the given equation will be perpendicular to each other, when

$$\text{Coefficient of } x^2 + \text{coefficient of } y^2 = 0$$

Now,

$$3a + a^2 - 2 = 0$$

On solving the above quadratic equation,

$$a = \frac{-3 \pm \sqrt{17}}{2}$$

There are two values.

48. The equation of the line is given as $ax + by = 1$ and the equation of the curve is given as, $x^2 + y^2 - x - y - 1 = 0$

By homogenising both equations,

$$x^2 + y^2 - x(ax + by) - y(ax + by) - (ax + by)^2 = 0$$

$$x^2 + y^2 - ax^2 - bxy - axy - by^2 - (a^2x^2 + b^2y^2 + 2abxy) = 0$$

$$x^2(1 - a - a^2) + xy(-a - b - 2ab) + y^2(1 - b - b^2) = 0$$

Since it is given that the pair of lines joining the origin and the points of intersection of the line and curve are at right angles.

Coefficient of x^2 + coefficient of $y^2 = 0$

$$1 - a - a^2 + 1 - b - b^2 = 0$$

$$a^2 + b^2 + a + b - 2 = 0$$

The locus of the circle from the above equation (a, b)

Now,

$$x^2 + y^2 + x + y - 2 = 0$$

On comparing in equation (i) with $x^2 + y^2 + 2gx + 2fy + c = 0$,

The radius is,

$$\begin{aligned} r &= \sqrt{g^2 + f^2 - c} \\ &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + 2} \\ &= \sqrt{\frac{1}{4} + \frac{1}{4} + 2} \\ &= \sqrt{\frac{5}{2}} \end{aligned}$$

49. The area of the circle is $A = 9\pi$

The given equations are,

$$x + 2y - 5 = 0$$

$$2x - 3y + 4 = 0$$

From above equations, the value of x and y are 1 and 2 respectively.

The point of intersection is $(x, y) = (1, 2)$.

The area is,

$$\pi r^2 = 9\pi$$

$$r^2 = 9$$

The equation of the circle is,

$$(x-1)^2 + (y-2)^2 = 9$$

$$x^2 + y^2 - 2x - 4y - 4 = 0$$

50. The equation of circles are given as,

$$x^2 + y^2 = b^2$$

$$(x-a)^2 + y^2 = b^2$$

The equation of the common tangent for 1st is,

$$y = mx - b\sqrt{1+m^2}$$

The equation of the common tangent for 2nd circle is,

$$y = m(x-a) + b\sqrt{1+m^2}$$

Equate the common tangents,

$$mx - b\sqrt{1+m^2} = mx - am + b\sqrt{1+m^2}$$

$$2b\sqrt{1+m^2} = am$$

$$a^2m^2 = 4b^2 + 4b^2m^2$$

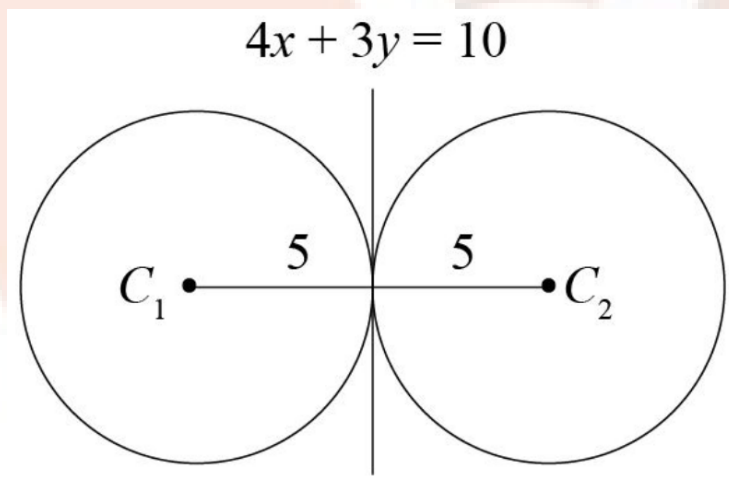
$$a^2m^2 = 4b^2 + 4b^2m^2$$

$$m^2(a^2 - 4b^2) = 4b^2$$

The value of m is,

$$m = \frac{2b}{\sqrt{a^2 - 4b^2}}$$

51. The figure below represents the two circles with the common tangent.



The slope of the common tangent,

$$m = -\frac{4}{3}$$

The slope of the line perpendicular to tangent is,

$$m' = \tan \theta = \frac{3}{4}$$

$$\text{Therefore, } \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}$$

Now,

$$\frac{x-1}{4/5} = \frac{y-1}{3/5} = \pm 5$$

Now,

$$x = \left(\pm 5 \times \frac{4}{5} + 1 \right), y = \left(\pm 5 \times \frac{3}{5} + 2 \right)$$

$$x = (5, -3), y = (5, -1)$$

The coordinates of $C_1(5, 5)$ and $C_2(-3, -1)$.

The equations of the required circles is,

$$(x-5)^2 + (y-5)^2 = 5^2$$

$$x^2 + y^2 - 10x - 10y + 25 = 0$$

And

$$(x+3)^2 + (y+1)^2 = 5^2$$

$$x^2 + y^2 + 6x + 2y - 15 = 0$$

52. The equation of the circles is given as,

$$x^2 + y^2 + 4x - 5 = 0$$

$$x^2 + y^2 + 4\lambda y - 5 = 0$$

On comparing in equations with $x^2 + y^2 + 2gx + 2fy + c = 0$,

$$g_1 = 2, f_1 = 0, c_1 = -5$$

$$g_2 = 0, f_2 = \lambda, c_2 = -4$$

The angle between the circles is,

$$\cos \theta = \frac{2g_1g_2 + 2f_1f_2 - c_1 - c_2}{2\sqrt{g_1^2 + f_1^2 + c_1}\sqrt{g_2^2 + f_2^2 + c_2}}$$

$$\cos \frac{\pi}{3} = \frac{2 \times 2 \times 0 + 2 \times 0 \times \lambda - 5 - 4}{2\sqrt{4 + 0 + 5}\sqrt{0 + \lambda^2 + 4}}$$

$$\frac{1}{2} = \frac{3}{2\sqrt{\lambda^2 + 4}}$$

After simplification,

$$\lambda = \pm 5$$

53. The equation of a circle passing through the points of intersection of the circles,

$$(x^2 + y^2 - 4x - 6y - 12) + \lambda(x^2 + y^2 + 6x + 4y - 12) = 0$$

$$x^2(1 + \lambda) + y^2(1 + \lambda) + x(6\lambda - 4) + y(4\lambda - 4) - 12\lambda - 12 = 0$$

Rewrite the above equation.

$$x^2 + y^2 + \frac{x(6\lambda - 4)}{(1 + \lambda)} + \frac{y(4\lambda - 6)}{(1 + \lambda)} - 12 = 0$$

On comparing in equations with $x^2 + y^2 + 2gx + 2fy + c = 0$,

$$g = \frac{(3\lambda - 2)}{(1 + \lambda)}, f = \frac{(2\lambda - 3)}{(1 + \lambda)}, c = -12$$

It is given that the radius is $\sqrt{13}$.

$$13 = \frac{(3\lambda - 2)^2}{(1 + \lambda)^2} + \frac{(2\lambda - 3)^2}{(1 + \lambda)^2} + 12$$

$$13(1 + \lambda)^2 = (3\lambda - 2)^2 + (2\lambda - 3)^2 + 12(1 + \lambda)^2$$

After simplification,

$$13\lambda^2 + 13 + 26\lambda = 25\lambda^2 + 25$$

$$12\lambda^2 - 26\lambda + 12 = 0$$

$$6\lambda^2 - 13\lambda + 6 = 0$$

$$(2\lambda - 3)(3\lambda - 2) = 0$$

$$\lambda = \frac{3}{2}, \frac{2}{3}$$

The required equation when $\lambda = \frac{3}{2}$,

$$x^2 \left(1 + \frac{3}{2}\right) + y^2 \left(1 + \frac{3}{2}\right) + x \left(6 \times \frac{3}{2} - 4\right) + y \left(4 \times \frac{3}{2} - 6\right) - 12 \times \frac{3}{2} - 12 = 0$$

Further simplification,

$$x^2 + y^2 + 2x - 12 = 0$$

The required equation when $\lambda = \frac{2}{3}$,

$$x^2 \left(1 + \frac{2}{3}\right) + y^2 \left(1 + \frac{2}{3}\right) + x \left(6 \times \frac{2}{3} - 4\right) + y \left(4 \times \frac{2}{3} - 6\right) - 12 \times \frac{2}{3} - 12 = 0$$

Further simplification,

$$x^2 + y^2 - 2x - 12 = 0$$

54. The given equation of parabola,

$$y^2 = 4x$$

The equation of normal at point $P(5, 0)$,

$$y = -tx + 2t + t^3$$

$$0 = -5t + 2t + t^3$$

$$3t = t^3$$

$$t^2 = 3$$

$$t = \pm\sqrt{3}, 0$$

Here, $a = 1$

Co normal point

$$P(at_1^2, 2at_1) = (3, 2\sqrt{3})$$

$$Q(at_2^2, 2at_2) = (3, -2\sqrt{3})$$

$$R(at_3^2, 2at_3) = (0, 0)$$

The centroid is,

$$C = \left(\frac{3+3+0}{3}, \frac{2\sqrt{3}-2\sqrt{3}+0}{3} \right)$$

$$C = \left(\frac{6}{3}, 0 \right) = (2, 0)$$

55. The equation of the parabola,

$$y^2 = 4x$$

The equation of the normal to the parabola,

$$y^2 = 4ax \text{ at } (am^2, -2am) \text{ is } y = mx - 2am - am^3,$$

Here, $a = 1$, the equation of the normal becomes,

$$y = mx - 2m - m^3 \quad \text{..... (I)}$$

Since the normal is perpendicular to the line,

$$x + 3y + 1 = 0$$

Thus,

$$m_1 m_2 = -1$$

$$m \left(\frac{-1}{3} \right) = -1$$

$$m = 3$$

Substitute the values in equation (I).

$$y = 3x - 2(3) - 3^3$$

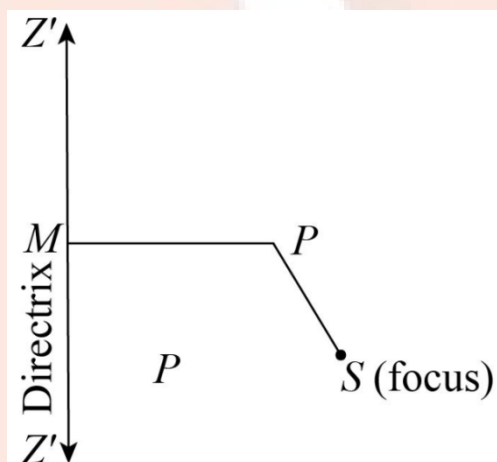
$$y = 3x - 6 - 27$$

$$3x - y = 33$$

56. The given equation of the ellipse is,

$$7x^2 + 16y^2 = 112$$

$$\frac{x^2}{16} + \frac{y^2}{7} = 1$$



The eccentricity of the ellipse is,

$$e = \sqrt{1 - \frac{7}{16}} = \sqrt{\frac{9}{16}} = \frac{3}{4}$$

And

$$\frac{SP}{PM} = e$$

$$\frac{SP}{PM} = \frac{3}{4}$$

57. The given equation of the ellipse is,

$$\frac{x^2}{9} + \frac{y^2}{5} = 1$$

The eccentricity of the ellipse is,

$$e = \sqrt{1 - \frac{5}{9}} = \frac{2}{3}$$

The coordinates of the end points of latusrectum are

$$L\left(2, \frac{5}{3}\right), M\left(-2, \frac{5}{3}\right), M'\left(-2, -\frac{5}{3}\right) \text{ and } L'\left(2, -\frac{5}{3}\right)$$

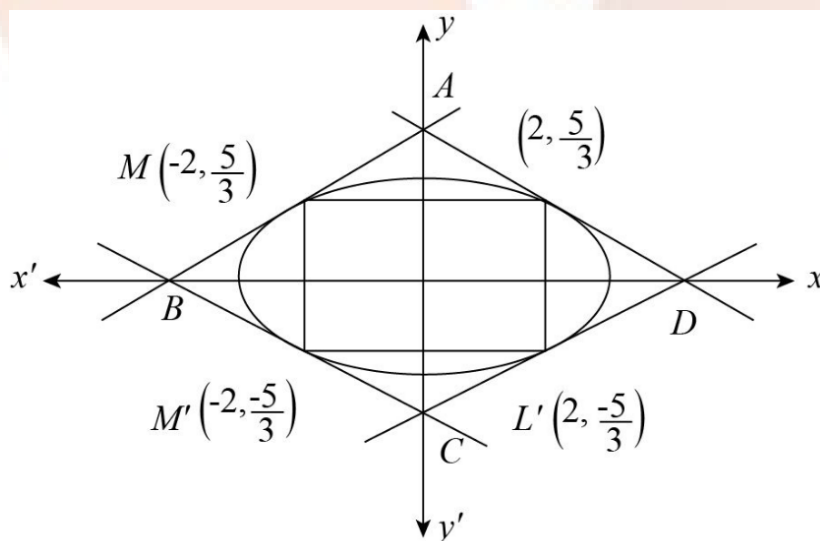
The equations of tangents at these points are,

$$2x + 3y - 9 = 0$$

$$-2x + 3y - 9 = 0$$

$$2x + 3y + 9 = 0$$

$$-2x + 3y + 9 = 0$$



By solving above equations,

$$A(0,3)$$

$$B\left(-\frac{9}{2},0\right)$$

$$C(0,-3)$$

$$D\left(\frac{9}{2},0\right)$$

And

$$AC = \sqrt{(3+3)^2} = 6$$

$$BD = \sqrt{\left(\frac{9}{2} + \frac{9}{2}\right)^2} = 9$$

The area of the quadrilateral is,

$$\begin{aligned} A &= \frac{1}{2} \times 9 \times 6 \\ &= 27 \text{ sq units} \end{aligned}$$

58. Consider the equation of the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

It is given that the centre of the hyperbola is (0,0) and its transverse axis is along x-axis and its length is 12.

$$2a = 12$$

$$a = 6$$

Substitute this value in equation of hyperbola.

$$\frac{x^2}{6^2} - \frac{y^2}{b^2} = 1$$

The above equation is passes through point (8,2)

$$\frac{8^2}{6^2} - \frac{2^2}{b^2} = 1$$

$$\frac{64}{36} - \frac{4}{b^2} = 1$$

After simplification,

$$b = \frac{6}{\sqrt{7}}$$

The equation of hyperbola becomes,

$$\frac{x^2}{6^2} - \frac{y^2}{\left(\frac{6}{\sqrt{7}}\right)^2} = 1$$

$$\frac{x^2}{36} - \frac{7y^2}{36} = 1$$

The eccentricity of hyperbola is,

$$e = \sqrt{1 + \frac{a^2}{b^2}}$$

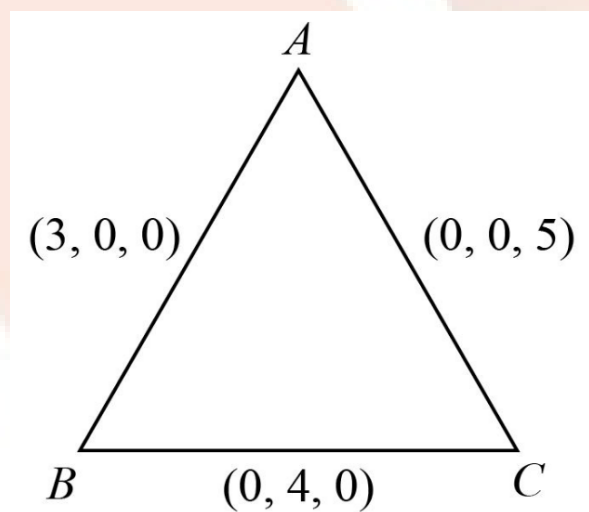
$$e = \sqrt{1 + \frac{6^2}{(6/\sqrt{7})^2}}$$

$$e = \sqrt{\frac{8}{7}}$$

$$e = \frac{2\sqrt{2}}{\sqrt{7}}$$

59. The mid points of the sides are given as,

AB , BC and CA are $(3, 0, 0)$, $(0, 4, 0)$, $(0, 0, 5)$.



The coordinates of A , B and C

$$A = (3, -4, 5)$$

$$B = (3, 4, -5)$$

$$C = (-3, 4, 5)$$

And

$$\begin{aligned}AB^2 &= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \\&= (3 - 3)^2 + (4 + 4)^2 + (-5 - 5)^2 \\&= 8^2 + 10^2 \\&= 164\end{aligned}$$

Similarly,

$$\begin{aligned}BC^2 &= (-3 - 3)^2 + (4 - 4)^2 + (5 + 5)^2 \\&= 36 + 100 \\&= 136\end{aligned}$$

Similarly,

$$\begin{aligned}CA^2 &= (-3 - 3)^2 + (4 - 4)^2 + (5 + 5)^2 \\&= 36 + 64 \\&= 100\end{aligned}$$

$$\begin{aligned}AB^2 + BC^2 + CD^2 &= 164 + 136 + 100 \\&= 400\end{aligned}$$

60. It is given that the direction ratios of first line are 2,2,1 and the direction ratios of second line joining the points (3,1,4) and (7,2,12) are,

$$(7 - 3, 2 - 1, 12 - 4)$$

$$(4, 1, 8)$$

The angle between the two lines is,

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$\cos \theta = \frac{2 \times 4 + 2 \times 1 + 1 \times 8}{\sqrt{9} \sqrt{81}} = \frac{18}{27}$$

$$\theta = \cos^{-1} \left(\frac{2}{3} \right)$$

61. It is given that the points,

$$A(-2, 1, 3), B(1, 1, 1), C(2, 3, 4)$$

The position vectors are,

$$\overrightarrow{AB} = (\hat{i} + \hat{j} + \hat{k}) - (-2\hat{i} + \hat{j} + \hat{k}) = 3\hat{i} - 2\hat{j}$$

$$\overrightarrow{BC} = (2\hat{i} + 3\hat{j} + 4\hat{k}) - (\hat{i} + \hat{j} + \hat{k}) = \hat{i} + 2\hat{j} + 3\hat{k}$$

The normal of the plane,

$$\vec{n} = \overrightarrow{AB} \times \overrightarrow{BC}$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & -2 \\ 1 & 2 & 3 \end{vmatrix}$$

By solving the above matrix,

$$\vec{n} = 4\hat{i} - 11\hat{j} + 6\hat{k}$$

The equation of the plane is,

$$4(x-2) - 11(y-3) + 6(z-4) = 0$$

$$4x - 11y - 6z + 1 = 0$$

If $ax + by + cz + d = 0$ is the equation of plane, then the normal form is,

$$\left\{ \left(\frac{a}{\sqrt{a^2 + b^2 + c^2}} \right) x + \left(\frac{b}{\sqrt{a^2 + b^2 + c^2}} \right) y + \left(\frac{c}{\sqrt{a^2 + b^2 + c^2}} \right) z + \left(\frac{d}{\sqrt{a^2 + b^2 + c^2}} \right) \right\} = 0$$

Substitute the value and simplify the equation.

$$\left\{ \frac{4x}{\sqrt{173}} - \frac{11y}{\sqrt{173}} + \frac{6z}{\sqrt{173}} + \frac{1}{\sqrt{173}} \right\} = 0$$

$$\left(\frac{-4}{\sqrt{173}} \right) x + \left(\frac{11}{\sqrt{173}} \right) y + \left(\frac{-6}{\sqrt{173}} \right) z = \frac{1}{\sqrt{173}}$$

62. It is given that,

$$\alpha = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x}$$

Apply L hospital rule,

$$\alpha = \lim_{x \rightarrow 0} \frac{2^x + x \cdot 2^x \log 2 - 1}{\sin x}$$

Again, apply L hospital rule,

$$\alpha = \lim_{x \rightarrow 0} \frac{2^x \log x + 2^x \log 2 + x \cdot 2^x (\log 2)^2}{\cos x}$$

$$\alpha = \log 2 + \log 2$$

$$\alpha = 2 \log 2 \quad \dots\dots (I)$$

And

$$\beta = \lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{\sqrt{1+x^2} - \sqrt{1-x^2}}$$

Apply L hospital rule,

$$\beta = \lim_{x \rightarrow 0} \frac{2^x + x \cdot 2^x \log 2 - 1}{\left(\frac{2x}{2\sqrt{1+x^2}} - \frac{2x}{2\sqrt{1-x^2}} \right)}$$

Again, apply L hospital rule,

$$\beta = \lim_{x \rightarrow 0} \frac{2^x \log 2 + 2^x \log 2 + x \cdot 2^x (\log 2)^2}{\left(\sqrt{1+x^2} - \frac{2x}{2\sqrt{1+x^2}} \right) + \left(\sqrt{1-x^2} + \frac{2x}{2\sqrt{1-x^2}} \right)}$$

$$\beta = \frac{2 \log 2}{(1-0) + (1+0)} = \frac{2 \log 2}{2}$$

$$\beta = \log 2 \quad \dots\dots (II)$$

From equation (I) and (II),

$$\alpha = 2\beta$$

63. The series is given as,

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \left(\frac{1}{3 \cdot 7} + \frac{1}{7 \cdot 11} + \frac{1}{11 \cdot 15} + \dots + (n \text{ terms}) \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{4} \left(\frac{4}{3 \cdot 7} + \frac{4}{7 \cdot 11} + \frac{4}{11 \cdot 15} \dots \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{4} \left(\frac{4}{3 \cdot 7} + \frac{4}{7 \cdot 11} + \frac{4}{11 \cdot 15} \dots \frac{1}{n} + \frac{1}{n} - \frac{1}{n+4} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{4} \left(\frac{1}{3} - \frac{1}{n+4} \right) \\
 &= \lim_{n \rightarrow \infty} \frac{1}{12} - \lim_{n \rightarrow \infty} \frac{1}{4(n+4)} \\
 &= \frac{1}{12} - \lim_{n \rightarrow \infty} \frac{1}{4 \left(1 + \frac{4}{n} \right)} \\
 &= \frac{1}{12} - 0 = \frac{1}{12}
 \end{aligned}$$

64. It is given that,

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + ax + b} - x \right)$$

Rationalize the above expression,

$$= \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + ax + b} - x \times \frac{\sqrt{x^2 + ax + b} + x}{\sqrt{x^2 + ax + b} + x} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{x^2 + ax + b - x}{\sqrt{x^2 + ax + b} + x} \right)$$

$$= \lim_{x \rightarrow \infty} \left(\frac{ax + b}{\sqrt{x^2 + ax + b} + x} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x \left(a + \frac{b}{x} \right)}{x \left(\sqrt{x^2 + \frac{a}{x} + \frac{b}{x^2}} + 1 \right)}$$

$$= \frac{a}{1+1} = \frac{a}{2}$$

It depends on a .

65. It is given that the function is differentiable at all points,

By first principle,

$$\text{LHD (at } x = 1) = \text{RHD (at } x = 1)$$

$$\lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1}$$

$$\lim_{x \rightarrow 1^-} \frac{\alpha x^2 - \beta - \alpha + \beta}{x - 1} = \lim_{x \rightarrow 1^+} \frac{-\frac{1}{x} + 1}{x - 1}$$

$$\lim_{x \rightarrow 1^-} \frac{\alpha(x^2 - 1)}{x - 1} = \lim_{x \rightarrow 1^+} \left(\frac{x - 1}{x} \right)$$

$$\lim_{x \rightarrow 1^-} \frac{\alpha(x + 1)(x - 1)}{x - 1} = \lim_{x \rightarrow 1^+} \frac{1}{x}$$

$$\lim_{x \rightarrow 1^-} \alpha(x + 1) = \lim_{x \rightarrow 1^+} \frac{1}{x}$$

$$\alpha(1 + 1) = 1$$

$$\alpha = 1/2$$

Every differentiable function is also continuous,

$$\text{RHL} = \text{LHL}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$\frac{1}{2}(1)^2 - \beta = -\frac{1}{1}$$

$$\beta = \frac{1}{2} + 1$$

$$\beta = 3/2$$

66. The equation is given as,

$$y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right)$$

Substitute $x = \cos 2\theta$ (I)

$$y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}} \right)$$

$$y = \sin^2 \left(\cot^{-1} \sqrt{\frac{2\cos^2 \theta}{2\sin^2 \theta}} \right)$$

$$y = \sin^2 (\cot^{-1} (\cot \theta))$$

$$y = \sin^2 \theta \quad \text{..... (II)}$$

Differentiate the above equation with respect θ ,

$$\frac{dy}{d\theta} = 2 \sin \theta \cos \theta$$

$$\frac{dy}{d\theta} = \sin 2\theta \quad \text{..... (III)}$$

Differentiate the above equation (I) with respect θ ,

$$\frac{dx}{d\theta} = -2 \sin 2\theta \quad \text{..... (IV)}$$

From equation (III) and (IV)

$$\frac{dy/d\theta}{dx/d\theta} = \frac{\sin 2\theta}{-2\sin 2\theta}$$

$$\frac{dy}{dx} = \frac{-1}{2}$$

67. It is given that,

$$a \neq b, x \neq n\pi, n \in \mathbb{Z} \text{ and } y^2 = a^2 \cos^2 x + b^2 \sin^2 x$$

Differentiate the above equation with respect to x ,

$$2y \frac{dy}{dx} = -a^2 \sin 2x + b^2 \sin 2x$$

$$2y \frac{dy}{dx} = (b^2 - a^2) \sin 2x$$

Again differentiate the above equation with respect to x ,

$$2 \left(y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right) = 2(b^2 - a^2) \cos 2x$$

$$y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = (b^2 - a^2) \cos 2x$$

Multiply y^2 on both sides,

$$y^3 \frac{d^2 y}{dx^2} + y^2 \left(\frac{dy}{dx} \right)^2 = y^2 (b^2 - a^2) \cos 2x$$

$$y^3 \frac{d^2 y}{dx^2} = y^2 (b^2 - a^2) \cos 2x - y^2 \left(\frac{dy}{dx} \right)^2$$

Add y^4 in the above equation.

$$y^4 + y^3 \frac{d^2 y}{dx^2} = y^4 + y^2 (b^2 - a^2) \cos 2x - y^2 \left(\frac{dy}{dx} \right)^2 + y^4$$

Substitute the values in the above equation.

$$\begin{aligned} y^4 + y^3 \frac{d^2 y}{dx^2} &= y^4 + y^2 (b^2 - a^2) \cos 2x - y^2 \left(\frac{dy}{dx} \right)^2 + y^4 \\ &= \left\{ (a^2 \cos^2 x + b^2 \sin^2 x)(b^2 - a^2)(\cos^2 x - \sin^2 x) - \right. \\ &\quad \left. (b^2 - a^2 \sin x \cos x)^2 + (a^2 \cos^2 x + b^2 \sin^2 x)^2 \right\} \\ &= a^2 b^2 \cos^4 x + a^2 b^2 \sin^2 x + 2a^2 b^2 \sin^2 x \cos^2 x \\ &= a^2 b^2 (\sin^2 x + \cos^2 x)^2 \\ &= a^2 b^2 \end{aligned}$$

Or it can be written as,

$$\frac{d^2 y}{dx^2} + y = \frac{1}{y} \left(\frac{ab}{y} \right)^2$$

68. It is given that $2y = 3x - 1$ is tangent to $y^2 = ax^3 + b$ at point $(1,1)$ with slope $\frac{3}{2}$.

Differentiate the above equation with respect to x ,

$$2y\left(\frac{dy}{dx}\right) = 3ax^2$$

$$\frac{dy}{dx} = \frac{3ax^2}{2y}$$

At (1,1),

$$\frac{dy}{dx} = \frac{3a(1)^2}{2(1)} = \frac{3a}{2}$$

The slope is,

$$\frac{dy}{dx} = \frac{3}{2}$$

$$\frac{3a}{2} = \frac{3}{2}$$

$$a = 1$$

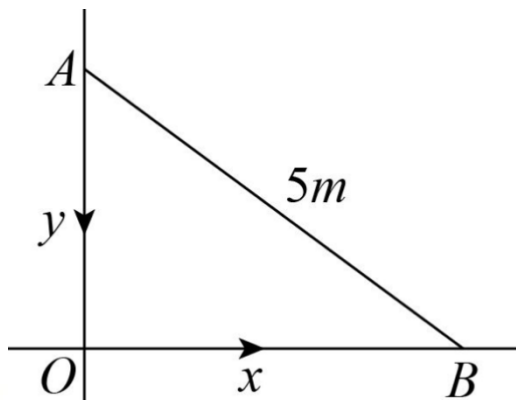
The point (1,1) lies on $y^2 = ax^3 + b$

$$1^2 = 1 \times 1^3 + b$$

$$b = 0$$

The point is (1,0)

69. It is given that the length of the ladder is: $AB = 5 \text{ m}$



From the geometry,

$$x^2 + y^2 = 5^2$$

Differentiate the above equation.

$$2x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} + y \frac{dy}{dx} = 0 \quad \text{..... (I)}$$

It is given that,

$$\frac{dx}{dt} = 3, \frac{dy}{dt} = -4$$

Substitute the values in the equation (I).

$$3x - 4y = 0$$

$$3\sqrt{25 - y^2} - 4y = 0$$

$$3\sqrt{25 - y^2} = 4y$$

Squaring both sides,

$$9(25 - y^2) = 16y^2$$

By solving,

$$y = 3$$

70. Consider

$$f(x) = ax^2 + bx + c$$

$$f(2) = 4a + 2b + c = 2 \quad \dots\dots (I)$$

$$f(3) = 9a + 3b + c = 5 \quad \dots\dots (II)$$

$$f(4) = 16a + 4b + c = 10 \quad \dots\dots (III)$$

Subtract equation (I) from equation (II),

$$5a + b = 3 \quad \dots\dots (IV)$$

Subtract equation (II) from equation (III),

$$7a + b = 5 \quad \dots\dots (V)$$

Subtract equation (IV) from equation (V),

$$2a = 2$$

$$a = 1$$

Substitute the value in equation (IV),

$$5 \times 1 + b = 3$$

$$b = -2$$

Now,

$$f'(x) = 2x - 2$$

$$f''(x) = 2$$

Since, $f''(x) > 0$ for some $x \in (2, 4)$

71. It is given that,

$$f(x) = x^2 e^{2x}$$

Differentiate the above equation.

$$f'(x) = 2e^{2x}x^2 + 2xe^{2x}$$

$$0 = 2e^{2x}(x^2 + x)$$

$$x^2 + x = 0$$

$$x = 0, -1$$

The maxima and minima is obtained at $-2, -1, 0, 2$ as the function bound.

Thus,

$$f(-2) = (-2)^2 e^{-4} = 4e^{-4}$$

$$f(-1) = (-1)^2 e^{-2}$$

$$f(0) = 0$$

$$f(2) = (2)^2 e^4 = 4e^4$$

Thus,

$$p = 4e^4 \text{ and } q = 0$$

$$pe^{-4} + qe^4 = 4e^4(e^{-4}) + 0 = 4e^0 = 4$$

72. Consider

$$\begin{aligned} I &= \int \frac{x + \sin x}{1 + \cos x} dx \\ &= \int \frac{x}{1 + \cos x} dx + \int \frac{\sin x}{1 + \cos x} dx \end{aligned}$$

Consider $I = I_1 + I_2$ (I)

Now,

$$I_1 = \int \frac{x}{1 + \cos x} dx = \int \frac{xdx}{1 + \left(\frac{1 - \tan^2 \frac{x}{2}}{1 - \tan^2 \frac{x}{2}} \right)}$$

$$= \int \frac{x \sec^2 \frac{x}{2} dx}{2}$$

$$= \int \frac{x}{2} \sec^2 \frac{x}{2} dx$$

$$= 2 \frac{x}{2} \tan \frac{x}{2} - 2 \log \left| \sec \frac{x}{2} \right| + c_1$$

$$= x \tan \frac{x}{2} - \log \left| \sec^2 \frac{x}{2} \right| + c_1$$

Now,

$$I_2 = \int \frac{\sin x}{1 + \cos x} dx$$

$$I_2 = -\log|1 + \cos x| + c_2$$

$$I_2 = \log \left| \frac{1}{1 + \cos x} \right| + c_2$$

$$I_2 = \log \left| \frac{1}{1 + \cos^2 \frac{x}{2}} \right| + c_2$$

$$I_2 = \log \left| \sec^2 \frac{x}{2} \right| + c_2$$

Now,

Substitute the value in equation (I).

$$I = x \tan \frac{x}{2} - \log \left| \sec^2 \frac{x}{2} \right| + c_1 + \log \left| \sec^2 \frac{x}{2} \right| + c_2$$

$$I = x \tan \frac{x}{2} + c$$

73. The integral expression is given as,

$$I = \int x^2 \left[\sqrt{2} \left(\sin \frac{\pi}{4} + x \right) + e^x \right] dx$$

$$I = \int x^2 \left[\sqrt{2} \left(\sin \frac{\pi}{4} \cos x + \sin x \cos \frac{\pi}{4} \right) + e^x \right] dx$$

$$I = \int x^2 \left[\sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x \right) + e^x \right] dx$$

Further simplify,

$$I = \int x^2 [\cos x + \sin x + e^x] dx$$

$$I = \int x^2 (\cos x + \sin x) dx + \int x^2 e^x dx$$

$$I = \left\{ \begin{aligned} &(x^2 + 2x - 2) \sin x - \\ &(x^2 - 2x - 2) \cos x \end{aligned} \right\} + \{x^2 e^x - 2x e^x + 2e^x\} + c$$

$$I = (x^2 + 2x - 2) \sin x + (-x^2 + 2x + 2) \cos x + (x^2 - 2x + 2) e^x + c$$

74. The integral expression is given as,

$$I = \int \frac{dx}{(x+1)^2 (x^2 + 1)}$$

Consider

$$\frac{1}{(x+1)^2 (x^2 + 1)} = \frac{A}{(x+1)^2} + \frac{B}{(x^2 + 1)} + \frac{Cx + D}{x^2 + 1}$$

$$1 = A(x^2 + 1) + B(x+1)(x^2 + 1) + (Cx + D)(x+1)^2$$

Substitute $x = -1$,

$$A = \frac{1}{2}$$

Substitute $x^2 = -1$

$$1 = (Cx + D)(x^2 + 2x + 1)$$

$$1 = (Cx + D)2x$$

$$1 = -2C + 2Dx$$

Substitute $x = 0$,

$$-\frac{1}{2} = C, D = 0$$

Comparing coefficient of x^3 ,

$$0 = B + C$$

$$B = \frac{1}{2}$$

Now,

$$\begin{aligned} I &= \int \left[\frac{1}{2} \frac{1}{(x+1)^2} + \frac{1}{2} \frac{1}{x+1} - \frac{1}{2} \frac{x}{x^2+1} \right] dx \\ &= \frac{1}{2} \left[-\frac{1}{x+1} + \log(x+1) - \frac{1}{2} \log(x^2+1) \right] + c \\ &= \log \sqrt{x+1} - \frac{1}{2} \log \sqrt{x^2+1} - \frac{1}{2(x+1)} + c \end{aligned}$$

75. The integral expression is given as,

$$\begin{aligned}
 I_n &= \int (\sin x + \cos x)^n dx \\
 &= \int (\sin x + \cos x)^{n-1} \cdot (\sin x + \cos x) dx \\
 &= (\sin x + \cos x)^{n-1} (\sin x - \cos x) + (n-1) \int (\sin x + \cos x)^{n-2} (\sin x - \cos x) dx \\
 &= (\sin x + \cos x)^{n-1} (\sin x - \cos x) + (n-1) \int (\sin x + \cos x)^{n-2} 2 - (\sin x + \cos x)^{n-2} dx \\
 &= (\sin x + \cos x)^{n-1} (\sin x - \cos x) + 2(n-1) I_{n-2} - (n-1) I_n \\
 n I_n - 2(n-1) I_{n-2} &= (\sin x + \cos x)^{n-1} (\sin x - \cos x) + c
 \end{aligned}$$

76. The expression is given as,

$$\begin{aligned}
 I &= \lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \dots + \sqrt{n}}{n^{3/2}} \\
 &= \lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \dots + \sqrt{n}}{n\sqrt{n}} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \dots + \sqrt{\frac{n}{n}} \right] \\
 &= \sum_{r=1}^n \lim_{n \rightarrow \infty} \frac{1}{n} \sqrt{\frac{1}{n}} \\
 &= \int_0^1 \sqrt{x} dx \\
 &= \frac{2}{3} \left[x^{3/2} \right]_0^1 = \frac{2}{3}
 \end{aligned}$$

77. The formula for the integration,

$$I_m = \int_0^{\infty} e^{-x} \sin^m x dx, \text{ where } m > 2 \text{ then,}$$

$$I_m = \frac{m(m-1)}{1+m^2} I_{m-2}$$

Now,

$$I_6 = \int_0^{\infty} e^{-x} \sin 6x dx$$

$$I_6 = \frac{6(5)}{1+6^2} I_4$$

$$I_6 = \frac{30}{37} I_4$$

Similarly,

$$I_4 = \int_0^{\infty} e^{-x} \sin 4x dx$$

$$I_4 = \frac{4(3)}{1+4^2} I_2$$

$$I_4 = \frac{12}{17} I_2$$

Similarly,

$$I_6 = \int_0^{\infty} e^{-x} \sin 6x dx$$

$$I_2 = \frac{2(1)}{1+2^2} I_0$$

$$I_2 = \frac{2}{5} I_0$$

And

$$I_0 = \int_0^{\infty} e^{-x} \sin^0 x dx$$

$$I_0 = \left[-e^{-x} \right]_0^{\infty} = 1$$

The value of I_6 is,

$$I_6 = \frac{30}{37} \times \frac{12}{17} \times \frac{2}{5} \times 1 = \frac{144}{629}$$

78. The equation is given as,

$$y = x^2 + 2x + 1$$

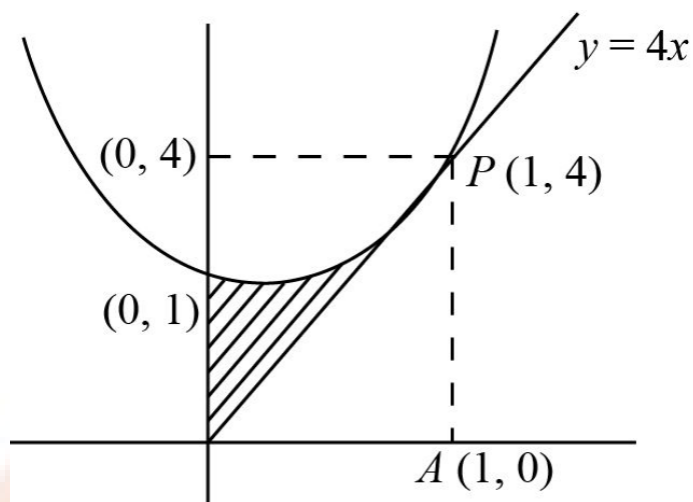
Differentiate the above equation with respect to x ,

$$\frac{dy}{dx} = 2x + 2$$

At $(1, 4)$

$$\frac{dy}{dx} = 2(1) + 2 = 4$$

The figure below represents the area bounded.



The bounded area is,

$$A = \int_0^1 y dx - \frac{1}{2}(OA \times AP)$$

$$A = \int_0^1 (x^2 + 2x + 1) dx - \frac{1}{2}(1 \times 4)$$

$$A = \left(\frac{x^3}{3} + x^2 + x \right)_0^1 - 2$$

$$A = \left(\frac{1}{3} + 1^2 + 1 - 0 \right) - 2$$

$$A = \frac{1}{3}$$

79. The differential equation is given as,

$$y = e^x (a \cos x + b \sin x)$$

Differentiate the above equation,

$$\frac{dy}{dx} = e^x (a \cos x + b \sin x) + e^x (-a \sin x + b \cos x)$$

$$\frac{dy}{dx} = y + e^x (-a \sin x + b \cos x) \quad \dots (I)$$

Again differentiate the above equation,

$$\frac{d^2 y}{dx^2} = \frac{dy}{dx} + e^x (-a \sin x + b \cos x) + e^x (-a \cos x - b \sin x)$$

$$\frac{d^2 y}{dx^2} = \frac{dy}{dx} - e^x (a \cos x + b \sin x) + \frac{dy}{dx} - y$$

$$\frac{d^2 y}{dx^2} = 2 \frac{dy}{dx} - y - e^x (a \cos x + b \sin x)$$

$$\frac{d^2 y}{dx^2} = 2 \frac{dy}{dx} - y - y$$

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$$

80. The solution is given as,

$$y = A(x)e^{-\int P dx}$$

The above equation is linear differential equation.

The integral factor is,

$$IF = e^{\int P(x) dx}$$

The solution is,

$$y(IF) = \int Q(x) \times IF dx$$

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx$$

Substitute the values in the above equation from given expression.

$$A(x) = \int Q(x)e^{\int P(x) dx} dx$$

$$A'(x) = Q(x)e^{\int P(x) dx}$$

AP EAMCET-2019-Engineering-21 Apr-Shift-2

Solutions – Physics

81. The division of the main scale: N

$$N_{\text{MSD}} = (N + 1)_{\text{VSD}}$$

$$1 \text{ VSD} = \frac{N}{N + 1} \text{ MSD}$$

The least count of vernier calliper is,

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \frac{N}{N + 1} \text{ MSD}$$

$$= \frac{N + 1 - N}{N + 1} \text{ MSD}$$

$$= \frac{1}{N + 1} \text{ MSD}$$

It is given that 1 MSD is a , Thus,

$$\text{LC} = \frac{a}{N + 1}$$

82. (A),

At mean position,

The velocity of the body is maximum that is $A\omega$.

(B),

The total energy during SHM is,

$$TE = \frac{1}{2}kA^2$$

The kinetic energy is,

$$KE = \frac{1}{2}k(A^2 - x^2)$$

It is given that, $KE = \frac{3}{4}TE$

Substitute the values in the above equation.

$$\frac{1}{2}k(A^2 - x^2) = \frac{3}{4} \times \frac{1}{2}kA^2$$

Simplify the above equation.

$$x = \pm \frac{A}{2}$$

(C),

The total energy during SHM is,

$$TE = \frac{1}{2}kA^2$$

The potential energy is,

$$PE = \frac{1}{2}kx^2$$

It is given that, $PE = \frac{3}{4}TE$

Substitute the values in the above equation.

$$\frac{1}{2}kx^2 = \frac{3}{4} \times \frac{1}{2}kA^2$$

Simplify the above equation.

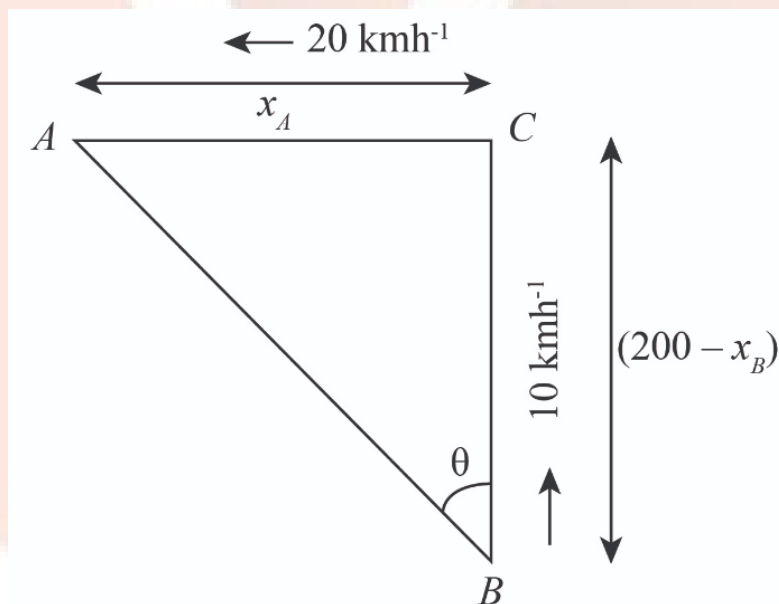
$$x = \pm \frac{\sqrt{3}}{2} A$$

(D),

The acceleration is maximum at extreme position.

Thus, option (3) is correct.

83. The figure below represents the ships A and B.



The time required to travel by ship A from A to C is,

$$t = \frac{x_A}{v_A}$$

$$t = \frac{x_A}{20}$$

The time required to travel by ship B is,

$$t = \frac{x_B}{v_B}$$

$$t = \frac{x_B}{10}$$

From above two equations,

$$\frac{x_A}{20} = \frac{x_B}{10}$$

$$x_A = 2x_B$$

The shortest distance is,

$$AB^2 = x_A^2 + (200 - x_B)^2$$

$$AB = \sqrt{(2x_A)^2 + (200 - x_B)^2}$$

$$AB = \sqrt{5x_B^2 - 400x_B + 40000} \quad \text{..... (I)}$$

$$\text{For the shortest distance, } \frac{d}{dx_B}(AB) = 0$$

$$0 = \frac{1}{2\sqrt{5x_B^2 - 400x_B + 40000}}(10x_B - 400) \quad \text{..... (II)}$$

$$x_B = 40 \text{ m}$$

Again differentiate equation (II).

$$\frac{d^2}{dx_B^2}(AB) > 0_{\text{at } x=40 \text{ m}}$$

The distance always greater than zero

Substitute the value in equation (I).

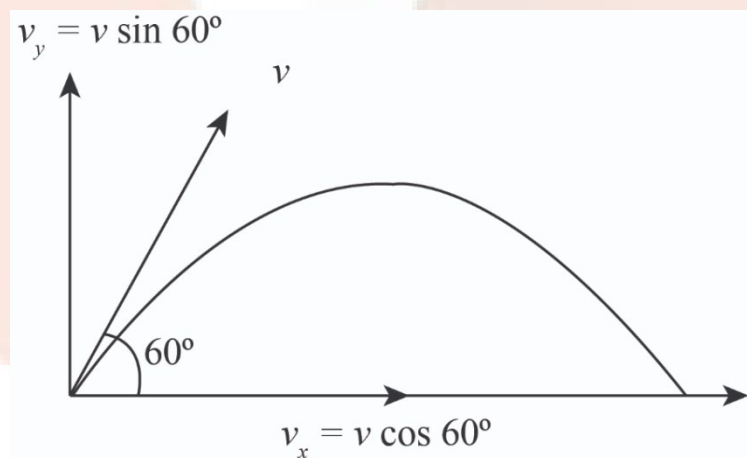
$$AB = \sqrt{5 \times 40^2 - 400 \times 40 + 40000}$$

$$AB = 80\sqrt{5} \text{ km}$$

The time after which the distance between the ships is shortest is,

$$t = \frac{x_B}{10} = \frac{40}{10} = 4 \text{ hr}$$

84. The figure below represents the projectile motion.



The vertical component is given as: 40 m/s.

$$v_y = v \cos 60^\circ$$

$$40 = \frac{\sqrt{3}}{2} v$$

$$v = \frac{80}{\sqrt{3}}$$

The expression for the time of flight is,

$$T = \frac{2v \sin 60^\circ}{10}$$

Substitute the values in the above equation.

$$T = \frac{2 \left(\frac{80}{\sqrt{3}} \right) \sin 60^\circ}{10}$$

$$T = 8 \text{ s}$$

At time, $t = \frac{T}{4} = \frac{8}{4} = 2 \text{ s}$, the velocity is,

$$V_x = \frac{80}{\sqrt{3}} = \frac{40}{\sqrt{3}} \text{ m/s}$$

And from the equation of motion,

$$V_y = v_y - gt$$

$$V_y = 40 - (10) \times 2$$

$$V_y = 20 \text{ m/s}$$

Thus, the velocity at $t = \frac{T}{4}$ is,

$$\begin{aligned} V &= \sqrt{V_x^2 + V_y^2} \\ &= \sqrt{\left(\frac{40}{\sqrt{3}}\right)^2 + 20^2} \\ &= 30.54 \text{ m/s} \end{aligned}$$

85. The linear density of the rope is, $\rho = 0.5 \text{ kg/m}$

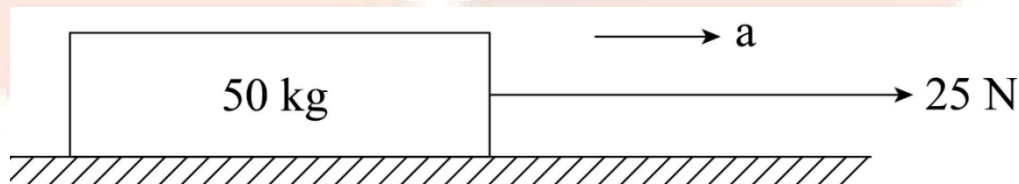
The mass of the rope is,

$$\begin{aligned} m &= \rho \times l \\ &= 0.5 \times 4 \\ &= 2 \text{ kg} \end{aligned}$$

The total mass of the system is,

$$M = 48 + 2 = 50 \text{ kg}$$

The figure below represents the free body diagram.



$$\begin{aligned} a &= \frac{F}{M} \\ &= \frac{25}{50} = 0.5 \text{ m/s}^2 \end{aligned}$$

The force acting on the block is,

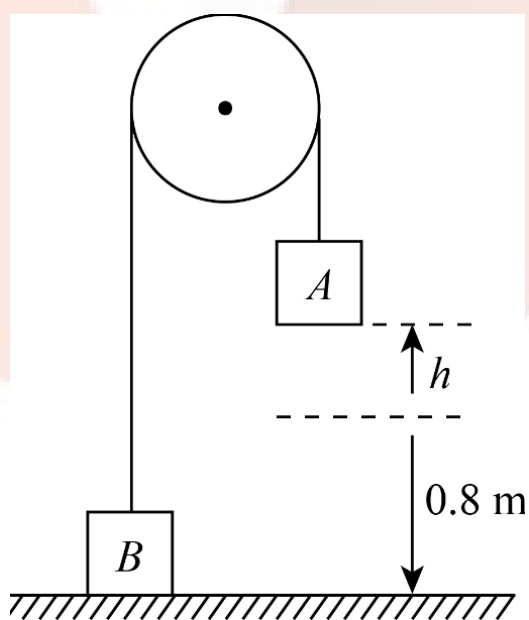
$$\begin{aligned}f &= ma \\&= 48 \times 0.5 \\&= 24 \text{ N}\end{aligned}$$

86. It is given that, $m_A = 1.5 \text{ kg}$, $m_B = 0.5 \text{ kg}$

The common acceleration of two masses is,

$$\begin{aligned}a &= \left(\frac{m_A - m_B}{m_A + m_B} \right) g \\&= \left(\frac{1.5 - 0.5}{1.5 + 0.5} \right) \times 10 \\&= 5 \text{ m/s}^2\end{aligned}$$

The figure below represents the block A is lifted until block touches the ground.



From the equation of motion,

$$v^2 = u^2 + 2gh$$

$$v^2 = 0 + 2(10)h$$

$$v^2 = 20h$$

For initial condition, the block A falls at the height of 0.8 m,

$$v_i^2 = u_i^2 - 2a(0.8)$$

$$0 = v^2 - 2(5)(0.8)$$

Substitute the values in the above equation.

$$v^2 = 2(5)(0.8)$$

$$20h = 2(5)(0.8)$$

$$h = 0.4 \text{ m}$$

The maximum height reached by the block B is

$$h_{\max} = 0.8 + 0.4 = 1.2 \text{ m}$$

$$h_{\max} = 120 \text{ cm}$$

87. The total energy at height 30 m is,

$$TE = mgh$$

$$TE = mg(30)$$

$$TE = 30mg$$

The velocity at $PE = 2KE$,

$$TE = KE + PE$$

$$30mg = \frac{1}{2}mv^2 + 2\left(\frac{1}{2}mv^2\right)$$

$$30m(10) = \frac{3}{2}mv^2$$

Simplify the above equation.

$$v = \sqrt{200} = 10\sqrt{2} \text{ m/s}$$

88. The force on the body is given as,

$$\vec{F} = -5x^4\hat{i} \text{ N}$$

The work done by the force is,

$$W = \int_2^{-2} F dx$$

$$W = \int_2^{-2} (-5x^4) dx$$

$$W = -5 \left[\frac{x^5}{5} \right]_2^{-2}$$

$$W = -5 \left(\frac{(-2)^5}{5} - \frac{(2)^5}{5} \right)$$

$$W = 64 \text{ J}$$

89. The torque required to stop the wheel is,

$$\tau = I\alpha$$

$$\tau = I\left(\frac{\omega}{t}\right)$$

Substitute the values in the above equation.

$$\tau = 2\left(\frac{50 \times \frac{2\pi}{60}}{60 \text{ sec}}\right)$$

$$\tau = \frac{\pi}{18} \text{ N} \cdot \text{m}$$

90. The kinetic energy during SHM is,

$$KE = \frac{1}{2}k(A^2 - x^2)$$

Or

$$KE = \frac{1}{2}k(\theta_0^2 - \theta^2)$$

From the above equation, the reduced kinetic energy (ΔE) is,

$$\Delta E \propto \theta^2$$

91. It is given that the amplitude of the motion, $A = 2x_0$

The time required to distance from compressed position to mean position is $\frac{T}{4}$

The displacement from mean position,

$$y = A \sin \omega t \text{ and}$$

$$x_0 = 2x_0 \sin \omega t$$

$$x_0 = 2x_0 \sin \frac{2\pi}{T} t$$

$$\sin \frac{\pi}{6} = \sin \frac{2\pi}{T} t$$

$$t = \frac{T}{12}$$

Thus, the time taken to hit the wall,

$$\frac{T}{4} + \frac{T}{12} = \frac{T}{3}$$

Or,

The expression for the time period is,

$$T = 2\pi \sqrt{\frac{m}{k}}$$

So, the time taken to hit the wall is calculated as,

$$\frac{T}{3} = \frac{2\pi}{3} \sqrt{\frac{m}{k}}$$

92. The expression for the amplitude of damped oscillator,

$$A = A_0 e^{-\lambda t}$$

For 1 min,

$$\frac{A_0}{2} = A_0 e^{-\lambda}$$

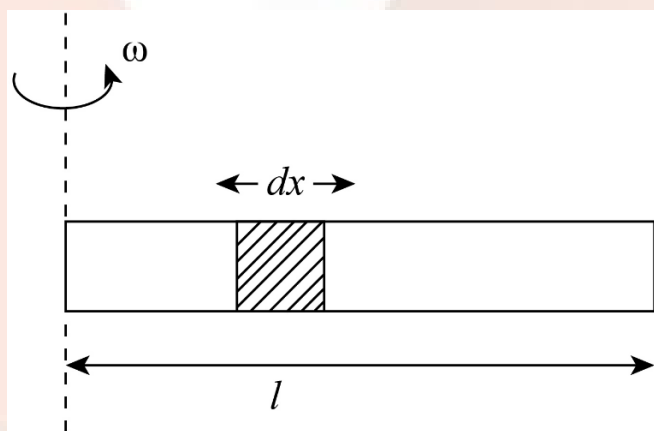
$$e^{-\lambda} = \frac{1}{2}$$

For 3 min,

$$\frac{A_0}{x} = \frac{A_0}{e^{(-\lambda)^3}} = \frac{1}{(2)^3}$$

$$x = 8$$

93. The figure below represents the uniform rod.



The centrifugal force per unit area of rod for elemental area,

$$F = \int_0^l \frac{m\omega^2 dx}{A}$$

$$F = \int_0^l \frac{(\rho Ax)\omega^2 dx}{A}$$

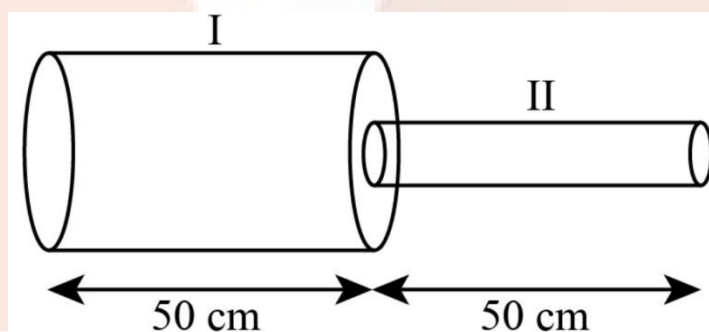
$$F = \rho\omega^2 \int_0^l (x) dx$$

Solve further,

$$F = \rho\omega^2 \left[\frac{x^2}{2} \right]_0^l$$

$$F = \frac{\rho\omega^2 l^2}{2}$$

94. The figure below represents the capillary tubes of same lengths but different radii.



From the Poiseuille's equation, the expression for the water resistance,

$$R = \frac{8\eta l}{\pi r^4}$$

$$R \propto \frac{1}{r^4}$$

The ratio of water resistance in tube I and II,

$$\frac{R_I}{R_{II}} = \frac{(r_{II})^4}{(r_I)^4} = \frac{16 \times 10^{-12}}{16 \times 16 \times 10^{-12}}$$

$$16R_I = R_{II}$$

Let $R_I = R$, $R_{II} = 16R$

Thus, the pressure difference between ends of first tube is,

$$\frac{p}{R + 16R} \times R = \frac{p}{17}$$

95. From the phenomenon of thermal expansion, plates P and Q are uniformly heated through the same range of temperature.

Thus, both the plates expand outwards.

Hence, both distance x and y decrease.

96. The expression for the rate of heat flow is,

$$\frac{dq}{dt} = \frac{msdT}{dt}$$

Substitute the given values in the above equation.

$$30 = \frac{\left(\frac{250}{1000}\right)s(47 - 46.9)}{10}$$

Simplify the above equation.

$$s = 12000 \text{ J/kg} \cdot \text{K}$$

97. The expression for the heat supplied at constant volume,

$$Q = nC_v dT$$

$$Q = \frac{m}{M} C_v dT$$

Substitute the given values in the above equation.

$$Q = \frac{35}{32} \left(\frac{5}{2} R \right) (80)$$

$$Q = \frac{35}{32} \left(\frac{5}{2} \times 8.31 \right) (80)$$

$$Q = 1817.812 \text{ J}$$

$$Q \approx 1.81 \text{ kJ}$$

98. The formula for the efficiency of Carnot engine,

$$\eta = \left(1 - \frac{T_2}{T_1} \right) \times 100$$

$$\text{For } \eta = 40\%, T_1 = 500 \text{ K},$$

$$\frac{40}{100} = \left(1 - \frac{T_2}{500}\right) \times 100$$

$$T_2 = 300 \text{ K}$$

For $\eta = 40\%$, $T_1 = 300 \text{ K}$ (the source temperature of Carnot engine for the same sink temp),

$$\frac{60}{100} = 1 - \frac{T_2}{300}$$

$$T_2 = 750 \text{ K}$$

99. It is given that, $\frac{R}{C_v} = 0.4$

From the formula,

$$\gamma - 1 = \frac{R}{C_v}$$

$$\gamma - 1 = 0.4$$

$$\gamma = 1.4$$

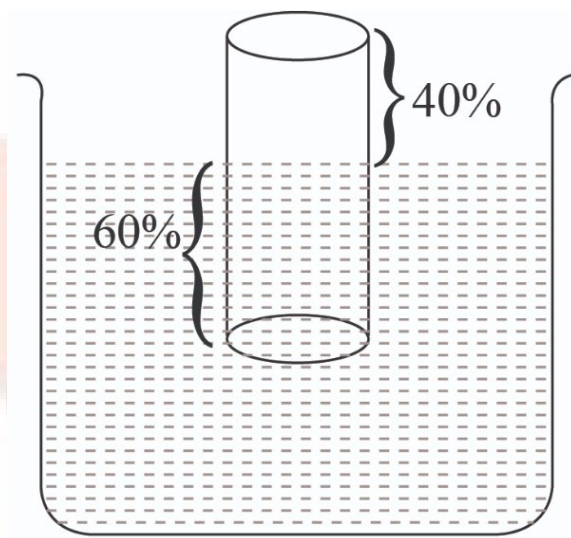
Thus, the gas is diatomic.

100. Consider the length of the tube be L

The fundamental frequency for the open organ pipe is,

$$f = \frac{V}{2L}$$

The figure below represents the 60% of tube is dipped vertically in water.



The length of the closed organ pipe is,

$$l' = \frac{40}{100} L$$

$$l' = \frac{2L}{5}$$

The fundamental frequency of air column is,

$$f' = \frac{V}{4l'}$$

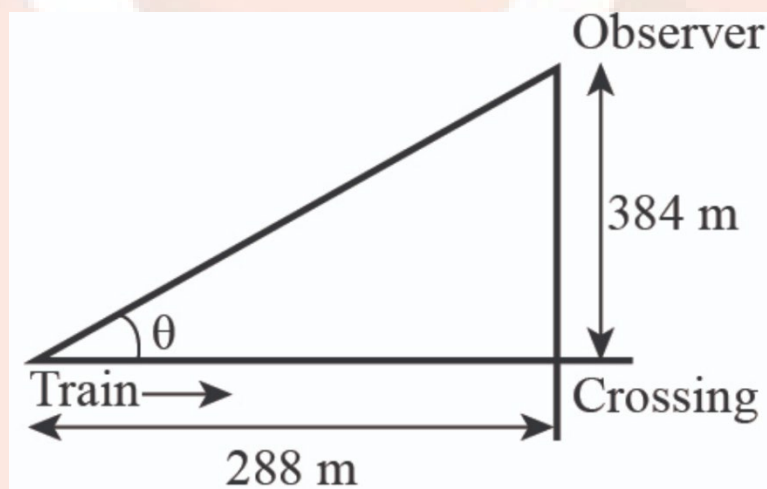
Substitute the values in the above equation.

$$f' = \frac{V}{4\left(\frac{2L}{5}\right)}$$

$$f' = \frac{5}{4}\left(\frac{V}{2L}\right)$$

$$f' = \frac{5}{4}f$$

101. The figure below represents the observer, train and crossing.



The speed of the observer,

$$u = 120 \cos \theta$$

$$u = 120 \times \frac{288}{\sqrt{384^2 + 288^2}}$$

$$u = 120 \times \frac{3}{5} \text{ km/h}$$

$$u = 120 \times \frac{3}{5} \times \frac{5}{18} \text{ m/s}$$

$$u = 20 \text{ m/s}$$

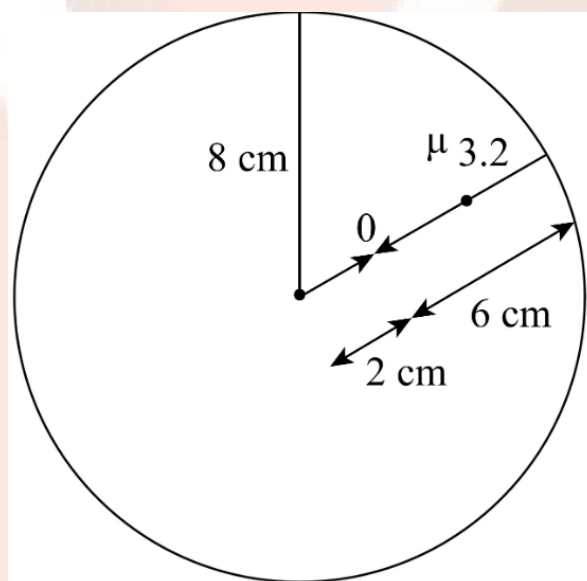
The formula for the frequency heard by observer, when a sound source is moving towards stationary observer,

$$f' = \left(\frac{v}{v - u} \right) f$$

$$f' = \left(\frac{340}{340 - 20} \right) 576$$

$$f' = 612 \text{ Hz}$$

102. The figure below represents transparent solid sphere.



From the figure,

$$u = -6 \text{ cm}$$

$$v = -3.2 \text{ cm}$$

The formula for the refractive indexes of first and second medium with respect to air,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \left(\frac{\mu_2 - \mu_1}{R} \right)$$

Substitute the values in the above equation.

$$\frac{1}{-3.2} - \frac{\mu}{-6} = \left(\frac{1 - \mu}{8} \right)$$

Simplify the above equation.

$$\mu = 1.5$$

103. The fringe width when Young's double slit experiment setup is immersed in water,

$$y = \frac{\lambda D}{\mu d}$$

Substitute the given values in the above equation.

$$y = \frac{6300 \times 10^{-10} \times 1.33}{1.33 \times 10^{-3}}$$

$$y = 6.3 \times 10^{-4} \text{ m}$$

$$y = 0.63 \text{ mm}$$

104. The centripetal force is equal to the electrostatic force, when an electron revolves around a proton.

$$\frac{m_e v^2}{r} = \frac{k q_1 q_2}{r^2}$$

Rewrite the above equation.

$$m_e v^2 r = k q_1 q_2$$

$$m_e (r\omega)^2 r = k q q$$

$$\omega = \sqrt{\frac{kq^2}{m_e r^3}}$$

Substitute the given values in the above equation.

$$\omega = \sqrt{\frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{9.1 \times 10^{-31} \times (0.53 \times 10^{-10})^3}}$$

$$\omega = 4.1 \times 10^{16} \text{ s}^{-1}$$

The radial acceleration of electron can be calculated as,

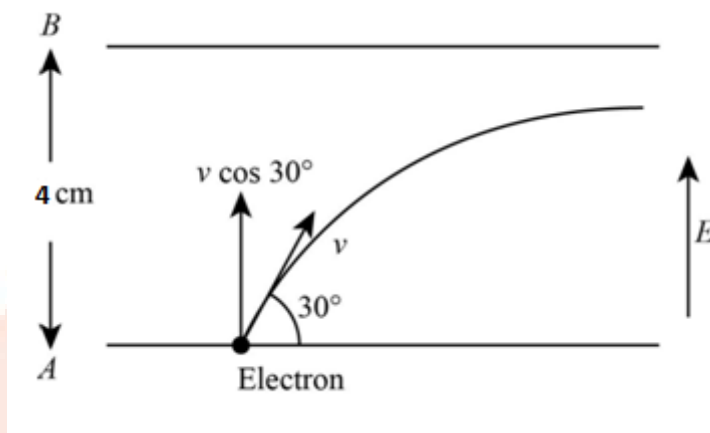
$$a_r = r\omega^2$$

$$a_r = 0.53 \times 10^{-10} \times (4.1 \times 10^{16})^2$$

$$a_r = 8.9 \times 10^{22} \text{ m/s}^2$$

$$a_r \approx 9 \times 10^{22} \text{ m/s}^2$$

105. The figure below represents the electron between two parallel plates A and B.



The expression for the maximum height is,

$$h_{\max} = \frac{v^2 \sin^2 30^\circ}{2a}$$

$$\frac{4}{100} = \frac{v^2 \sin^2 30^\circ}{2a} \quad \dots\dots (I)$$

The acceleration of the electron is,

$$F = q_e E$$

$$ma = q_e E$$

$$a = \frac{q_e E}{m}$$

Substitute the values in equation (I).

$$\frac{4}{100} = \frac{v^2 \sin^2 30^\circ}{2 \left(\frac{q_e E}{m} \right)}$$

$$\frac{4}{100} = \frac{v^2 \left(\frac{1}{4} \right)}{2 \left(\frac{q_e E}{m} \right)}$$

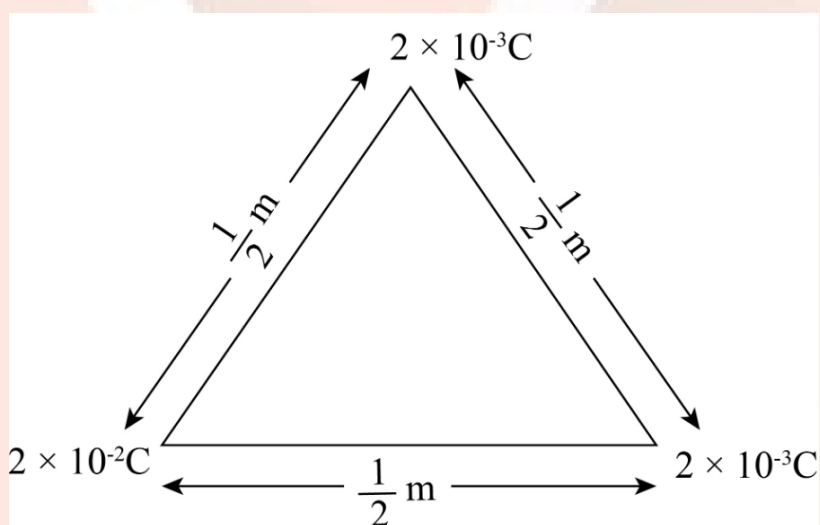
Rewrite the above equation.

$$v = \sqrt{\frac{4 \times 8}{100} \left(\frac{q_e E}{m_e} \right)}$$

$$v = \sqrt{\frac{4 \times 8}{100} \frac{1.6 \times 10^{-19} \times 45.5}{9.1 \times 10^{-31}}}$$

$$v = 1600 \text{ km/s}$$

106. The figure below represents the three points charges forming equilateral triangle.



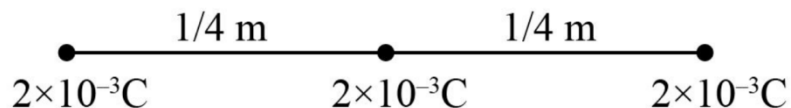
The energy system of three charges,

$$U = 3 \left(\frac{k q_1 q_2}{r} \right)$$

$$U = 3 \times \frac{k (2 \times 10^{-3} \times 2 \times 10^{-3})}{1/2}$$

$$U = 24k \times 10^{-6} \text{ J}$$

When one charge move to the midpoint of the line joining the other two charges,



The energy of the system,

$$U' = \frac{k(2 \times 2 \times 10^{-6})}{(1/4)} \times 2 + \frac{k(2 \times 2 \times 10^{-6})}{(1/2)}$$

$$U' = 40k \times 10^{-6} \text{ J}$$

The net energy,

$$\Delta U = U' - U$$

$$\Delta U = (40 - 24)k \times 10^{-6}$$

$$\Delta U = 16 \times 9 \times 10^3 \text{ J}$$

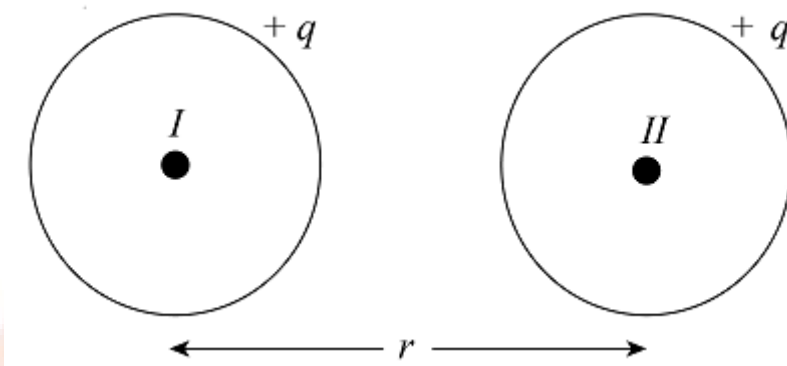
The energy supplied to the system at the rate of 2 kW,

$$2 \text{ kW} \times t = 16 \times 9 \times 10^3$$

$$2 \times 10^3 \times t = 16 \times 9 \times 10^3$$

$$t = 72 \text{ s}$$

107. The figure below represents the two identical spheres.



The force between the charged sphere is,

$$F = \frac{kqq}{r^2}$$

If 10% of electrons are transferred from one sphere to another, then the charge become on I sphere is

$$\left(q + \frac{q}{10} \right) = \frac{11}{10}q.$$

And the charge on II sphere is $\left(q - \frac{q}{10} \right) = \frac{9}{10}q.$

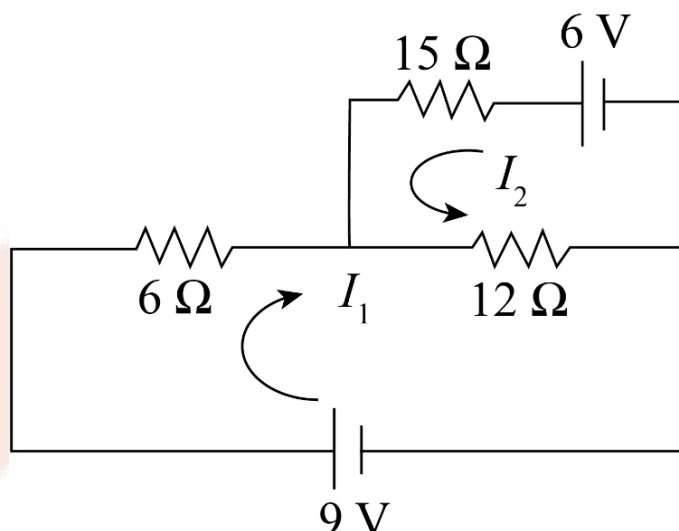
The force between the spheres is,

$$F' = \frac{k \left(\frac{11q}{10} \right) \left(\frac{9q}{10} \right)}{r^2}$$

$$F' = \frac{99}{100} \frac{kqq}{r^2}$$

$$F' = \frac{99}{100} F$$

108. The figure below represents the given circuit.



Apply Kirchhoff law in loop I,

$$-9 + 6I_1 + 12(I_1 + I_2) = 0$$

$$18I_1 + 12I_2 = 9$$

$$6I_1 + 4I_2 = 3 \quad \text{..... (I)}$$

Apply Kirchhoff law in loop II,

$$-6 + 15I_2 + 12(I_1 + I_2) = 0$$

$$12I_1 + 27I_2 = 6$$

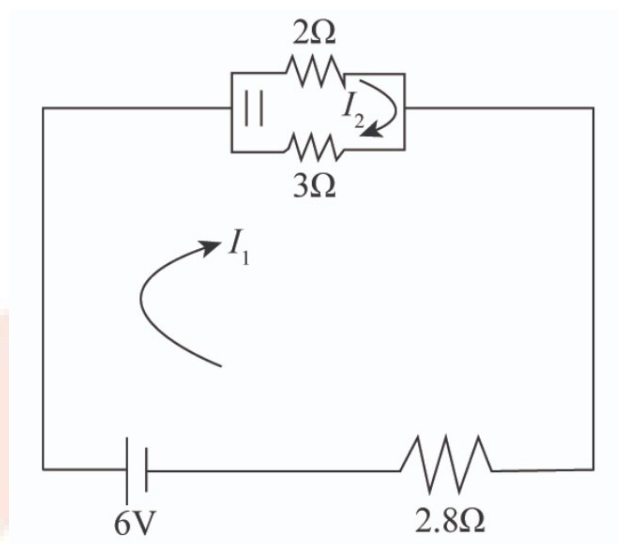
$$4I_1 + 9I_2 = 2 \quad \text{..... (II)}$$

From equation (I) and (II),

$$I_1 = 0.5 \text{ A}$$

$$I_2 = 0$$

109. At the steady state, the capacitor behaves as open circuit. The figure below represents the simple circuit,



Apply Kirchhoff law in loop I,

$$-6 + 3(I_1 + I_2) + 2.8I_1 = 0$$

$$5.8I_1 - 3I_2 = 6$$

..... (I)

Apply Kirchhoff law in loop II,

$$2I_2 + 3(I_2 - I_1) = 0$$

..... (II)

From equation (I) and (II)

$$I_2 = 0.9 \text{ A}$$

Thus, the current in 2Ω resistor is 0.9 A.

110. The expression for the magnetic field at center of a circular coil of n turns,

$$B = \frac{\mu_0 i}{2r} n$$

If length of the wire is doubled,

$$2(2\pi rn) = 2\pi r'n'$$

$$(2rn) = \frac{r}{3}n'$$

$$n' = 6n$$

For the same field at the centre,

$$\frac{\mu_0 in}{2r} = \frac{\mu_0 in'}{2r'}$$

$$\frac{\mu_0 in}{2r} = \frac{\mu_0 i(6n)}{2(r/3)}$$

$$i' = \frac{i}{18}$$

Volume will remain same after doubling the length of the coil.

$$l \times A = 2l \times A'$$

$$A = \frac{A}{2}$$

The resistance after doubling the length,

$$R' = \frac{\rho l'}{A'}$$

$$R' = \frac{\rho(2l)}{(A/2)}$$

$$R' = \frac{4\rho l}{A}$$

$$R' = 4R$$

The emf is,

$$E' = i'R'$$

$$E' = \left(\frac{i}{18}\right)(4R)$$

$$E' = \frac{2}{9}E$$

111. When the force is acting at right angle to the direction of motion of charge, then the work done is zero.

$$W = Fd \cos 90^\circ$$

$$W = 0$$

The electrostatic force is zero when a point charge moves in a circular path around another charge.

Therefore, the electrostatic force acts at right angle to the direction of motion of the charge.

112. Given:

$$x_{m_1} = 1.5 \times 10^{-2}$$

$$T_1 = (173^\circ\text{C} + 273) = 100 \text{ K}$$

From Curie's Law,

$$x_m \propto \frac{1}{T}$$

$$\frac{x_{m_1}}{x_{m_2}} = \frac{T_2}{T_1}$$

Substitute the values in the above equation.

$$\frac{1.5 \times 10^{-2}}{0.5 \times 10^{-2}} = \frac{T_2}{100}$$

$$T_2 = 300 \text{ K}$$

$$T_2 = 300 \text{ K} - 273$$

$$T_2 = 27^\circ\text{C}$$

The change in temperature is,

$$\begin{aligned}\Delta T &= T_2 - T_1 \\ &= 27^\circ\text{C} - (-173^\circ\text{C}) \\ &= 200^\circ\text{C}\end{aligned}$$

113. At time, $\Delta t = \frac{1}{10} = 0.1 \text{ s},$

Coil turned from $\theta = 0$ to 180°

The expression for the change in flux is,

$$\Delta \theta = \theta_f - \theta_i$$

$$\Delta \theta = NBA \cos 180^\circ - NBA \cos 0^\circ$$

$$\Delta \theta = -2NBA$$

The emf induced is,

$$E = -\frac{\Delta\theta}{\Delta t}$$

$$E = -\left(\frac{-2NBA}{\Delta t}\right)$$

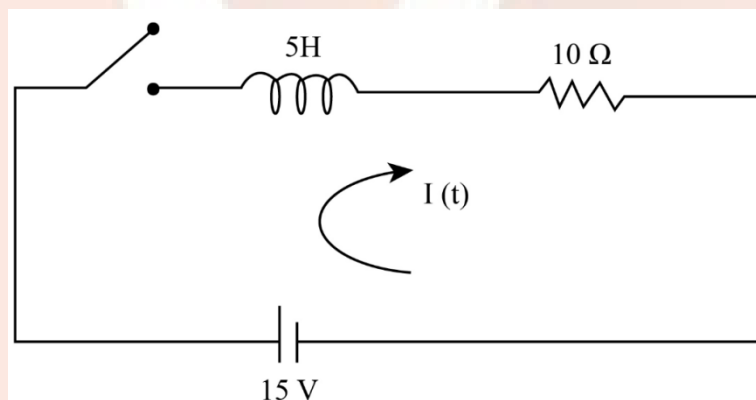
$$E = \frac{2NBA}{\Delta t}$$

Substitute the values in the above equation.

$$E = \frac{2 \times 1000 \times 0.4 \times 10^{-4} \times 0.05}{0.1}$$

$$= 0.04 \text{ V}$$

114. The figure below represents the given circuit.

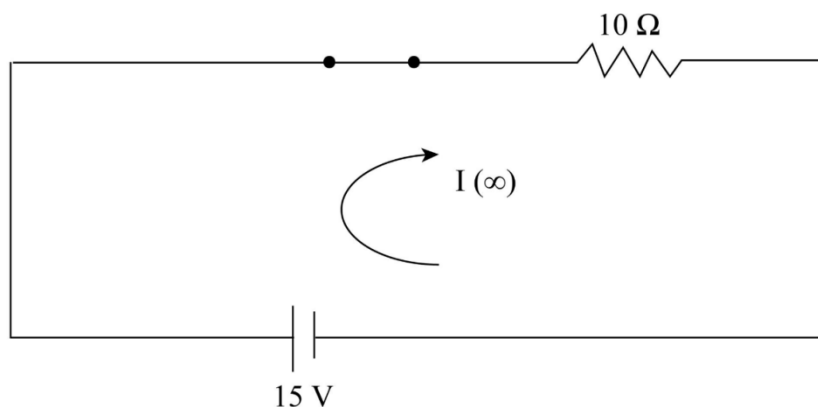


The current in the circuit is,

$$I(t) = I(\infty) - [I(\infty) - I(0)]e^{-t/\tau} \quad \text{..... (I)}$$

The time constant, $\tau = \frac{L}{R} = \frac{5}{10} = 0.5 \text{ s}$

At $t \rightarrow \infty$, inductor behaves as short circuit hence circuit will be,



$$I(\infty) = \frac{15}{10} = 1.5 \text{ A} \quad \text{..... (II)}$$

At $t \rightarrow 0$, inductor behaves as open circuit, $I(0) = 0$

$$I(t) = 1.5 - [1.5 - 0]e^{\frac{-t}{(1/2t)}}$$

At $t = 1 \text{ sec}$

$$I(1) = 1.5(1 - e^{-2}) \quad \text{..... (III)}$$

From equation (II) and (III),

$$\frac{I(\infty)}{I(0)} = \frac{1.5}{1.5(1 - e^{-2})}$$

$$\frac{I(\infty)}{I(0)} = \frac{e^2}{e^2 - 1}$$

115. The expression for the electromagnetic wave is,

$$I = \frac{1}{2} \epsilon_0 E_0^2 c$$

Rewrite the above equation.

$$E_0 = \sqrt{\frac{2I}{\epsilon_0 c}}$$

Substitute the values in the above equation.

$$E_0 = \sqrt{\frac{2 \times 53.1}{8.85 \times 10^{-12} \times 3 \times 10^8}}$$

$$E_0 = 200 \text{ NC}^{-1}$$

116. The expression for the circular path in a uniform magnetic fields is,

$$R = \frac{mv}{qB}$$

$$mv = RqB$$

From de Broglie wavelength,

$$\lambda = \frac{h}{mv}$$

Substitute the expression in the above equation.

$$\lambda = \frac{h}{Rq_{\alpha}B}$$

$$\lambda = \frac{h}{(+2C)BR}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{2 \times 1.6 \times 10^{-19} \times 0.125 \times 1 \times 10^{-2}}$$

$$\lambda = 1.65 \times 10^{-12} \text{ m}$$

117. The expression for the momentum of the photon,

$$p = \frac{h}{\lambda}$$

$$mv = \frac{h}{\lambda}$$

$$v = \frac{h}{m\lambda}$$

From the formula for the Bohr's for H-atom,

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Substitute the given values in the above equation.

$$\frac{1}{\lambda} = (1.097 \times 10^7) \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

$$\lambda = 4.86 \times 10^{-7} \text{ m}$$

The recoil speed of photon is,

$$v = \frac{6.62 \times 10^{-34}}{1.6 \times 10^{-27} \times 4.86 \times 10^{-7}}$$

$$v = 0.8513 \text{ m/s}$$

$$v = 0.814 \text{ m/s}$$

118. The total binding energy,

$$N^{14} = 7.5 \times 14 \text{ MeV}$$

$$N^{15} = 7.7 \times 15 \text{ MeV}$$

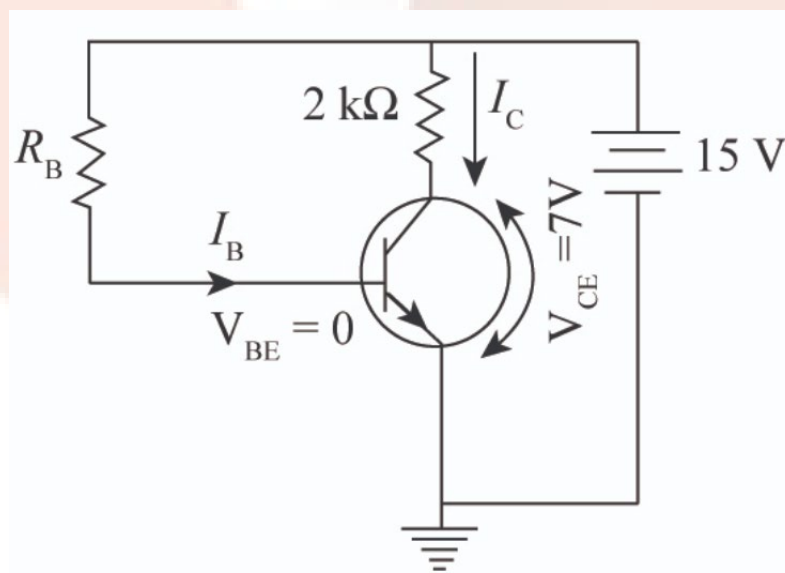
The energy required to remove a neutron from N^{15} ,

$$E = \text{BE}_t N^{15} - \text{BE}_t N^{14}$$

$$E = (7.7 \times 15 - 7.5 \times 14) \text{ MeV}$$

$$E = 10.5 \text{ MeV}$$

119. The figure below represents the common emitter circuit,



Apply Kirchhoff law,

$$-15 + 2I_C + V_{CE} = 0$$

$$-15 + 2I_C + 7 = 0$$

$$I_C = 4 \text{ mA}$$

The common emitter gain is,

$$\beta = \frac{I_C}{I_B}$$

$$100 = \frac{4 \times 10^{-3}}{I_B}$$

$$I_B = 0.04 \text{ mA}$$

120. The sky wave acts as a radio wave, thus, the suitable frequency suitable for beyond the horizon communication using the sky wave is 10 MHz or (10^7 Hz).

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121. The kinetic energy is given as,

$$h(\nu - \nu_0) = \frac{1}{2} m_e v^2$$

Here, the minimum energy required to eject an electron is $h\nu_0$ and the energy at which photons strike is $h\nu$.

On solving the equation we get,

$$v = \sqrt{\frac{2h(\nu - \nu_0)}{m_e}}$$

122. The number of neutron is given as

$$N_n = a + 0.32a = 1.32a$$

We know, the number of element is sum of number of protons and number of neutrons.

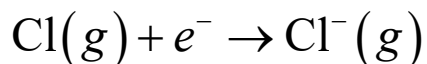
Substitute the given value in above expression we get,

$$1.81 = 1.32a$$

$$a = 78.02 \approx 78$$

The element which acquires the proton of 78 is platinum.

123. The reaction of chlorine on gain of electron is given as,



It is given the ground state of $\text{Cl}(g)$ is $x \text{ kJmol}^{-1}$

It is assumed the ground state of $\text{Cl}^-(g)$ is $y \text{ kJmol}^{-1}$.

The electron gain enthalpy is given as,

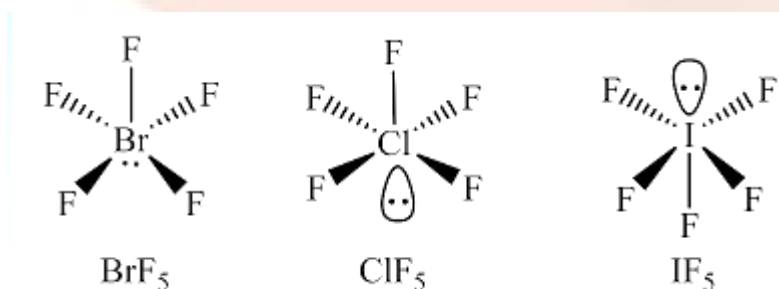
$$\Delta e_g H = E_p - E_R$$

$$-349 \text{ kJ/mol} = y - x$$

$$y = (x - 349) \text{ kJ/mol}$$

124. The molecules with different geometries and central atom with different hybridization are SF_4 , XeF_4 and CF_4 , their hybridization and geometry is sp^3d , sp^3d^2 and sp^3 , see-saw shape, square planar shape and tetrahedral shape respectively.

125. The molecules having shape of square pyramidal is 3. This is shown below:



126. The average kinetic energy is given as,

$$E = \frac{3}{2}RT$$

Substitute the given value in above expression.

$$4 \times 10^3 \text{ J/mol} = \frac{3}{2} \times 8.314 \text{ J/Kmol} \times T$$

$$T = 320 \text{ K}$$

The expression for root mean square velocity is given as,

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Substitute the value in above expression.

$$5 \times 10^2 \text{ m/s} = \sqrt{\frac{3 \times 8.314 \text{ J/kmol} \times 320 \text{ K}}{M}}$$

$$M \approx 32 \text{ g/mol}$$

127. The oxidation state of sulphur in $\text{H}_2\text{S}_2\text{O}_8$ is,

$$x + 2(-2) + 1(-1) + 1(-1) = 0$$

$$x = +6$$

The oxidation state of sulphur in H_2SO_5 is,

$$x + 2(-2) + 1(-1) + 1(-1) = 0$$

$$x = +6$$

The oxidation state of sulphur in H_2SO_3 is,

$$x + 2(-1) + 1(-2) = 0$$

$$x = +4$$

The oxidation state of sulphur in H_2SO_4 is,

$$x + 2(-1) + 1(-2) = 0$$

$$x = +4$$

The oxidation state of sulphur in $\text{H}_2\text{S}_2\text{O}_7$ is,

$$x + 2(-2) + 1(-1) + 1(-1) = 0$$

$$x = +6$$

The oxidation state of sulphur in SO_2Cl_2 is,

$$x + 2(-2) + 1(-1) + 1(-1) = 0$$

$$x = +6$$

The oxidation state of sulphur in SOCl_2 is,

$$x + 1(-2) + 2(-1) = 0$$

$$x = +4$$

Thus, the compound having +6 O.S is 4.

128. The expression for Gibbs energy is given as,

$$\Delta G = \Delta H - T\Delta S$$

Substitute the given value in above expression.

$$\Delta G = -2.5 \times 10^3 - 298 \times 7.4$$

$$\Delta G = -4.7 \times 10^3 \text{ cal}$$

The value of Gibbs energy comes out to be negative, which shows the reaction is spontaneous in nature.

129. The equilibrium constant for the given reaction is,

$$K_c = \frac{[\text{CO}(g)]^2}{[\text{CO}_2(g)]}$$

Substitute the given value in above expression.

$$0.036 = \frac{[\text{CO}(g)]^2}{0.004}$$

$$[\text{CO}(g)]^2 = 0.036 \times 0.004$$

$$[\text{CO}(g)] = 1.2 \times 10^{-2} \text{ mol/L}$$

130. The Ph value for an acidic buffer is given as,

$$\text{pH} = \text{p}K_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{CH}_3\text{COONa}]}{[\text{CH}_3\text{COOH}]}$$

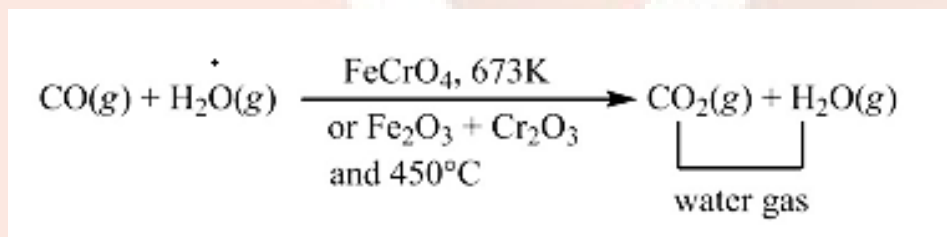
Substitute the value in above expression.

$$\text{pH} = 4.76 + \log \frac{1}{2}$$

$$\text{pH} = 4.76 + \log 5 - \log 10$$

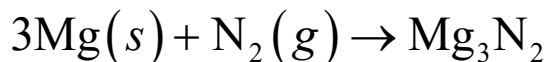
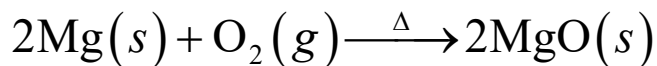
$$\text{pH} = 4.46$$

131. The water-gas shift reaction describes the reaction of carbon monoxide and water vapor to form carbon dioxide and hydrogen. In this type of reaction, syn gas is oxidized by mixing it with more steam at a temperature of 673 K in presence of Fe_2O_3 or Cr_2O_3 as a catalyst. The reaction is shown below.

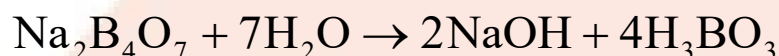


132. On burning the magnesium in air the products formed are magnesium oxide and magnesium nitride. Now, on hydrolysis of magnesium nitride the products come out to be ammonia and magnesium hydroxide. Thus, by Ostwald's process the compound D is NH_3 and compound C is Mg(OH)_2 .

The corresponding reactions are shown below:



133. The reaction used for preparation of boric acid is given as,

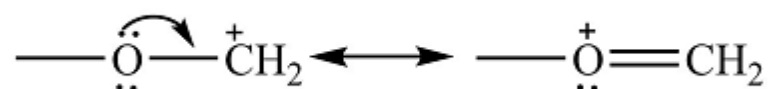


134. Ge, Sn and Pb are present in trace amounts, thus the statement I is correct.

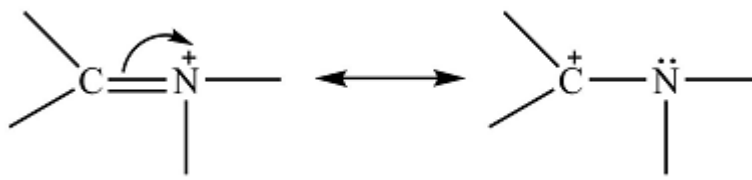
All the elements of group 14 are available in solid state. Thus, statements III is correct.

135. The process is known as 'Wacker's process' and the catalyst used in this process is palladium, which exists in the form of PdCl_2 .

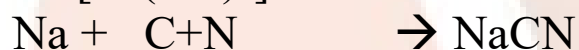
136. In figure A, the curved arrow shows the electrons are shifted from 'O' atom to adjacent bond. This resonating structures is shown below:



In fig B, the curved arrow shows shifting of π - bond between C and N atoms to adjacent atom N. In this case as the nitrogen is most electronegative, thus it pulls the electron towards itself.



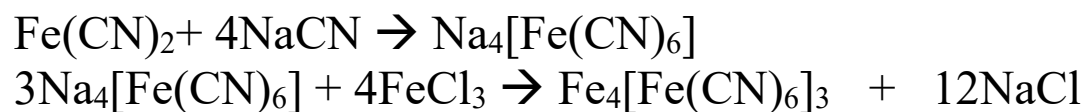
137. Lassaigne's test is used to detect the presence of N, S, Cl, Br, I in organic compounds using sodium fusion extract. In the case of nitrogen, Prussian blue colour is obtained due to the formation of $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$. Reactions involved in this are as follows-



(C&N
from
organic
compound

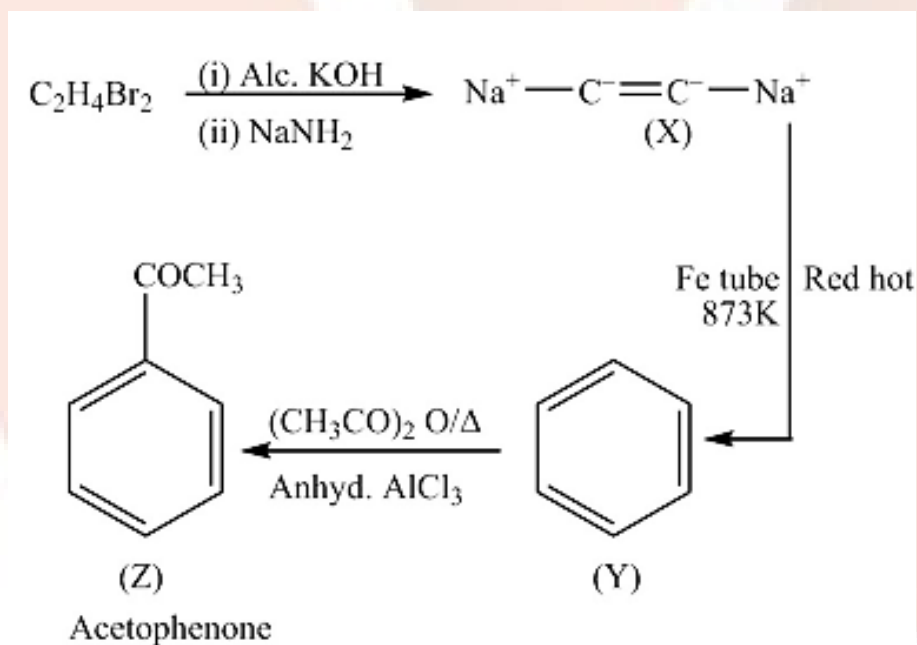


Obtained
from
sodium
extract



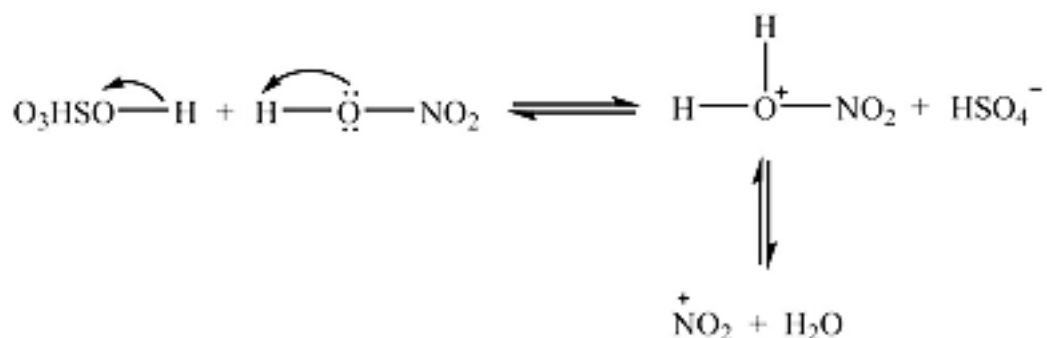
Ferric
ferrocyanide
(Prussian
blue
coloured
compound)

138. The sequence of the reaction is shown below:



139. The process in which equimolar mixture of nitric acid and sulphuric acid is termed as nitration mixture. The reaction

below shows the generation of nitronium ion from nitrating mixture,



140. Atoms of the element Y (as anions) make ccp and those of the element X (as cations) occupy all the octahedral voids.

Since, $\frac{1}{2}$ of the octahedral and $\frac{1}{2}$ of tetrahedral voids are occupied by the atoms of X.

The total number of atoms of X is,

$$N_X = \frac{1}{2}N + \frac{2N}{2} = \frac{3}{2}N$$

The ratio of X:Y is $\frac{3}{2}N : N = 3 : 2$

The formula comes out to be X_3Y_2 .

141. The expression for moles of CHCl_3 is given as,

$$n = \frac{m_{\text{CHCl}_3}}{M_{\text{CHCl}_3}}$$

Here, the mass of CHCl_3 is m_{CHCl_3} and its molar mass is M_{CHCl_3} .

Substitute the given value in above formula we get,

$$n = \frac{59.75 \text{ g}}{119.5 \text{ g} \cdot \text{mol}^{-1}} = 0.5 \text{ mol}$$

The expression for moles of CH_2Cl_2 is given as,

$$n = \frac{m_{\text{CH}_2\text{Cl}_2}}{M_{\text{CH}_2\text{Cl}_2}}$$

Here, the mass of CH_2Cl_2 is $m_{\text{CH}_2\text{Cl}_2}$ and its molar mass is $M_{\text{CH}_2\text{Cl}_2}$.

Substitute the given value in above formula we get,

$$n = \frac{21.25 \text{ g}}{85 \text{ g} \cdot \text{mol}^{-1}} = 0.25 \text{ mol}$$

The mole fraction of CH_2Cl_2 is given as:

$$\chi_{\text{CH}_2\text{Cl}_2} = \frac{n_{\text{CH}_2\text{Cl}_2}}{n_{\text{CH}_2\text{Cl}_2} + n_{\text{CHCl}_3}}$$

$$\chi_{\text{CH}_2\text{Cl}_2} = \frac{0.25}{0.25 + 0.5} = 0.33$$

The mole fraction of CHCl_3 is given as:

$$\chi_{\text{CHCl}_3} = 1 - \chi_{\text{CH}_2\text{Cl}_2}$$

$$\chi_{\text{CHCl}_3} = 1 - 0.33 = 0.67$$

The mole fraction of CHCl_3 in vapour phase is given as,

$$y_{\text{CHCl}_3} = \frac{p_{\text{CHCl}_3}}{p_1^0 + (p_2^0 - p_1^0)x_2}$$

$$y_{\text{CHCl}_3} = \frac{134 \text{ mm}}{200 + (415 - 300)0.33}$$

$$y_{\text{CHCl}_3} = 0.494$$

The mole fraction of CH_2Cl_2 in vapour phase is given as,

$$y_{\text{CH}_2\text{Cl}_2} = \frac{p_{\text{CH}_2\text{Cl}_2}}{p_1^0 + (p_2^0 - p_1^0)x_2}$$

$$y_{\text{CH}_2\text{Cl}_2} = \frac{136.95 \text{ mm}}{200 + (415 - 300)0.33}$$

$$y_{\text{CH}_2\text{Cl}_2} = 0.505$$

142. The molarity is given as,

$$\Delta T_b = iK_b \times m$$

$$m = \frac{\Delta T_b}{iK_b}$$

Substitute the given value in above expression.

$$m = \frac{0.01}{1.92 \times 0.52} \text{ mol/kg}$$

$$m = 0.01 \text{ mol/kg}$$

143. The Faraday's first law of electrolysis is given as,

$$w = \frac{W \times it}{96500 \text{ C} \cdot \text{mol}^{-1}}$$

Here, the equivalent weight of copper is given as W .

Substitute the given value in above expression.

$$0.4725 = \frac{63 \times i \times 15 \times 60}{2 \times 96500 \text{ C} \cdot \text{mol}^{-1}}$$

$$i = 1.608 \text{ A}$$

144. The expression for Arrhenius equation is given as,

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Substitute the given value in above expression.

$$\log \frac{2.60 \times 10^{-3}}{2.60 \times 10^{-5}} = \frac{E_a}{2.303 \times 8.314} \left[\frac{500 \text{ K} - 400 \text{ K}}{500 \text{ K} \times 400 \text{ K}} \right]$$

$$E_a = 76.6 \times 10^3 \text{ J/mol}$$

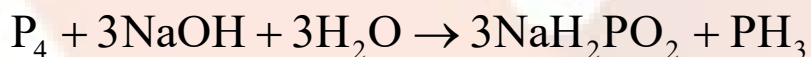
$$E_a = 76.6 \text{ kJ/mol}$$

145. The catalyst for the reactions is,

List-I Reaction	List-II Catalyst
A) $\text{N}_2(g) + 3\text{H}_2(g) \rightarrow 2\text{NH}_3(g)$	IV) Fe
B) $2\text{H}_2(g) + \text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(l)$	II) Pt
C) $\text{CO}(g) + 3\text{H}_2(g) \rightarrow \text{CH}_4(g) + \text{H}_2\text{O}(g)$	I) Ni
D) $\text{CO}(g) + 2\text{H}_2(g) \rightarrow \text{CH}_3\text{OH}(g)$	III) Zn-Cr ₂ O ₃

146. The vapour phase refining is the process of refining metal by converting it into its volatile compound and then, decomposing it to obtain a pure metal. The metals refined using this process is Ni, Ti, V, Th, Zr.

147. The reaction of phosphorous with concentrated NaOH is shown below:



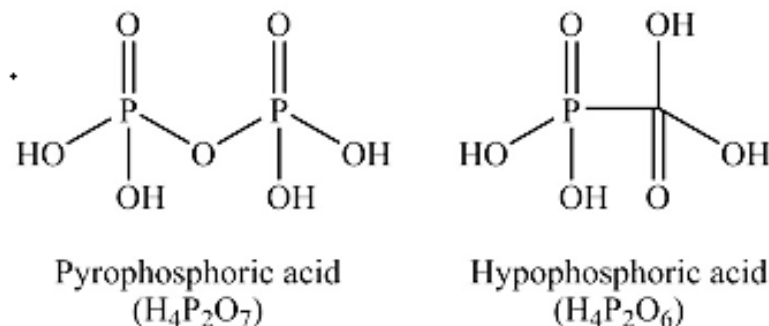
The oxidation state of phosphorous is given as:

$$1 + 2(1) + x + 2(-2) = 0$$

$$x = +1$$

Thus, the oxidation of P is +1.

148. The structure of pyrophosphoric acid and hypophosphoric acid is shown below,



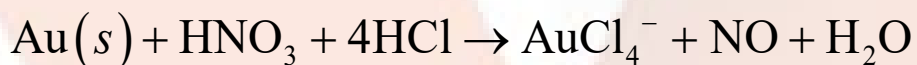
The number of P-OH bond in both cases is 4.

149. The reaction of sodium nitrite with sulphuric acid is shown below:



Here, the molecule X is NO.

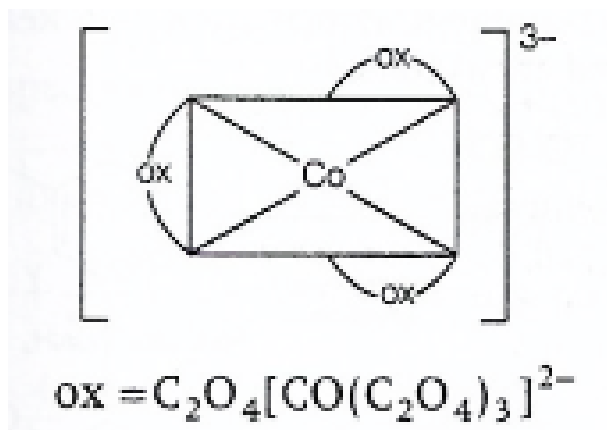
The reaction of gold with aqua-regia is shown below:



Here, the molecule Y is NO.

150. Among all, the compound $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ as the $\text{C}_2\text{O}_4^{2-}$ is a bidentate ligand which forms chelation and will be most stable.

The structure is shown below:



151. The molecule CuI_2 is not stable, the iodide ion in this will act as reducing agent that converts Cu(II) to Cu(I) and some of its part gets oxidized to iodine.

152. The correct match is,

List-I

List-II

A) Addition polymer

V) Polythene

B) Condensation polymer

I) Bakelite

C) Acrilan

IV) Vinyl cyanide

D) Rubber

II) 2-methyl-1, 3-butadiene

A-V, B-I, C-IV, D-II.

153. When DNA is hydrolysed adenine and thymine are obtained in equal quantities as these two are bonded by H- bonds.

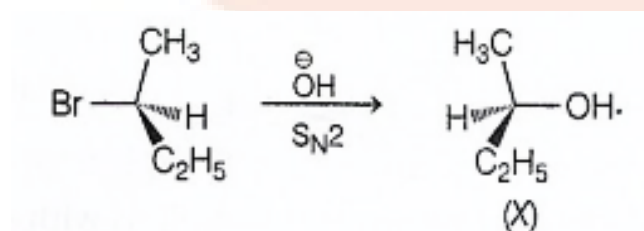
Addison disease is due to the abnormal functioning of adrenal cortex. It is due to deficiency of minerals corticoids.

Thus, the statements I and IV are correct.

154. The correct pair of medicine is shown below,

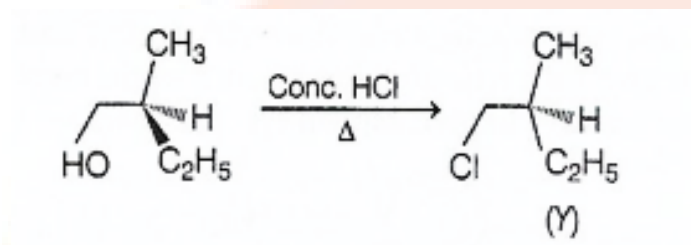
Chloramphenicol	Antibiotic
Nardil	Antidepressant
Histamine	Organic nitrogenous compound
Salvarsan	Antibiotic
Norethindrone	Antifertility drug
Alitame	Artificial sweetening agent

155. The reaction 1 is given as:



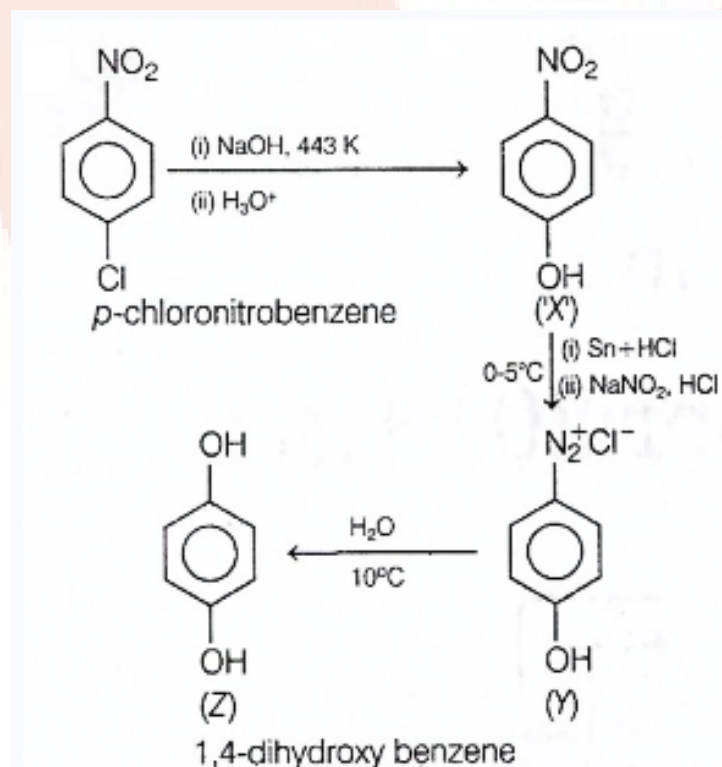
The reaction 1 is of S_N1 mechanism, in which the attack of Nu^+ occurs from the backside and the leaving group leaves from the front side.

The reaction 2 is given as:

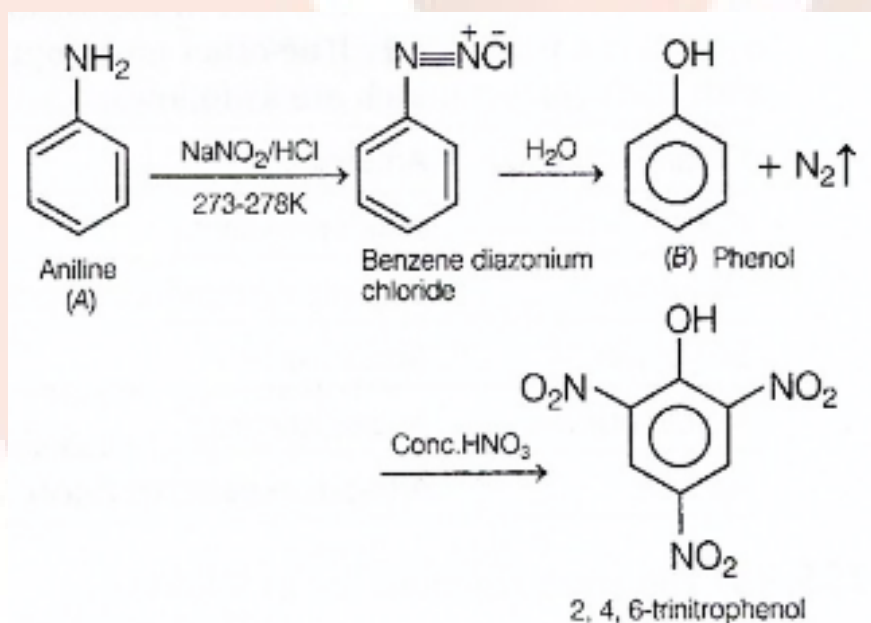


The reaction 2 is of S_N2 mechanism, in which no inversion of product takes place.

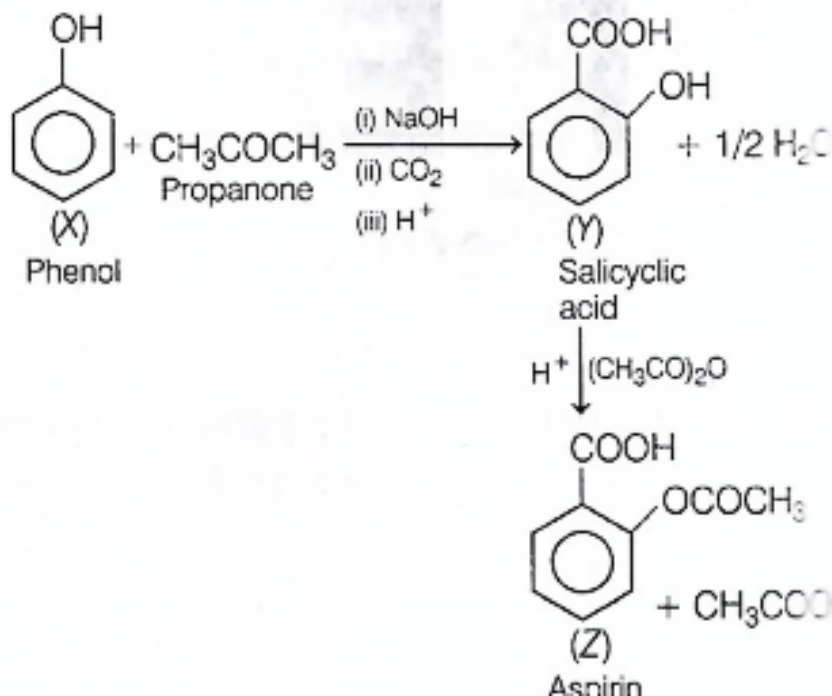
156. The sequence of reaction is shown below:



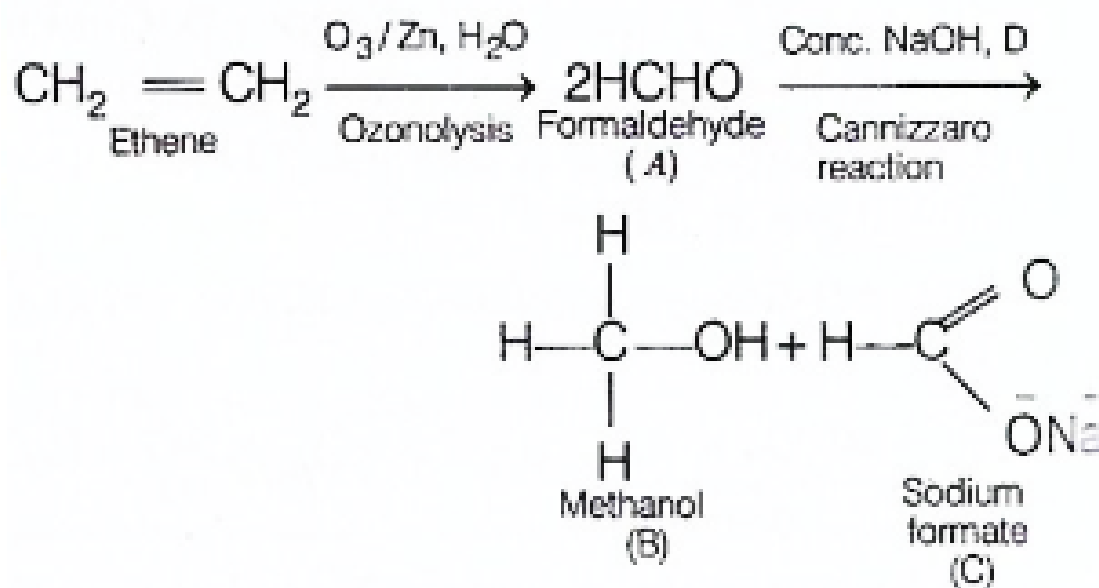
157. The reaction of aniline with NaNO_2/HCl at 273-278 K is shown below,



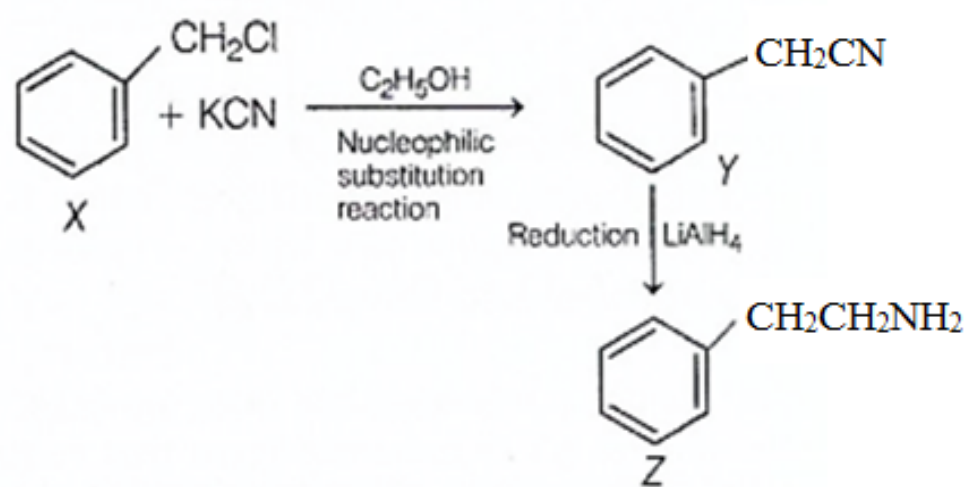
158. The sequence of reaction is shown below:



159. The sequence of reaction is given below:



160. The reaction of benzyl chloride with KCN is given as,



2-phenyl ethan-1-amine